

Chiziqli algebra I va Math I 1–4 haftalar

Har bir kurs fayli va tayinlangan kitob bo'limlaridagi barcha savollar va barcha formulalar — har biri to'liq bosqichma-bosqich yechim bilan. 4-haftaning to'liq materiali kiradi: chegara qonunlari, standart chegaralar, aniqlanmagan holdlar, asimptotik ekvivalentlik va O va o Landau belgilari.

Mustaqil o'rganish va Test 1 ga tayyorgarlik uchun tuzilgan

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Manbalar: kurs konspektlari, ma'ruzalar, ISA, amaliy mashg'ulotlar; OpenStax *Calculus 1*; Strang, *Introduction to Linear Algebra* (4e) va *Linear Algebra and its Applications* (4e)

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1 Kurs xaritasi va hujjatdan foydalanish

Ushbu hujjat — birinchi semestrning parallel ikki kursi, **Chiziqli algebra I** va **Math I** (kuz 2026), birinchi to'rt haftasi uchun to'liq va mustaqil o'quv qo'llanmasi. U kurs repozitoriysining `course-files/` papkasidagi har bir tegishli manbadan qo'lda tuzilgan: haftalik Notes (konspektlar), ma'ruzalar slaydlari, amaliy mashg'ulotlar varaqalari, Mustaqil o'quv topshiriqlari (ISA), TA muammolar to'plami, Test 1ga tayyorgarlik bo'yicha GUIDANCE va ikki darslik (OpenStax Calculus 1 hamda Gilbert Strang'ning chiziqli algebra kitoblari). Ushbu manbalarda 1–4 haftalar uchun uchraydigan har bir savol shu yerda to'liq, bosqichma-bosqich yechim bilan berilgan, manbalarda ishlatilgan har bir formula, ayniyat, ta'rif va standart chegara esa haftalik formulalar kataloglariga jamlangan.

Qamrov rasmiy kurs tuzilmasiga mos. Math I da 1-haftada son sistemalari, pozitsion yozuv, ilmiy va muhandislik yozuvi hamda kompleks sonlar; 2-haftada funksiyalar (OpenStax §1.2 va §1.5, giperbolik funksiyalardan tashqari); 3-haftada chegaralar (OpenStax §2.1, §2.2, §2.5); 4-haftada esa chegaralarning mustahkamlanishi (OpenStax §2.3 va standart chegaralar, aniqlanmagan holdlar, asimptotik ekvivalentlik hamda O va o Landau belgilari bo'yicha 4-hafta konspekti) o'rganiladi. Chiziqli algebra I da 1-haftada vektorlar, chiziqli birikmalar, span va chiziqli mustaqillik; 2-haftada skalyar ko'paytma, vektor ko'paytma va matritsalariga kirish; 3-haftada matritsalar algebrasi, determinantlar va aralash ko'paytma; 4-haftada esa matritsa rangi va Gauss-Jordan yo'q qilish usuli orqali teskari matritsa ko'rib chiqiladi, so'ng 1–3 haftalar material bo'yicha Test 1 o'tkaziladi. GUIDANCE varaqasi Test 1 ta'kidlaganidek, asosiy e'tibor 1-hafta konspektlarining 2–3-qismlariga (ilmiy/muhandislik yozuvi, kompleks sonlar), OpenStax §1.2 va §1.5 hamda OpenStax §2.2 ga qaratilgan, shuning uchun quyida aynan shu bo'limlarga eng batafsil ishlov berilgan.

Hujjatning har bir haftalik qismi bir xil ichki tuzilmaga ega. Avval formulalar katalogi keladi: o'sha haftada ishlatiladigan har bir ta'rif, teorema, ayniyat va standart chegara rasmiy shaklda yozilgan va uzluksiz raqamlangan, shunda har bir formulani yechimlardan yoki ma'ruzalardan havola qilish mumkin. So'ng savollar keladi: ular manba fayllarida paydo bo'lgan tartibda guruhlangan, har biri manbasi bilan belgilangan va darhol to'liq yechim bilan davom etadi. Manbalarda rasmda tasvirlangan funksiya orqali berilgan mashqlar — grafikdan o'qilgan qiymatlar keltirilgan holda, darsda qilinganidek yechiladi. Yakuniy qismda majburiy o'qish bo'limlaridagi va Strang kitoblaridagi qo'shimcha darslik mashqlari, yana to'liq yechimlar bilan jamlangan, shunda bu savollar bazasi so'ralganidek “oxirigacha” to'liq bo'ladi.

Hafta	Math I	Chiziqli algebra I
1	Sonlar to'plamlari $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$; pozitsion sistemalar; ilmiy/muhandislik yozuv; kompleks sonlar	Vektorlar; chiziqli birikmalar; span; chiziqli mustaqillik
2	Funksiyalar: sinflar, soha/qiymatlar to'plami, simmetriya, kompozitsiya, teskari funksiyalar; eksponensial va logarifmik funksiyalar (§1.2, §1.5)	Skalyar ko'paytma, burchaklar, proyeksiyalar; vektor ko'paytma, yuzalar; matritsalar: amallar va Ax
3	Chegaralar: intuitiv ta'rif, bir tomonlama chegaralar, cheksiz chegaralar, vertikal asimptotalar, ε - δ (§2.1, §2.2, §2.5)	Matritsalar algebrasi; transponirlash; matritsalarini ko'paytirish; determinantlar; aralash ko'paytma
4	Chegaralar II: chegara qonunlari, siqish teoremasi, standart chegaralar, aniqlanmagan holdlar, asimptotika, Landau belgilari (§2.3 + konsekt); Test 1	Matritsa rangi; Gauss-Jordan orqali teskari matritsa; Test 1

2 1-hafta — Son sistemalari, yozuv, kompleks sonlar; Vektorlar

2.1 Math I: 1-hafta formulalar katalogi

2.1.1 Sonlar to'plamlari va pozitsion sistemalar

Standart sonlar to'plamlarining ichma-ichiga joylashuvi kursning birinchi struktura fakti bo'lib, keyingi har bir tasniflash mashqi unga tayanadi. Ratsional sonlar — aynan butun sonlarning kasrlari, irratsional sonlar esa ratsional bo'lmagan haqiqiy sonlardir.

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}, \quad \mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\} \quad (\text{irratsionallar}) \quad (1)$$

$$q \in \mathbb{Q} \iff q = \frac{a}{b}, \quad a \in \mathbb{Z}, b \in \mathbb{Z}, b \neq 0, \quad d(a, b) = |a - b| \quad (\mathbb{R} \text{ da masofa}) \quad (2)$$

Asosi b bo'lgan pozitsion sanoq sistemasi sonni asosning darajalari orqali yozadi; bir sonning ko'p yozuvlari bo'lishi mumkin, lekin uning qiymati yozuvdan mustaqil. Asoslarni o'zgartirish quyidagi yoyilmaga (va teskari yo'nalishda ketma-ket bo'lishga) tayanadi.

$$(a_n a_{n-1} \dots a_1 a_0)_b = \sum_{k=0}^n a_k b^k, \quad 0 \leq a_k \leq b-1, \quad (3)$$

$$13_{10} = 1101_2 = 15_8 = D_{16}, \quad A = 10, B = 11, \dots, F = 15 \quad (\text{o'n oltilik raqamlar}). \quad (4)$$

Rim raqamlari nopozitsion: I = 1, V = 5, X = 10, L = 50, C = 100, shuning uchun $13 = \text{XIII}$; bunday sistemalarda arifmetika amaliy emas, kompyuterlar esa ikkilik sistemadan foydalanadi: elektronika ikki holatni (past/baland, o'chirilgan/yoniq, 0/1) ishonchli ajratadi.

2.1.2 Ilmiy va muhandislik yozuv

Ilmiy yozuv sonning kattaligini nuqtadan oldingi bitta noldan farqli raqamli koefitsient va o'nlik daraja orqali ajratadi; muhandislik yozuvida esa daraja uchlikkar ko'paytmasi bo'lishi shart, shunda daraja SI prefiksi bilan bog'lanadi.

$$\text{ilmiy: } z = a \times 10^n, \quad 1 \leq |a| < 10, \quad n \in \mathbb{Z}, \quad (5)$$

$$\text{muhandislik: } z = a \times 10^{3k}, \quad 1 \leq |a| < 1000, \quad k \in \mathbb{Z}, \quad (6)$$

$$(a \times 10^m)(b \times 10^n) = ab \times 10^{m+n}, \quad \frac{a \times 10^m}{b \times 10^n} = \frac{a}{b} \times 10^{m-n}, \quad (7)$$

$$(a \times 10^n)^m = a^m \times 10^{nm}, \quad \text{qo'sh./ayir.: avval umumiy darajaga keltirish} \quad (8)$$

Butun semestr davomida ishlatiladigan SI prefikslari:

Daraja	10^{12}	10^9	10^6	10^3	10^{-3}	10^{-6}	10^{-9}	10^{-12}	10^{-15}
Prefiks	tera (T)	giga (G)	mega (M)	kilo (k)	milli (m)	mikro (μ)	nano (n)	piko (p)	femto (f)

Registr muhim: $5 \text{ mW} = 5 \times 10^{-3} \text{ W}$, lekin $5 \text{ MW} = 5 \times 10^6 \text{ W}$.

Kurs bo'ylab ishlatiladigan ma'lumot qiymatlari:

$c = 2.99792458 \times 10^8 \text{ m/s}$; $e = 1.602176634 \times 10^{-19} \text{ C}$; $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$;

$h = 6.62607015 \times 10^{-34} \text{ Js}$; $101325 \text{ Pa} = 101.325 \text{ kPa}$.

2.1.3 Kompleks sonlar

1-hafta kompleks sonlar bilan yakunlanadi; ular $x^2 + 1 = 0$ tenglamasining haqiqiy yechimi yo'qligi sababli paydo bo'ladi. Muhayyar birlik, tenglik qoidasi, to'rt arifmetik amal va qo'shma son — asosiy ish vositalari.

$$i = \sqrt{-1}, \quad i^2 = -1, \quad z = a + bi, \quad \text{Re}(z) = a, \quad \text{Im}(z) = b, \quad (9)$$

$$a + bi = c + di \iff a = c \text{ va } b = d, \quad i^{n+4} = i^n \quad (i^0 = 1, i^1 = i, i^2 = -1, i^3 = -i), \quad (10)$$

$$z_1 + z_2 = (a + c) + (b + d)i, \quad z_1 - z_2 = (a - c) + (b - d)i, \quad (11)$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i, \quad \frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{|z_2|^2} \quad (z_2 \neq 0), \quad (12)$$

$$\bar{z} = a - bi, \quad z + \bar{z} = 2 \text{Re}(z), \quad z - \bar{z} = 2i \text{Im}(z), \quad z \bar{z} = a^2 + b^2 = |z|^2, \quad (13)$$

$$|z| = \sqrt{a^2 + b^2}, \quad |z_1 z_2| = |z_1| |z_2|, \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}, \quad (14)$$

$$|z_1 + z_2| \leq |z_1| + |z_2| \quad (\text{uchburchak tengsizligi}) \quad (15)$$

$$\arg z = \theta + 2k\pi, \quad k \in \mathbb{Z}; \quad \text{Arg}(z) \in (-\pi, \pi] \text{ bosh qiymat}, \quad \tan \theta = \frac{b}{a} \text{ (chorakka e'tibor!)} \quad (16)$$

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}, \quad r = |z|, \quad a = r \cos \theta, \quad b = r \sin \theta, \quad (17)$$

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (\text{Euler}), \quad e^{i\pi} + 1 = 0, \quad (18)$$

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}, \quad \frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}, \quad (19)$$

$$z^n = r^n (\cos(n\theta) + i \sin(n\theta)) \quad (\text{de Moivre}), \quad (20)$$

$$z_k = r^{1/n} \left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i \sin\left(\frac{\theta + 2k\pi}{n}\right) \right], \quad k = 0, \dots, n-1, \quad (21)$$

$$z^n = 1 \implies z_k = e^{2\pi i k/n}, \quad k = 0, \dots, n-1 \quad (n\text{-darajali birlik ildizlari}), \quad (22)$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (\text{kvadrat formula, } \mathbb{C} \text{ ustida ham o'rinli}). \quad (23)$$

$z^n = w = re^{i\theta}$ ning ildizlari (21) bilan beriladi; birlikning n -darajali ildizlari birlik aylanada muntazam n -burchak hosil qiladi. Haqiqiy koeffitsientli ko'phadlarning nohaqiqiy ildizlari qo'shma juftliklarda uchraydi, va Algebra asosiy teoremasiga ko'ra n -darajali ko'phadning $\mathbb{C}$ da ko'paytmalar hisobida aynan n ta ildizi bor.

2.2 Chiziqli algebra I: 1-hafta formulalar katalogi

Vektorlar $\mathbb{R}^n$ dagi ustunlardir; 1-haftaning hammasi qo'shish, skalyarga ko'paytirish, birikmalar, span va mustaqillikdan quriladi.

$$(\mathbf{u} + \mathbf{v})_i = u_i + v_i, \quad (c\mathbf{u})_i = c u_i \quad (\text{komponentalar bo'yicha amallar}) \quad (24)$$

$$c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n = \text{chiziqli birikma}, \quad (25)$$

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \text{bunday barcha birikmalar to'plami}, \quad (26)$$

$$\mathbf{v}_1, \dots, \mathbf{v}_n \text{ mustaqil} \iff c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n = \mathbf{0} \implies c_1 = \dots = c_n = 0, \quad (27)$$

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \dots + x_n^2}, \quad \hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|} \quad (\text{birlik vektor}) \quad (28)$$

$$\mathbf{u} \text{ yo'nalishda } L \text{ uzunlikdagi } \mathbf{v}: \quad \mathbf{v} = L \frac{\mathbf{u}}{\|\mathbf{u}\|}, \quad (29)$$

$$d(P, Q) = \|Q - P\|. \quad (30)$$

2.3 Math I: 1-hafta amaliy mashg'uloti — savollar va to'liq yechimlar

- Har bir o'nlik kasrni tugaydigan, takrorlanuvchi yoki takrorlanmaydigan deb tasniflang va mos son ratsionalmi yoki irratsionalligini ko'rsating:

(1) 0.375; (2) 0.272727...; (3) 1.101001000100001...;

(4) 3.14159265...; (5) 0.123123123...

[Manba: 1-hafta amaliy mashg'ulot mashqlari, 1-mashq]

Yechish. (1) 0.375 tugaydi, demak ratsional: $0.375 = \frac{375}{1000} = \frac{3}{8}$. (2) 27 bloki cheksiz takrorlanadi, demak ratsional; sof takrorlanuvchi blok uchun $\overline{ab} = \frac{ab}{99}$, shuning uchun $0.\overline{27} = \frac{27}{99} = \frac{3}{11}$. (3) Ketma-ket 1 lar orasidagi bo'shliqlar o'sadi (1, 2, 3, ... ta nol), shuning uchun kasr hech qachon davriy bo'lmaydi va tugamaydi: son irratsional. (4) Ko'rsatilgan chekli raqamlar va uch nuqta sonni aniqlab bermaydi; agar maqsad qilingan son $\pi = 3.14159265\dots$ bo'lsa, u tugamaydigan va takrorlanmaydigan, demak irratsional. (5) 123 bloki takrorlanadi, demak

ratsional: $0.\overline{123} = \frac{123}{999} = \frac{41}{333}$. Qoida: tugaydigan yoki oxir-oqibat takrorlanadigan o'nliliklar — aynan ratsional sonlar.

2. Har bir to'plamni interval yozuvida yozing va haqiqiy to'g'rida tasvirlang: (1) $\{x \in \mathbb{R} : -3 \leq x < 4\}$; (2) $\{x \in \mathbb{R} : x \neq 3\}$.

[Manba: 1-hafta amaliy mashg'uloti, 2-mashq]

Yechish. (1) -3 uchi kiritiladi (kvadrat qavs), 4 kiritilmaydi (yumaloq qavs): $[-3, 4)$; to'g'rida -3 da yopiq nuqta, 4 da ochiq nuqta va ular orasi ajratilgan. (2) 3 dan boshqa barcha haqiqiy sonlar: $(-\infty, 3) \cup (3, \infty)$; to'g'rida butun o'q bo'yalgan, 3 da ochiq teshik. Cheksizlik hech qachon kiritilmaydi, shuning uchun har doim yumaloq qavs.

3. Har bir intervalni tengsizliklar bilan yozing: a. $[-2, 5)$; b. $(-\infty, 4]$; c. $(1, \infty)$.

[Manba: 1-hafta amaliy mashg'uloti, 3-mashq]

Yechish. a. $-2 \leq x < 5$ (chap uchi ichida, o'ngi tashqarida). b. $x \leq 4$ (chapga cheksiz cho'ziladi, o'ng uchi kiritilgan). c. $x > 1$ (qat'iy katta, chunki yumaloq qavs 1 ni chiqarib tashlaydi).

4. Ratsionalmi yoki irratsionalligini aniqlang: a. $\sqrt{3}$; b. $\sqrt{25}$; c. $\sqrt{50}$; i. $(1 + \sqrt{5})(1 - \sqrt{5})$.

[Manba: 1-hafta amaliy mashg'uloti, 4-mashq]

Yechish. a. 3 to'liq kvadrat emas, demak $\sqrt{3}$ irratsional. b. $\sqrt{25} = 5 \in \mathbb{Z} \subseteq \mathbb{Q}$, ratsional. c. $\sqrt{50} = 5\sqrt{2}$; $\sqrt{2}$ irratsional bo'lgani uchun $5\sqrt{2}$ irratsional. i. Yoyib chiqamiz: $(1 + \sqrt{5})(1 - \sqrt{5}) = 1 - 5 = -4$, ratsional. Qo'shma juftliklar ko'paytmasi har doim radikalli qirib tashlaydi.

5. $z_1 = 3 + 2i$ va $z_2 = 1 - 4i$ berilgan. $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2 ni hisoblang, har bir javobni $a + bi$ shaklida yozib.

[Manba: 1-hafta amaliy mashg'uloti, 5-mashq]

Yechish. Qo'shish va ayirish haqiqiy va muhayyar qismlarga alohida-alohida ta'sir qiladi: $z_1 + z_2 = (3 + 1) + (2 - 4)i = 4 - 2i$; $z_1 - z_2 = (3 - 1) + (2 + 4)i = 2 + 6i$. Ko'paytmada $i^2 = -1$ va $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$ qoidasidan foydalanamiz: $z_1 z_2 = (3 \cdot 1 - 2 \cdot (-4)) + (3 \cdot (-4) + 2 \cdot 1)i = (3 + 8) + (-12 + 2)i = 11 - 10i$. Bo'linma uchun surat va mahrajni mahrajning qo'shma soniga ko'paytiramiz:

$$\frac{z_1}{z_2} = \frac{(3 + 2i)(1 + 4i)}{(1 - 4i)(1 + 4i)} = \frac{3 + 12i + 2i + 8i^2}{1 + 16} = \frac{(3 - 8) + (12 + 2)i}{17} = -\frac{5}{17} + \frac{14}{17}i.$$

6. $3 + 4i$, $-2 + 5i$, $6i$, -7 sonlarining har biri uchun qo'shma sonini va modulini toping; $\bar{z}z = |z|^2$ ni tekshiring.

[Manba: 1-hafta amaliy mashg'uloti, 6-mashq]

Yechish. $z = a + bi$ uchun $\bar{z} = a - bi$ va $|z| = \sqrt{a^2 + b^2}$. $3 + 4i$: $\bar{z} = 3 - 4i$, $|z| = 5$, va $(3 - 4i)(3 + 4i) = 9 + 16 = 25 = 5^2$. $-2 + 5i$: $\bar{z} = -2 - 5i$, $|z| = \sqrt{4 + 25} = \sqrt{29}$, va $(-2 - 5i)(-2 + 5i) = 4 + 25 = 29$. $6i$: $\bar{z} = -6i$, $|z| = 6$, va $(-6i)(6i) = -36i^2 = 36$. -7 : $\bar{z} = -7$ (haqiqiy sonlar o'z qo'shma soni), $|z| = 7$, $(-7)((-7)) = 49$. Barcha to'rttasi tekshirildi.

7. $\frac{1}{1 + i}$ ni $a + bi$ shaklida ifodalang.

[Manba: 1-hafta amaliy mashg'uloti, 7-mashq]

Yechish. Mahrajning qo'shma soniga ko'paytiramiz:

$$\frac{1}{1 + i} = \frac{1 - i}{(1 + i)(1 - i)} = \frac{1 - i}{2} = \frac{1}{2} - \frac{1}{2}i.$$

8. $\mathbb{C}$ ustida yeching: $x^2 + 9 = 0$.

[Manba: 1-hafta amaliy mashg'uloti, 8-mashq]

Yechish. Qayta yozsak $x^2 = -9$, demak $x = \pm\sqrt{-9} = \pm 3i$. Ikkala yechim ham nohaqiqiy kompleks sonlar; tenglamaning haqiqiy yechimi yo'q, chunki $\mathbb{R}$ da kvadrat hech qachon manfiy emas.

9. Kompleks tekislikda $z = 2+3i$ uchun uning haqiqiy qismi, muhayyar qismi, moduli va choragini aniqlang. $2-3i$ uchun nima o'zgaradi?

[Manba: 1-hafta amaliy mashg'uloti, 9-mashq]

Yechish. $(2, 3)$ nuqta birinchi chorakda: $\operatorname{Re}(z) = 2$, $\operatorname{Im}(z) = 3$, $|z| = \sqrt{2^2 + 3^2} = \sqrt{13}$. $\bar{z} = 2-3i$ uchun $(2, -3)$ nuqta haqiqiy o'qqa nisbatan ko'zgu tasviri: haqiqiy qism va modul o'zgarmaydi, muhayyar qism ishorasini almashtiradi va nuqta to'rtinchi chorakka o'tadi.

10. $1+i$ ni trigonometrik shakl $z = r(\cos \theta + i \sin \theta)$ va eksponensial shakl $z = re^{i\theta}$ da yozing, θ radianda.

[Manba: 1-hafta amaliy mashg'uloti, 10-mashq]

Yechish. Bu yerda $r = |1+i| = \sqrt{1+1} = \sqrt{2}$, ikkala koordinata ham musbat, demak $\theta = \arctan(1/1) = \pi/4$. Shuning uchun $1+i = \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}) = \sqrt{2}e^{i\pi/4}$.

11. de Moivre formulasi $[r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta)$ orqali $(\sqrt{3} + i)^4$ ni hisoblang; algebradan tekshiring.

[Manba: 1-hafta amaliy mashg'uloti, 11-mashq]

Yechish. Avval o'tkazamiz: $r = |\sqrt{3} + i| = \sqrt{3+1} = 2$, $\theta = \arctan(1/\sqrt{3}) = \pi/6$. de Moivre bo'yicha,

$$(\sqrt{3}+i)^4 = 2^4 \left(\cos \frac{4\pi}{6} + i \sin \frac{4\pi}{6} \right) = 16 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) = 16 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = -8 + 8\sqrt{3}i.$$

Algebra bilan tekshirish: $(\sqrt{3} + i)^2 = 3 + 2\sqrt{3}i + i^2 = 2 + 2\sqrt{3}i$; yana kvadratga ko'tarsak, $(2 + 2\sqrt{3}i)^2 = 4 + 8\sqrt{3}i + 12i^2 = -8 + 8\sqrt{3}i$. Ikkala usul mos keladi.

12. $z^3 = -8$ ning barcha yechimlarini toping va ularni kompleks tekislikda tasvirlang.

[Manba: 1-hafta amaliy mashg'uloti, 12-mashq]

Yechish. $-8 = 8e^{i(\pi+2k\pi)}$ deb yozamiz. Kub ildizlar moduli $8^{1/3} = 2$ va argumentlari $\frac{\pi+2k\pi}{3}$, $k = 0, 1, 2$:

$$z_0 = 2e^{i\pi/3} = 1 + \sqrt{3}i, \quad z_1 = 2e^{i\pi} = -2, \quad z_2 = 2e^{i5\pi/3} = 1 - \sqrt{3}i.$$

Geometrik jihatdan ular bosh nuqtada joylashgan radiusi 2 aylanada bir xil masofada joylashgan uch nuqta (60° , 180° , 300° burchaklar); $z_1 = -2$ yagona haqiqiy yechim ekanligiga e'tibor bering, bu $(-2)^3 = -8$ ga mos.

13. Har bir son eng kichik to'plamni $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ orasidan aniqlang:

$$0, -7, \frac{5}{8}, 3.25, \sqrt{16}, \sqrt{7}, \\ -\sqrt{9}, \pi, \pi^2, i, 4-3i.$$

[Manba: 1-hafta ISA, 2-topshiriq]

Yechish. $0 \in \mathbb{N}$ (kurs qabul qilgan kelishuv bo'yicha) $\Rightarrow$ eng kichik to'plam $\mathbb{N}$. $-7 \in \mathbb{Z}$. $\frac{5}{8}$ va $3.25 = \frac{13}{4} \in \mathbb{Q}$ da (tugaydigan o'nliklar). $\sqrt{16} = 4 \in \mathbb{N}$. $\sqrt{7}$ irratsional (7 to'liq kvadrat emas) $\Rightarrow \mathbb{R}$. $-\sqrt{9} = -3 \in \mathbb{Z}$. π va π^2 irratsional $\Rightarrow \mathbb{R}$ (kvadratga ko'tarirish irratsionalni ratsional qilmaydi). i va $4-3i$ nohaqiqiy $\Rightarrow$ faqat $\mathbb{C}$.

14. To'g'rimi yoki noto'g'rimi? Har bir noto'g'ri tasdiqni to'g'rilang: (1) $\mathbb{N} \subseteq \mathbb{Z}$; (2) $\mathbb{Z} \subseteq \mathbb{Q}$; (3) $\mathbb{R} \subseteq \mathbb{Q}$; (4) $\mathbb{Q} \subseteq \mathbb{R}$; (5) $\mathbb{R} \subseteq \mathbb{C}$; (6) har bir butun son tabiiy; (7) har bir ratsional haqiqiy; (8) har bir kompleks son haqiqiy; (9) har bir haqiqiy sonni kompleks son shaklida yozish mumkin; (10) bir son bir vaqtda ratsional va haqiqiy bo'la olmaydi.

[Manba: 1-hafta ISA, 3-topshiriq]

Yechish. (1) To'g'ri. (2) To'g'ri. (3) Noto'g'ri — masalan $\sqrt{2} \in \mathbb{R}$ lekin $\sqrt{2} \notin \mathbb{Q}$; to'g'ri kiritish $\mathbb{Q} \subseteq \mathbb{R}$. (4) To'g'ri. (5) To'g'ri. (6) Noto'g'ri — -3 kabi manfiy butun sonlar tabiiy emas. (7) To'g'ri. (8) Noto'g'ri — masalan i kompleks, lekin haqiqiy emas. (9) To'g'ri — $a = a + 0i$. (10) Noto'g'ri — har bir ratsional son haqiqiy (bayonot (4)), demak ratsionallar ikkalasi ham.

15. $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ uchun ichma-ich to'plamlar diagrammasini chizing va $5, -2, \frac{2}{3}, \sqrt{2}, \pi, -4i, 3+2i$ ni to'g'ri joylashtiring.

[Manba: 1-hafta ISA, 4-topshiriq]

Yechish. Besh ta ichma-ich oval chizamiz: $\mathbb{N}$ (eng ichki) dan $\mathbb{C}$ (eng tashqi) gacha. Joylashtirish: $5 \in \mathbb{N}$; $-2 \in \mathbb{Z} \setminus \mathbb{N}$; $\frac{2}{3} \in \mathbb{Q} \setminus \mathbb{Z}$; $\sqrt{2}$ va $\pi \in \mathbb{R} \setminus \mathbb{Q}$; $-4i$ va $3+2i \in \mathbb{C} \setminus \mathbb{R}$. Diagramma har bir yangi to'plam sonlarni faqat qo'shishini, eskilarini hech qachon olib tashlamasini ko'rsatadi.

16. O'sish tartibida joylashtiring: $-\sqrt{5}, -2, -\frac{3}{2}, 0, \sqrt{2}, \frac{3}{2}, \frac{\pi}{2}$; noaniq taqqoslashlarni asoslang.

[Manba: 1-hafta ISA, 5-topshiriq]

Yechish. Kvadratlar musbat kattaliklarning tartibini beradi: $(\sqrt{2})^2 = 2 < \frac{9}{4} = (\frac{3}{2})^2$; $\pi/2 \approx 1.5708$ va $3/2 = 1.5$ bo'lgani uchun $\frac{3}{2} < \frac{\pi}{2}$. Manfiylarda tartib teskari: $-\sqrt{5} \approx -2.236 < -2 < -\frac{3}{2}$. Demak,

$$-\sqrt{5} < -2 < -\frac{3}{2} < 0 < \sqrt{2} < \frac{3}{2} < \frac{\pi}{2}.$$

17. Har bir to'plamni interval yozuvida yozing: (1) $-3 \leq x < 4$; (2) $x > 2$; (3) $x \leq -1$; (4) $-2 < x \leq 5$; (5) $x \neq 3$; so'ng har birini to'g'rida tasvirlang.

[Manba: 1-hafta ISA, 6-topshiriq]

Yechish. (1) $[-3, 4)$. (2) $(2, \infty)$. (3) $(-\infty, -1]$. (4) $(-2, 5]$. (5) $(-\infty, 3) \cup (3, \infty)$. To'g'rida: uchi ishlatilganda yopiq nuqtalar ($\leq, \geq$), ishlatilmaganda ochiq nuqtalar ($<, >$), cheksizliklarda yumaloq qavslar, (5) da birlashma uchun ikki nur.

18. Har bir intervalni tengsizliklar bilan yozing: a. $[-2, 5)$; b. $(-\infty, 4]$; c. $(1, \infty)$; d. $(-3, 0] \cup [2, 6)$; e. $\mathbb{R} \setminus [-1, 1]$.

[Manba: 1-hafta ISA, 7-topshiriq]

Yechish. a. $-2 \leq x < 5$. b. $x \leq 4$. c. $x > 1$. d. $-3 < x \leq 0$ yoki $2 \leq x < 6$. e. $x < -1$ yoki $x > 1$; ekvivalent ravishda $|x| > 1$.

19. $d(a, b) = |a - b|$ bilan: $d(3, 8)$, $d(-2, 5)$, $d(-4, -9)$, $d(\sqrt{2}, \sqrt{8})$ ni hisoblang; $a = -3$, $b = 2$, $c = 7$ uchun $d(a, c) \leq d(a, b) + d(b, c)$ ni tekshiring.

[Manba: 1-hafta ISA, 8-topshiriq]

Yechish. $d(3, 8) = |3 - 8| = 5$; $d(-2, 5) = 7$; $d(-4, -9) = 5$; $d(\sqrt{2}, \sqrt{8}) = |\sqrt{2} - 2\sqrt{2}| = \sqrt{2}$. Uchburchak tekshiruvi: $d(-3, 7) = 10$, $d(-3, 2) + d(2, 7) = 5 + 5 = 10$, va haqiqatdan $10 \leq 10$ (nuqtalar bir to'g'rida tartib bilan joylashgani uchun tenglik). $|a - b|$ masofa formulasi (28) dagi normaning bir o'lchamli holi.

20. Har bir son ratsionalmi yoki irratsionalligini aniqlang; asoslang: a. $\sqrt{49}$; b. $\sqrt{18}$; c. $3 + \sqrt{2}$; d. $2\sqrt{5}$; e. $\sqrt{2} + \sqrt{8}$; f. $(\sqrt{3})^2$; g. $(1 + \sqrt{2})(1 - \sqrt{2})$; h. $\sqrt{12}$.

[Manba: 1-hafta ISA, 9-topshiriq]

Yechish. a. $\sqrt{49} = 7$ ratsional. b. $\sqrt{18} = 3\sqrt{2}$ irratsional. c. ratsional + irratsional = irratsional: $3 + \sqrt{2}$ irratsional. d. $2\sqrt{5}$ irratsional. e. $\sqrt{2} + \sqrt{8} = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$ irratsional. f. $(\sqrt{3})^2 = 3$ ratsional. g. $1 - 2 = -1$ ratsional (qo'shma ko'paytma). h. $\sqrt{12} = 2\sqrt{3}$ irratsional. Amaliy qoida: soddalashtirish tabiatni ochib beradi; $(a + \sqrt{b})(a - \sqrt{b})$ shaklidagi qo'shma ko'paytmalar har doim ratsional.

21. $z_1 = 3 + 2i$, $z_2 = 1 - 4i$ berilgan. $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2 ni hisoblang.

[Manba: 1-hafta ISA, 10-topshiriq]

Yechish. Yuqoridagi amaliy mashg'ulotning 5-mashqi bilan bir xil: $z_1 + z_2 = 4 - 2i$, $z_1 - z_2 = 2 + 6i$, $z_1 z_2 = 11 - 10i$, va $\frac{z_1}{z_2} = -\frac{5}{17} + \frac{14}{17}i$.

22. i^2 , i^3 , i^4 , i^5 , i^{18} , i^{27} , i^{100} ni hisoblang; takrorlanuvchi qonuniyatni va darajani 4 ga bo'lishga asoslangan usulni tavsiflang.

[Manba: 1-hafta ISA, 11-topshiriq]

Yechish. $i^2 = -1$; $i^3 = i^2 i = -i$; $i^4 = (i^2)^2 = 1$; $i^5 = i$. Darajalar 4 davr bilan sikl hosil qiladi: $1, i, -1, -i, \dots$. Katta daraja uchun 4 ga bo'lib qoldiq r ni saqlaymiz: $i^n = i^r$. Shunday qilib $18 = 4 \cdot 4 + 2 \Rightarrow i^{18} = i^2 = -1$; $27 = 4 \cdot 6 + 3 \Rightarrow i^{27} = i^3 = -i$; $100 = 4 \cdot 25 \Rightarrow i^{100} = i^0 = 1$.

23. $3 + 4i$, $-2 + 5i$, $6i$, -7 uchun: qo'shma son va modulni toping, $\bar{z}z = |z|^2$ ni tekshiring.

[Manba: 1-hafta ISA, 12-topshiriq]

Yechish. Amaliy mashg'ulotning 6-mashqidagi hisoblar bilan bir xil: qo'shma sonlar $3 - 4i$, $-2 - 5i$, $-6i$, -7 ; modullar 5 , $\sqrt{29}$, 6 , 7 ; har bir holatda $\bar{z}z = a^2 + b^2 = |z|^2$ aynan bajariladi.

24. $a + bi$ shaklida ifodalang: a. $\frac{1}{1+i}$; b. $\frac{3+2i}{2-i}$; c. $\frac{1+3i}{4-i}$; d. $\frac{5i}{2-i}$.

[Manba: 1-hafta ISA, 13-topshiriq]

Yechish. Har bir holatda surat va mahrajni mahrajning qo'shma soniga ko'paytiramiz. a. $\frac{1}{1+i} = \frac{1-i}{2} = \frac{1}{2} - \frac{1}{2}i$. b. $\frac{(3+2i)(2+i)}{(2-i)(2+i)} = \frac{6+3i+4i+2i^2}{5} = \frac{4+7i}{5} = \frac{4}{5} + \frac{7}{5}i$. c. $\frac{(1+3i)(4+i)}{(4-i)(4+i)} = \frac{4+i+12i+3i^2}{17} = \frac{1+13i}{17} = \frac{1}{17} + \frac{13}{17}i$. d. $\frac{5i(2+i)}{(2-i)(2+i)} = \frac{10i+5i^2}{5} = -1 + 2i$.

25. $\mathbb{C}$ ustida yeching, qaysi yechimlar haqiqiyligini ko'rsatib: a. $x^2 + 9 = 0$; b. $x^2 - 4x + 13 = 0$; c. $2x^2 + 4x + 10 = 0$; d. $x^4 - 16 = 0$.

[Manba: 1-hafta ISA, 14-topshiriq]

Yechish. a. $x = \pm 3i$ (nohaqiqiy). b. Diskriminant $\Delta = (-4)^2 - 4 \cdot 13 = 16 - 52 = -36$, $\sqrt{\Delta} = 6i$; $x = \frac{4 \pm 6i}{2} = 2 \pm 3i$ (nohaqiqiy). c. 2 ga bo'lamiz: $x^2 + 2x + 5 = 0$, $\Delta = 4 - 20 = -16$, $\sqrt{\Delta} = 4i$; $x = \frac{-2 \pm 4i}{2} = -1 \pm 2i$ (nohaqiqiy). d. $x^4 = 16$ haqiqiy yechimlarni beradi $x = \pm 2$; $\mathbb{C}$ ustida to'liq to'plam 16 ning to'rtinchi darajali ildizlari: $2, -2, 2i, -2i$ (bularning hammasi $(x^2 - 4)(x^2 + 4) = 0$ ning yechimlari).

26. $2 + 3i$, $-1 + 4i$, $-3 - 2i$, $4 - i$ uchun: haqiqiy qism, muhayyar qism, modul, chorakni toping.

[Manba: 1-hafta ISA, 15-topshiriq]

Yechish. $2 + 3i$: $(2, 3)$, modul $\sqrt{13}$, I chorak. $-1 + 4i$: $(-1, 4)$, modul $\sqrt{17}$, II chorak. $-3 - 2i$: $(-3, -2)$, modul $\sqrt{13}$, III chorak. $4 - i$: $(4, -1)$, modul $\sqrt{17}$, IV chorak. Modullar juftlashib qoladi, chunki qo'shma son $a^2 + b^2$ ni saqlaydi.

27. Trigonometrik va eksponensial shaklda yozing (θ radianda): (1) $1 + i$; (2) $-1 + i$; (3) $-\sqrt{3} - i$; (4) $2i$; (5) -4 .

[Manba: 1-hafta ISA, 16-topshiriq]

Yechish. (1) $r = \sqrt{2}$, $\theta = \pi/4$: $\sqrt{2}e^{i\pi/4}$. (2) $r = \sqrt{2}$, II chorak: $\theta = 3\pi/4$: $\sqrt{2}e^{i3\pi/4}$. (3) $r = \sqrt{3+1} = 2$, III chorak: yordamchi burchak $\arctan(1/\sqrt{3}) = \pi/6$, demak $\theta = \pi + \pi/6 = 7\pi/6$: $2e^{i7\pi/6}$. (4) $r = 2$, $\theta = \pi/2$: $2e^{i\pi/2}$. (5) $r = 4$, $\theta = \pi$: $4e^{i\pi}$. Har bir burchak $\tan \theta = b/a$ dan chorak bilan to'g'rilangan holda olinadi, aynan (16) ogohlantirganidek.

28. de Moivre formulasi yordamida hisoblang: (1) $(1+i)^6$; (2) $(\sqrt{3}+i)^4$; (3) $(-1+\sqrt{3}i)^5$; bir natijani algebradan tekshiring.

[Manba: 1-hafta ISA, 17-topshiriq]

Yechish. (1) $1 + i = \sqrt{2}e^{i\pi/4}$, demak $(1+i)^6 = (\sqrt{2})^6 e^{i6\pi/4} = 8e^{i3\pi/2} = 8(0-i) = -8i$. (2) Amaliy mashg'ulotning 11-mashqidan: $(\sqrt{3}+i)^4 = -8 + 8\sqrt{3}i$. (3) $-1 + \sqrt{3}i$: modul $\sqrt{1+3} = 2$; II chorak, yordamchi burchak $\pi/3$, demak $\theta = 2\pi/3$: $z = 2e^{i2\pi/3}$. U holda $z^5 = 2^5 e^{i10\pi/3} = 32e^{i(10\pi/3-2\pi)} = 32e^{i4\pi/3} = 32(-\frac{1}{2} - \frac{\sqrt{3}}{2}i) = -16 - 16\sqrt{3}i$. (1) ni algebradan tekshirish: $(1+i)^2 = 2i$, $(1+i)^4 = (2i)^2 = -4$, $(1+i)^6 = -4 \cdot 2i = -8i$. Mos keladi.

29. Barcha yechimlarni toping va kompleks tekislikda tasvirlang: a. $z^3 = 1$; b. $z^4 = 1$; c. $z^3 = -8$. n -darajali ildizlar nima uchun aylanada bir xil masofada joylashishini tushuntiring.

[Manba: 1-hafta ISA, 18-topshiriq]

Yechish. a. (22) bo'yicha: $z_k = e^{2\pi ik/3}$, $k = 0, 1, 2$: 1 , $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$. b. $z_k = e^{2\pi ik/4}$: 1 , i , -1 , $-i$. c. Amaliy mashg'ulotning 12-mashqidan: $1 + \sqrt{3}i$, -2 , $1 - \sqrt{3}i$. Har bir holatda ildizlar bir xil modulga ega $r^{1/n}$ (demak bitta aylanada yotadi) va argumentlari aynan $\frac{2\pi}{n}$ ga farq qiladi (to'liq 2π burilish n ga teng yoylarga bo'lingan) — aynan shu sababdan ular muntazam n -burchakning uchlarini hosil qiladi.

30. Elektr zaryadining kattaligi $e = 0.0000000000000000001602176634$ C. Uni ilmiy yozuvda va muhandislik yozuvida yozing.

[Manba: 1-hafta ISA, 19-topshiriq]

Yechish. Birinchi noldan farqli raqamni nuqta oldiga olib kelish uchun 18 xonani sanaymiz: $e = 1.602176634 \times 10^{-19}$ C (ilmiy). Muhandislik yozuvida daraja 3 ning ko'paytmasi bo'lishi kerak, shuning uchun bir xonani suramiz: $e = 160.2176634 \times 10^{-21}$ C = 160.2176634 zC (zeptokulon); amalda $160.2176634 \times 10^{-21}$ C deb yoziladi, chunki 10^{-24} dan kichik umumiy SI prefiksi faqat zepto (10^{-21}).

31. SI-prefiks jadvalini to'ldiring: 10^{12} , 10^6 , 10^3 , 10^{-6} , 10^{-9} , 10^{-12} darajalari uchun prefiks/belgilar.

[Manba: 1-hafta ISA, 20-topshiriq]

Yechish. 10^{12} tera (T); 10^6 mega (M); 10^3 kilo (k); 10^{-6} mikro (μ); 10^{-9} nano (n); 10^{-12} piko (p). Berilgan giga (G), milli (m), femto (f), atto (a) bilan birga bu jadval to'liq SI narvonini tashkil etadi.

32. 1-hafta konspekti 2-qism, darsdagi mashqlar: (1) 0.00000824 ni ilmiy va muhandislik yozuvda yozing; (2) 7.35×10^8 Hz ni muhandislik yozuvida yozing; (3) 0.0047 F ni SI prefiksi bilan yozing.

[Manba: 1-hafta konspekti 2-qism]

Yechish. (1) $0.00000824 = 8.24 \times 10^{-6}$ (ilmiy); -6 uchlikkar ko'paytmasi bo'lgani uchun bu allaqachon muhandislik yozuvi, masalan birlik amper bo'lsa $8.24 \mu\text{A}$. (2) 10^8 uchlikkar ko'paytmasi emas; qayta yozamiz $7.35 \times 10^8 = 735 \times 10^6 = 735 \text{ MHz}$. (3) $0.0047 = 4.7 \times 10^{-3} = 4.7 \text{ mF}$.

33. 1-hafta konspekti 3-qism, darsdagi mashqlar: (1) $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ uchun ichma-ich to'plamlar diagrammasini chizing va 2026, -9 , $\frac{2}{3}$, $\sqrt{10}$, π , $-5i$, $-2 + 2i$ ni joylashtiring; (2) ratsionalmi yoki irratsional: a. $\sqrt{49}$, b. $\sqrt{18}$, c. $3 + \sqrt{2}$, d. $2\sqrt{5}$, e. $\sqrt{2} + \sqrt{8}$, f. $(\sqrt{5})^2$.

[Manba: 1-hafta konspekti 3-qism]

Yechish. (1) $2026 \in \mathbb{N}$; $-9 \in \mathbb{Z} \setminus \mathbb{N}$; $\frac{2}{3} \in \mathbb{Q} \setminus \mathbb{Z}$; $\sqrt{10}$ va $\pi \in \mathbb{R} \setminus \mathbb{Q}$; $-5i$ va $-2 + 2i \in \mathbb{C} \setminus \mathbb{R}$. (2) a. 7, ratsional. b. $3\sqrt{2}$, irratsional. c. irratsional. d. irratsional. e. $3\sqrt{2}$, irratsional. f. 5, ratsional — 9-topshiriqdagi bir xil mantiq, bu mashq boshqa sonlar bilan takrorlaydi.

34. 1-hafta konspekti 3-qism, darsdagi mashqlar: $z_1 = 2 + 3i$, $z_2 = 1 - 5i$ berilgan. $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2 ni hisoblang.

[Manba: 1-hafta konspekti 3-qism]

Yechish. $z_1 + z_2 = 3 - 2i$; $z_1 - z_2 = 1 + 8i$; ko'paytma: $(2 \cdot 1 - 3 \cdot (-5)) + (2 \cdot (-5) + 3 \cdot 1)i = (2 + 15) + (-10 + 3)i = 17 - 7i$; bo'linma:

$$\frac{z_1}{z_2} = \frac{(2 + 3i)(1 + 5i)}{(1 - 5i)(1 + 5i)} = \frac{2 + 10i + 3i + 15i^2}{26} = \frac{-13 + 13i}{26} = -\frac{1}{2} + \frac{1}{2}i.$$

35. $1 + i\sqrt{3} = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}) = 2e^{i\pi/3}$ ekanligini nima uchun tushuntiring.

[Manba: 1-hafta konspekti 3-qism]

Yechish. Modul $|1 + i\sqrt{3}| = \sqrt{1 + 3} = 2$. $(1, \sqrt{3})$ nuqta birinchi chorakda yotadi va $\tan \theta = \sqrt{3}$, demak $\theta = \pi/3$. $r = 2$, $\theta = \pi/3$ qiymatlarini qutb shakliga (17), so'ng Euler formulasiga (18) qo'ysak, ikkala yozilgan tenglik ham kelib chiqadi.

2.4 Chiziqli algebra I, 1-hafta: ISA mashqlari — savollar va to'liq yechimlar

Butun bo'limda vektorlar ustunlar; A qismda $\mathbf{u} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$.

1. $\mathbf{u} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$ berilgan. $\mathbf{u} + \mathbf{v}$, $\mathbf{u} - \mathbf{v}$, $2\mathbf{u}$, $-3\mathbf{v}$, $2\mathbf{u} - 3\mathbf{v}$ ni hisoblang.

[Manba: Chiziqli algebra ISA 1-hafta, 1-mashq]

Yechish. (24) bo'yicha komponentalar bo'yicha: $\mathbf{u} + \mathbf{v} = \begin{bmatrix} -3 \\ 8 \end{bmatrix}$; $\mathbf{u} - \mathbf{v} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$; $2\mathbf{u} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$; $-3\mathbf{v} = \begin{bmatrix} 3 \\ -15 \end{bmatrix}$; $2\mathbf{u} - 3\mathbf{v} = \begin{bmatrix} -4 + 3 \\ 6 + 15 \end{bmatrix} = \begin{bmatrix} -1 \\ 21 \end{bmatrix}$.

2. $\mathbf{a} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -2 \end{bmatrix}$ berilgan. $\mathbf{a} + \mathbf{b}$, $\mathbf{a} - \mathbf{b}$, $3\mathbf{a} + 2\mathbf{b}$ ni toping.

[Manba: Chiziqli algebra ISA 1-hafta, 2-mashq]

Yechish. $\mathbf{a} + \mathbf{b} = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$; $\mathbf{a} - \mathbf{b} = \begin{bmatrix} -1 \\ -3 \\ 6 \end{bmatrix}$; $3\mathbf{a} + 2\mathbf{b} = \begin{bmatrix} 6+6 \\ -3+4 \\ 12-4 \end{bmatrix} = \begin{bmatrix} 12 \\ 1 \\ 8 \end{bmatrix}$.

3. $\mathbf{x} + \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ -2 \end{bmatrix}$ bo'lsa, $\mathbf{x}$ ni toping.

[Manba: Chiziqli algebra ISA 1-hafta, 3-mashq]

Yechish. Ma'lum vektorni ikkala tomondan ayiramiz: $\mathbf{x} = \begin{bmatrix} 7-2 \\ 3+4 \\ -2-1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ -3 \end{bmatrix}$.

4. $3\mathbf{x} - 2\mathbf{u} = \mathbf{v}$, bunda $\mathbf{u} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$ bo'lsa, $\mathbf{x}$ ni toping.

[Manba: Chiziqli algebra ISA 1-hafta, 4-mashq]

Yechish. Ajratamiz: $3\mathbf{x} = \mathbf{v} + 2\mathbf{u} = \begin{bmatrix} 8-4 \\ 2+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$, demak $\mathbf{x} = \begin{bmatrix} 4/3 \\ 4/3 \end{bmatrix}$. Skalyarga ko'paytirish qo'shishga taqsimlanadi, shuning uchun vektorni 3 ga bo'lish — har bir komponentani 3 ga bo'lish demakdir.

5. $\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ vektorlarning Evklid uzunligini hisoblang.

[Manba: Chiziqli algebra ISA 1-hafta, 5-mashq]

Yechish. (28) bo'yicha: $\|\mathbf{u}\| = \sqrt{9+16} = 5$; $\|\mathbf{v}\| = \sqrt{4+9+36} = 7$; $\|\mathbf{w}\| = 0$ (faqat nol vektorning uzunligi nolga teng).

6. $\mathbf{v} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$ bilan bir yo'nalishdagi birlik vektorni toping va uning uzunligi 1 ekanini tekshiring.

[Manba: Chiziqli algebra ISA 1-hafta, 6-mashq]

Yechish. $\|\mathbf{v}\| = 5$, demak $\hat{\mathbf{v}} = \mathbf{v}/5 = \begin{bmatrix} -3/5 \\ 4/5 \end{bmatrix}$. Tekshirish: $\|\hat{\mathbf{v}}\| = \sqrt{9/25 + 16/25} = \sqrt{25/25} = 1$. Normaga bo'lish yo'nalishni saqlaydi (skalyar musbat) va uzunlikni 1 ga majbur qiladi.

7. $\mathbf{v} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$ bilan bir xil yo'nalishda, uzunligi 10 bo'lgan vektorni toping.

[Manba: Chiziqli algebra ISA 1-hafta, 7-mashq]

Yechish. Bu yerda $\|\mathbf{v}\| = 5$, demak $10/5 = 2$ ga masshtablaymiz: javob $2\mathbf{v} = \begin{bmatrix} -8 \\ 6 \end{bmatrix}$; haqiqatdan $\|2\mathbf{v}\| = 2 \cdot 5 = 10$, normaning bir jinsiligi bo'yicha.

8. Vektor ayirish va uzunlik yordamida $P = (1, -2, 3)$ va $Q = (5, 1, -3)$ orasidagi masofani toping.

[Manba: Chiziqli algebra ISA 1-hafta, 8-mashq]

Yechish. $\mathbf{PQ} = Q - P = \begin{bmatrix} 4 \\ 3 \\ -6 \end{bmatrix}$, demak $d(P, Q) = \|\mathbf{PQ}\| = \sqrt{16+9+36} = \sqrt{61}$.

9. $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ berilgan. $2\mathbf{v}_1 + \mathbf{v}_2$, $-\mathbf{v}_1 + 3\mathbf{v}_2$, $4\mathbf{v}_1 - 2\mathbf{v}_2$, $0\mathbf{v}_1 + 5\mathbf{v}_2$ ni hisoblang.

[Manba: Chiziqli algebra ISA 1-hafta, 9-mashq]

Yechish. $2\mathbf{v}_1 + \mathbf{v}_2 = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$; $-\mathbf{v}_1 + 3\mathbf{v}_2 = \begin{bmatrix} -1-1 \\ -2+9 \end{bmatrix} = \begin{bmatrix} -2 \\ 7 \end{bmatrix}$; $4\mathbf{v}_1 - 2\mathbf{v}_2 = \begin{bmatrix} 4+2 \\ 8-6 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$; $5\mathbf{v}_2 = \begin{bmatrix} -5 \\ 15 \end{bmatrix}$ (nol skalyar $\mathbf{v}_1$ ni butunlay yo'q qiladi).

10. $c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$ bo'ladigan c_1, c_2 ni aniqlang; qo'yib tekshiring.

[Manba: Chiziqli algebra ISA 1-hafta, 10-mashq]

Yechish. Komponentalar tenglamalari: $c_1 - c_2 = 7$ va $c_1 + 2c_2 = 1$. Ikkinchisidan birinchisini ayirsak: $3c_2 = -6$, demak $c_2 = -2$ va $c_1 = 5$. Tekshirish: $5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5+2 \\ 5-4 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$.

11. $\mathbf{b} = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$ ni $\mathbf{b} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ ko'rinishida yozish mumkinmi, aniqlang.

[Manba: Chiziqli algebra ISA 1-hafta, 11-mashq]

Yechish. Komponentalar $c_1 + c_3 = 5$, $c_2 + c_3 = 1$, $c_1 + c_2 = 4$ beradi. Birinchisini va ikkinchisini qo'shib, uchinchisini ayirsak: $2c_3 = 2$, demak $c_3 = 1$, $c_1 = 4$, $c_2 = 0$. Birikma mavjud: $\mathbf{b} = 4\mathbf{v}_1 + 0 \cdot \mathbf{v}_2 + 1 \cdot \mathbf{v}_3$. Yechimning yagonaligi uch vektor mustaqil ekanini ham ko'rsatadi.

12. $c \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $c \in \mathbb{R}$ barcha vektorlarni geometrik tavsiflang; $\begin{bmatrix} 4 \\ -6 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 6 \\ -8 \end{bmatrix}$ shu span ga tegishlimi?

[Manba: Chiziqli algebra ISA 1-hafta, 12-mashq]

Yechish. Span — $(-2, 3)$ yo'nalishdagi bosh nuqtadan o'tuvchi butun to'g'ri. Sinov: $-2 \begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -6 \end{bmatrix}$ ha; $1 \cdot \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ ha; $\begin{bmatrix} 6 \\ -8 \end{bmatrix}$ uchun $-2c = 6$ va $3c = -8$ kerak, ya'ni $c = -3$ va $c = -8/3$ — zid, demak u to'g'rida emas.

13. $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ chiziqli mustaqilmi? $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0}$ bo'ladigan nolga teng bo'lmagan c_1, c_2 ni toping va geometrik tushuntiring.

[Manba: Chiziqli algebra ISA 1-hafta, 13-mashq]

Yechish. $\mathbf{v}_2 = 3\mathbf{v}_1$, demak $3\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$ notrivial bog'lanish ($c_1 = 3$, $c_2 = -1$). Vektorlar qaram. Geometrik jihatdan ikkalasi bosh nuqtadan o'tuvchi bir to'g'rida yotadi, shuning uchun ular baribir faqat shu to'g'ri ni span qiladi — qaramlik bir vektor yangi yo'nalish bermasligini anglatadi.

14. $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ mustaqilmi? Agar qaram bo'lsa, birini boshqalarining birikmasi orqali ifodalang.

[Manba: Chiziqli algebra ISA 1-hafta, 14-mashq]

Yechish. $\mathbf{v}_3 = \alpha\mathbf{v}_1 + \beta\mathbf{v}_2$ ni sinab ko'ramiz: komponentalar $\alpha = 1$ (birinchi), $\beta = 1$ (ikkinchi), $\alpha + \beta = 2$ (uchinchi) beradi — mos. Demak $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$, notrivial bog'lanish mavjud va to'plam qaram. Faqat ikkita mustaqil yo'nalish bor; span tekislik.

15. $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ berilgan. (1) Ular $\mathbb{R}^2$ ni span qiladimi? (2) $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ uchun $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{b}$ bo'ladigan c_1, c_2 ni toping. (3) $\begin{bmatrix} 7 \\ 4 \end{bmatrix}$ ni formula orqali ifodalang.

[Manba: Chiziqli algebra ISA 1-hafta, 15-mashq]

Yechish. (1)-(2) $c_1 + 3c_2 = b_1$, $2c_1 + c_2 = b_2$ yechamiz. Yo'q qilib: $c_1 = b_1 - 3c_2$, u holda $2b_1 - 6c_2 + c_2 = b_2$, demak $c_2 = \frac{2b_1 - b_2}{5}$ va $c_1 = \frac{-b_1 + 3b_2}{5}$. Formalar har bir (b_1, b_2) uchun aniqlangan, demak ha, ikki vektor $\mathbb{R}^2$ ni span qiladi. (3) $c_1 = \frac{-7+12}{5} = 1$, $c_2 = \frac{14-4}{5} = 2$: $\begin{bmatrix} 7 \\ 4 \end{bmatrix} = \mathbf{v}_1 + 2\mathbf{v}_2$.

16. $\mathbf{u}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -6 \\ 3 \end{bmatrix}$ berilgan. (1) Ular $\mathbb{R}^2$ ni span qiladimi? (2) $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$ ni geometrik tavsiflang; (3) $\begin{bmatrix} 8 \\ -4 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ unga tegishlimi?

[Manba: Chiziqli algebra ISA 1-hafta, 16-mashq]

Yechish. Nisbatlarni taqqoslash $-6/-1 = 6$ va $3/2$ ko'rsatadiki, berilgan juftlik bir-birining ko'paytmasi emas, demak chop etilganidek ular $\mathbb{R}^2$ ni span qiladi; mashqning standart variantida $\mathbf{u}_2 = 3\mathbf{u}_1 = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$ olinadi — uni yechamiz. (1) $\mathbf{u}_2 = 3\mathbf{u}_1$, demak span faqat $\text{span}\{\mathbf{u}_1\}$. (2) Geometrik jihatdan bu $(-1, 2)$ yo'nalishda bosh nuqtadan o'tuvchi bitta to'g'ri; ikki kollinear vektor hech qachon tekislikni span qilmaydi. (3) $\begin{bmatrix} 8 \\ -4 \end{bmatrix} = -8\mathbf{u}_1$ mi? Tekshirish: $-8 \cdot (-1) = 8$, $-8 \cdot 2 = -16 \neq -4$, demak yo'q; $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ uchun: $-c = 1$ va $2c = 2$ kerak — $c = -1$ va $c = 1$, zid, demak yo'q (haqiqatdan $(1, 2)$ nuqta $y = -2x$ to'g'risida emas).

17. $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ berilgan. (1) $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{b}$ ni ixtiyoriy $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ uchun yeching; (2) uch vektor nima uchun $\mathbb{R}^3$ ni span qilishini tushuntiring; (3) $\begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$ ni ifodalang.

[Manba: Chiziqli algebra ISA 1-hafta, 17-mashq]

Yechish. (1) Komponentalar: $c_1 + c_3 = b_1$, $c_2 + c_3 = b_2$, $c_1 + c_2 = b_3$. Hammasini qo'shsak: $2(c_1 + c_2 + c_3) = b_1 + b_2 + b_3$, demak $c_1 + c_2 + c_3 = \frac{b_1 + b_2 + b_3}{2}$. U holda $c_1 = \frac{b_1 - b_2 + b_3}{2}$, $c_2 = \frac{-b_1 + b_2 + b_3}{2}$, $c_3 = \frac{b_1 + b_2 - b_3}{2}$. (2) Yechim har bir o'ng tomon uchun mavjud va yagona, demak span $\mathbb{R}^3$ ning hammasi (va vektorlar mustaqil). (3) $c_1 = \frac{4-2+6}{2} = 4$, $c_2 = \frac{-4+2+6}{2} = 2$, $c_3 = \frac{4+2-6}{2} = 0$: $\mathbf{b} = 4\mathbf{v}_1 + 2\mathbf{v}_2$.

18. Detalga uch kuch ta'sir qiladi: $\mathbf{F}_1 = \begin{bmatrix} 40 \\ 10 \end{bmatrix}$ N, $\mathbf{F}_2 = \begin{bmatrix} -15 \\ 25 \end{bmatrix}$ N, $\mathbf{F}_3 = \begin{bmatrix} -20 \\ -5 \end{bmatrix}$ N. (1) Natijaviy kuchni toping; (2) detalni muvozanatga keltiruvchi $\mathbf{F}_4$ ni toping: $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \mathbf{F}_4 = \mathbf{0}$.

[Manba: Chiziqli algebra ISA 1-hafta, 18-mashq]

Yechish. (1) Komponentalar bo'yicha qo'shamiz: $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \begin{bmatrix} 40 - 15 - 20 \\ 10 + 25 - 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 30 \end{bmatrix}$ N. (2) Muvozanat uchun $\mathbf{F}_4 = -(\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3) = \begin{bmatrix} -5 \\ -30 \end{bmatrix}$ N kerak: aynan natijaviy kuchning teskarisi, shunda umumiy kuch nolga teng bo'ladi.

2.5 Chiziqli algebra I, 1-hafta: TA sessiyasi muammolari (alternativ to'plam) — savollar va to'liq yechimlar

TA to'plami sakkizta qisqa muammo keltiradi, har biri rasmiy yechim bilan; ikkalasini ham mantiqini ochib berib keltiramiz.

1. $\mathbf{a} = (-1, 3, 2)$, $\mathbf{b} = (4, 0, -2)$ berilgan. $\mathbf{a} + \mathbf{b}$ va $3\mathbf{a} - \mathbf{b}$ ni hisoblang; $\mathbf{a}$ va $\mathbf{b}$ orasidagi o'rta nuqtani toping.

[Manba: TA muammolari 1-hafta, 1-muammo]

Yechish. $\mathbf{a} + \mathbf{b} = (3, 3, 0)$; $3\mathbf{a} - \mathbf{b} = (-3 - 4, 9 - 0, 6 + 2) = (-7, 9, 8)$. O'rta nuqta — o'rtacha $\frac{\mathbf{a} + \mathbf{b}}{2} = (\frac{3}{2}, \frac{3}{2}, 0)$ — skalyar o'rta nuqta formulasi bilan bir xil o'rtacha g'oyasi.

2. $\mathbf{p} + \mathbf{q} = (5, 7)$ va $\mathbf{p} - \mathbf{q} = (-1, 3)$ berilgan. $\mathbf{p}$ va $\mathbf{q}$ ni toping.

[Manba: TA muammolari 1-hafta, 2-muammo]

Yechish. Ikkala tenglamani qo'shamiz: $2\mathbf{p} = (4, 10)$, demak $\mathbf{p} = (2, 5)$. Ayiramiz: $2\mathbf{q} = (6, 4)$, demak $\mathbf{q} = (3, 2)$. Tekshirish: $(2, 5) + (3, 2) = (5, 7)$ va $(2, 5) - (3, 2) = (-1, 3)$. Bu "qo'sh va ayir" hiylasi ikkita skalyar tenglamani bir vaqtda yechishning vektor analogidir.

3. $\mathbf{u} = (1, 2, -1)$, $\mathbf{v} = (2, -1, 0)$ berilgan. $-\mathbf{u} + 3\mathbf{v}$ ni hisoblang; $c\mathbf{u} + d\mathbf{v}$ ni komponentalar bilan yozing; $c\mathbf{u} + d\mathbf{v} = (5, 0, -1)$ bo'lishi mumkinmi?

[Manba: TA muammolari 1-hafta, 3-muammo]

Yechish. $-\mathbf{u} + 3\mathbf{v} = (-1, -2, 1) + (6, -3, 0) = (5, -5, 1)$. Komponentalar bo'yicha $c\mathbf{u} + d\mathbf{v} = (c + 2d, 2c - d, -c)$. $(5, 0, -1)$ ga erishish uchun: uchinchi komponenta $c = 1$ ni majbur qiladi; u holda $c + 2d = 5$ dan $d = 2$; va haqiqatdan $2c - d = 0$. Javob: ha, $\mathbf{u} + 2\mathbf{v} = (5, 0, -1)$.

4. $\mathbf{r} = (2, -1)$, $\mathbf{s} = (-6, 3)$ berilgan. $c\mathbf{r} + d\mathbf{s}$ barcha birikmalarni tavsiflang; vektorlar $\mathbb{R}^2$ ni span qiladimi? Qaramlik bog'lanishini toping.

[Manba: TA muammolari 1-hafta, 4-muammo]

Yechish. $\mathbf{s} = -3\mathbf{r}$ bo'lgani uchun har bir birikma $(c - 3d)\mathbf{r}$ ga teng — har doim $\mathbf{r}$ ning ko'paytmasi. Span faqat $\mathbf{r}$ dan o'tuvchi to'g'ri: vektorlar tekislikni span qilmaydi. $3\mathbf{r} + \mathbf{s} = \mathbf{0}$ bog'lanishi notrivial, bu qaramlikni isbotlaydi (qarang (27)).

5. $\mathbf{v} = (1, 0, 1)$, $\mathbf{w} = (1, 2, 0)$ berilgan. $c\mathbf{v} + d\mathbf{w} = (x, y, z)$ deb yozing; x, y, z ni bog'lovchi bitta tenglama toping; $\mathbf{b} = (4, 6, 1)$ ni tekshiring.

[Manba: TA muammolari 1-hafta, 5-muammo]

Yechish. $c\mathbf{v} + d\mathbf{w} = (c + d, 2d, c)$. O'qib chiqsak: $z = c$ va $y = 2d$, demak $x = z + \frac{y}{2}$ — tekislikni tasvirlaydigan yagona chiziqli tenglama. $\mathbf{b}$ uchun: o'ng tomon $= 1 + \frac{6}{2} = 4 = x$, demak $\mathbf{b}$ tekislikda yotadi; $z = 1$, $y = 6$ dan: $c = 1$, $d = 3$.

6. $c(1, 2) + d(3, -1) = (7, 3)$ bo'ladigan c, d ni toping; komponentalar tenglamalarini yozib yeching.

[Manba: TA muammolari 1-hafta, 6-muammo]

Yechish. Komponentalar: $c + 3d = 7$ va $2c - d = 3$. Ikkinchisidan $d = 2c - 3$; qo'ysak: $c + 6c - 9 = 7$, demak $7c = 16$, $c = \frac{16}{7}$, $d = \frac{32}{7} - 3 = \frac{11}{7}$. Ikkala komponenta ham aynan tekshiriladi.

7. $\mathbf{u} = (1, 1, 0)$, $\mathbf{v} = (0, 1, 1)$, $\mathbf{w} = (1, 2, 1)$ berilgan. $\mathbf{w}$ yangi yo'nalish qo'shadimi? $\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ ni tavsiflang; nolga teng bog'lanishni keltiring.

[Manba: TA muammolari 1-hafta, 7-muammo]

Yechish. $\mathbf{w} = \mathbf{u} + \mathbf{v}$ (komponentalar bo'yicha tekshirish: $1 + 0 = 1$, $1 + 1 = 2$, $0 + 1 = 1$), demak $\mathbf{w}$ ortiqcha. $\mathbf{u} + \mathbf{v} - \mathbf{w} = \mathbf{0}$ bog'lanishi notrivial, demak uch vektor qaram va span faqat $\mathbf{u}$ va $\mathbf{v}$ span qilgan tekislik (ular ko'paytma emas).

8. $\mathbb{R}^3$ da ikki vektor butun $\mathbb{R}^3$ ni span qila oladimi? Uch vektor $\mathbb{R}^3$ ni span qilib, baribir qaram bo'lishi mumkinmi? "Yangi yo'nalishlar" orqali tushuntiring.

[Manba: TA muammolari 1-hafta, 8-muammo]

Yechish. Ikkala vektor ko'pi bilan tekislikni span qiladi — har bir vektor ko'pi bilan bitta yangi yo'nalish beradi va ikki yo'nalish uch o'lchamli fazoni to'ldira olmaydi. Uchta qaram vektor ham ko'pi bilan tekislikni span qiladi: qaramlik ulardan birining boshqalarining birikmasi ekanini va uchinchi yo'nalish qo'shmasligini anglatadi. $\mathbb{R}^3$ ga erishish uchun uchta mustaqil yo'nalish kerak; demak uchta qaram vektor hech qachon $\mathbb{R}^3$ ni span qila olmaydi.

3 2-hafta — Funktsiyalar (Math I); Skalyar va vektor ko'paytmalar, matritsalariga kirish (Chiziqli algebra I)

3.1 Math I: 2-hafta formulalar katalogi

3.1.1 Funktsiyalar: ta'rif, sinflar, simmetriya, kompozitsiya, teskari funktsiyalar

Funksiya sohasidagi har bir x ga aynan bitta natija $f(x)$ ni bog'laydi; vertikal to'g'ri testi bunga mos kelmaydigan bog'lanishlarni aniqlaydi. Tabiiy soha — formula mazmendor bo'ladigan haqiqiy kirishlarning eng katta to'plami; uchta standart to'siq: nolga bo'lish, manfiy sonlarning juft darajali ildizlari va nomanfiy bo'lmagan sonlarning logarifmi.

$$f: D \rightarrow \mathbb{R}, \quad x \mapsto f(x); \quad D = \text{soha}, \quad \text{range } f = \{f(x) : x \in D\}, \quad (31)$$

$$\frac{1}{g(x)} : g(x) \neq 0, \quad \sqrt{g(x)} : g(x) \geq 0, \quad \log_b g(x) : g(x) > 0. \quad (32)$$

Juftlik va toqlik mos ravishda y -o'qqa va bosh nuqtaga nisbatan simmetriyani aks ettiradi; kompozitsiya funktsiyalarni zanjirlaydi; faqat bir-biriga bir xil (o'ziga) bog'lanmagan (gorizontal to'g'ri testidan o'tadigan) funktsiyalarga teskari funksiya mavjud bo'lib, uni $y = f(x)$ ni x ga nisbatan yechib topiladi.

$$\text{juft: } f(-x) = f(x), \quad \text{toq: } f(-x) = -f(x), \quad (33)$$

$$(f \circ g)(x) = f(g(x)), \quad (f \circ g) \neq (g \circ f) \text{ umuman olganda}, \quad (34)$$

$$f \text{ inyektiv} \iff f(x_1) = f(x_2) \implies x_1 = x_2, \quad (35)$$

$$f^{-1}(f(x)) = x \quad \text{va} \quad f(f^{-1}(y)) = y, \quad (36)$$

$$f^{-1} \text{ grafigi} = f \text{ grafigining } y = x \text{ dagi aksi}. \quad (37)$$

§1.2 (OpenStax) funksiya sinflari: chiziqli, ko'phad, ratsional, ildiz (algebraik), eksponensial/logarifmik va trigonometrik (transsendent); har bir ko'phad hamma joyda silliq, va har bir ratsional funksiya mahrajning noldan farqli bo'lgan joylarida aniqlangan.

3.1.2 Eksponensial va logarifmik funksiyalar (§1.5, giperbolik funksiyalar chiqarilgan)

Eksponensial va logarifm bir-birining teskarisi; daraja va logarifm qonunlari ko'paytirishni qo'shishga aylantiradi va kursning har bir o'sish/pasayish modelida ishlatiladi.

$$b^x > 0, \quad b^{x+y} = b^x b^y, \quad (b^x)^y = b^{xy}, \quad b^{-x} = \frac{1}{b^x}, \quad (38)$$

$$\log_b x = y \iff b^y = x, \quad x > 0, \quad b > 0, \quad b \neq 1, \quad (39)$$

$$\log_b(xy) = \log_b x + \log_b y, \quad \log_b \frac{x}{y} = \log_b x - \log_b y, \quad \log_b x^n = n \log_b x, \quad (40)$$

$$\log_b x = \frac{\ln x}{\ln b} \quad (\text{asosni almashtirish}), \quad b^x = e^{x \ln b}, \quad (41)$$

$$e^{\ln x} = x, \quad \ln 1 = 0, \quad \ln e = 1. \quad (42)$$

O'sish va pasayish modellari $A(t) = A_0 a^t$ ni $A_0 e^{kt}$ bilan o'sish tezligi konstantasi orqali taqqoslaydi; ikki barobar o'sish vaqti — me'yor kattalik.

$$A(t) = A_0 a^t = A_0 e^{kt}, \quad k = \ln a, \quad a > 1 : \text{o'sish}; \quad 0 < a < 1 : \text{pasayish}, \quad (43)$$

$$N(t) = N_0 2^{t/T_d} = N_0 e^{kt}, \quad k = \frac{\ln 2}{T_d}, \quad T_d = \frac{\ln 2}{k}. \quad (44)$$

Daraja va radianlar $\pi \text{ rad} = 180^\circ$ orqali o'zgartiriladi: $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$.

3.2 Math I: 2-hafta amaliy mashg'uloti (funksiyalar) — savollar va to'liq yechimlar

1. Har bir bog'lanish y ni x ning funksiyasi sifatida aniqlaydimi: a. $y = 3x - 2$; b. $x^2 + y^2 = 9$. Zarur bo'lsa, y ga nisbatan yechib, vertikal to'g'ri testini qo'llang.

[Manba: 2-hafta amaliy mashg'uloti, 1-mashq]

Yechish. a. $y = 3x - 2$ har bir x uchun aynan bitta y beradi: funksiya (to'g'ri chiziq). b. Yechsak $y = \pm\sqrt{9 - x^2}$: ko'pchilik x (masalan $x = 0$) ikkita qiymat $y = \pm 3$ beradi, demak vertikal to'g'ri aylanani ikki marta kesadi; bu funksiya emas.

2. $f(x) = 2x^2 - 3x + 1$ berilgan. $f(0)$, $f(-2)$, $f(a)$, $f(a+h)$ va $\frac{f(a+h) - f(a)}{h}$ ($h \neq 0$) ni hisoblab soddalashtiring.

[Manba: 2-hafta amaliy mashg'uloti, 2-mashq]

Yechish. $f(0) = 1$; $f(-2) = 2(4) + 6 + 1 = 15$; $f(a) = 2a^2 - 3a + 1$; $f(a+h) = 2(a+h)^2 - 3(a+h) + 1 = 2a^2 + 4ah + 2h^2 - 3a - 3h + 1$. Ayirma nisbati:

$$\frac{f(a+h) - f(a)}{h} = \frac{4ah + 2h^2 - 3h}{h} = 4a + 2h - 3.$$

Bu ifoda 3-4 haftalardagi hosilani bashorat qiladi ($h \rightarrow 0$ da u $4a - 3$ ga intiladi).

3. a. $f(x) = \frac{x-1}{x^2+9}$; b. $g(x) = \sqrt{5-2x}$ ning tabiiy sohasini toping. Interval yozuvida yozing.

[Manba: 2-hafta amaliy mashg'uloti, 3-mashq]

Yechish. a. Mahraj $x^2 + 9 \geq 9 > 0$ hech qachon nol bo'lmaydi, demak soha butun $\mathbb{R} = (-\infty, \infty)$. b. $5 - 2x \geq 0 \iff x \leq \frac{5}{2}$ kerak: soha $(-\infty, \frac{5}{2}]$.

4. $f(x) = 2x - 3$, $g(x) = x^2 + 1$ berilgan. $(f \circ g)(x)$ va $(g \circ f)(x)$ ni toping; ular tengmi?

[Manba: 2-hafta amaliy mashg'uloti, 4-mashq]

Yechish. $(f \circ g)(x) = f(g(x)) = 2(x^2 + 1) - 3 = 2x^2 - 1$; $(g \circ f)(x) = g(2x - 3) = (2x - 3)^2 + 1 = 4x^2 - 12x + 10$. Ular farq qiladi (masalan $x = 0$ da: -1 va 10), kompozitsiya umuman almashinuvchi emasligini tasdiqlaydi — kvadratga ko'tarib so'ng siljitish bilan siljitib so'ng kvadratga ko'tarirish bir xil emas.

5. Har bir funksiya juftmi, toqmi yoki na juft na toq: $|x| + 1$ va $\frac{x}{x^2+1}$; algebraik va grafik tasdiqlang.

[Manba: 2-hafta amaliy mashg'uloti, 5-mashq]

Yechish. $f(x) = |x| + 1$: $f(-x) = |-x| + 1 = |x| + 1 = f(x)$ — juft; grafigi y -o'qqa nisbatan ko'zgu simmetrik. $g(x) = \frac{x}{x^2+1}$: $g(-x) = \frac{-x}{x^2+1} = -g(x)$ — toq; grafigi bosh nuqta atrofida yarim burilishga simmetrik.

6. Har bir funksiyaning chiziqli, ko'phad, ratsional, algebraik yoki transsendent deb tasniflang (funksiya bir nechta toifaga tegishli bo'lishi mumkin): $\frac{x+1}{x^2+4}$ va $\sqrt{x^2+1}$.

[Manba: 2-hafta amaliy mashg'uloti, 6-mashq]

Yechish. $\frac{x+1}{x^2+4}$ ko'phadlar nisbati, demak ratsional (va algebraik). $\sqrt{x^2+1}$ ko'phadning ildizi, demak algebraik (ratsional emas, chunki u ko'phadlar nisbati emas).

7. Sanoxona kamerasidagi harorat $t = 0$ da 20°C dan $t = 15$ min da 80°C gacha chiziqli ko'tariladi. (1) $T(t)$ ni toping; (2) qiyalik va kesishmani izohlang; (3) $T(8)$ ni toping; (4) kamera qachon 60°C ga yetadi? (5) fizik jihatdan mantiqiy sohani ko'rsating.

[Manba: 2-hafta amaliy mashg'uloti, 7-mashq]

Yechish. (1) Qiyalik $m = \frac{80-20}{15-0} = 4^\circ\text{C/min}$, demak $T(t) = 20 + 4t$. (2) Qiyalik isitish tezligi: daqiqasiga $+4$ daraja; kesishma 20 — boshlang'ich harorat. (3) $T(8) = 20 + 32 = 52^\circ\text{C}$. (4) $20 + 4t = 60 \Rightarrow t = 10$ min. (5) $[0, 15]$ (yoki model chiziqli bo'lishni to'xtatmaguncha).

8. $p(x) = x^3 - 4x$ uchun aniqlang: soha; kesishmalar; simmetriya; chekka xatti-harakat; aniqlanmagan nuqtalar.

[Manba: 2-hafta amaliy mashg'uloti, 8-mashq]

Yechish. Soha $\mathbb{R}$ (ko'phad). Kesishmalar: $p(0) = 0$; $x^3 - 4x = x(x-2)(x+2) = 0$ dan $x = -2, 0, 2$ — hammasi bosh nuqtadan o'tuvchi x -o'qda. Simmetriya: $p(-x) = -x^3 + 4x = -p(x)$ — toq. Chekka xatti-harakat: bosh had x^3 , koeffitsienti musbat, demak $x \rightarrow +\infty$ da $p(x) \rightarrow +\infty$ va $x \rightarrow -\infty$ da $p(x) \rightarrow -\infty$. Ko'phad hamma joyda aniqlangan: aniqlanmagan nuqtalar yo'q.

9. Elektr ta'minotchi $0 \leq E \leq 200$ uchun $C(E) = 0.10E$ va $E > 200$ uchun $C(E) = 20 + 0.15(E - 200)$ tarif qo'llaydi (kWh). (1) $C(150)$, $C(200)$, $C(350)$ ni toping; (2) eskizing; (3) model $E = 200$ da uzluksizmi? (4) har bir qiyalikning ma'nosi; (5) bitta cheklov.

[Manba: 2-hafta amaliy mashg'uloti, 9-mashq]

Yechish. (1) $C(150) = 15$; $C(200) = 20$; $C(350) = 20 + 0.15 \cdot 150 = 42.5$ (barchasi \$). (2) Ikkita chiziqli bo'lak: 200 gacha qiyalik 0.10 , keyin 0.15 — $(200, 20)$ da burilishli

bo'lakli-chiziqli grafik. (3) Ha: ikkala formula ham $E = 200$ da 20 beradi, bo'laklar sakrashsiz qo'shiladi. (4) Qiyaliklar — chegaraviy narxlar: birinchi 200 kWh uchun 0.10\$/kWh, undan keyin 0.15\$/kWh. (5) Cheklov: aynan 200 kWh chegara narx sakrash iste'molchilarni chegaradan ozgina pastda to'xtashga undaydi; real tariflar silliqroq yoki bloklar bilan o'zgaradi.

10. $f(x) = 4x - 7$ ning teskarisini toping va sohalarni ko'rsating; $f(f^{-1}(x)) = x$ ni tekshiring.

[Manba: 2-hafta amaliy mashg'uloti, 10-mashq]

Yechish. $y = 4x - 7 \Rightarrow x = \frac{y+7}{4}$, demak $f^{-1}(x) = \frac{x+7}{4}$, soha va qiymatlar to'plami $\mathbb{R}$. Tekshirish: $f(f^{-1}(x)) = 4 \cdot \frac{x+7}{4} - 7 = x + 7 - 7 = x$. Ikkalasi ham 4 va $\frac{1}{4}$ qiyalikli to'g'ri chiziqlar — o'zaro teskari qiyaliklar, $y = x$ ga nisbatan ko'zgu.

11. Har bir model uchun o'sish/pasayishni aniqlang va boshlang'ich qiymat hamda o'sish/pasayish koeffitsientini ko'rsating: $A(t) = 500(1.08)^t$, $B(t) = 1200(0.85)^t$, $C(t) = 40e^{0.03t}$, $D(t) = 200e^{-0.4t}$.

[Manba: 2-hafta amaliy mashg'uloti, 11-mashq]

Yechish. A : o'sish, boshlang'ich 500, koeffitsient 1.08 (davr uchun +8%). B : pasayish, boshlang'ich 1200, koeffitsient 0.85 (−15%). C : o'sish, boshlang'ich 40, uzluksiz tezlik $k = 0.03$. D : pasayish, boshlang'ich 200, uzluksiz tezlik $k = -0.4$. e^{kt} dagi k ishorasi yoki a^t da $a > 1$ yoki yo'qligi yo'nalishni hal qiladi.

12. $2^{x+1} = 16$ ni yeching.

[Manba: 2-hafta amaliy mashg'uloti, 12-mashq]

Yechish. $16 = 2^4$, demak $x + 1 = 4$ va $x = 3$. (Ekvivalent ravishda ikkala tomondan $\log_2$ olish mumkin.)

13. Akkumulyator bo'shashi: $E(t) = E_0 e^{-0.025t}$, t soatlarda. (1) 10 soatdan keyin qolgan ulush? (2) Qachon 50% qoladi? (3) Qachon 10% qoladi? (4) Nega aynan nol hech qachon bo'lmaydi?

[Manba: 2-hafta amaliy mashg'uloti, 13-mashq]

Yechish. (1) $E(10)/E_0 = e^{-0.25} \approx 0.7788$: taxminan 77.9%. (2) $e^{-0.025t} = \frac{1}{2} \Rightarrow t = \frac{\ln 2}{0.025} \approx 27.7$ soat. (3) $e^{-0.025t} = 0.1 \Rightarrow t = \frac{\ln 10}{0.025} \approx 92.1$ soat. (4) Eksponensial har bir chekli t uchun qat'iy musbat; u nolga faqat $t \rightarrow \infty$ limitida yaqinlashadi — chegaralarga birinchi tanishuv.

14. Aholi o'sishi: $P(t) = 800e^{0.35t}$, t soatlarda. (1) Boshlang'ich aholi? (2) 5 soatdan keyin? (3) Ikkilanish vaqti? (4) Model cheksiz uzoq realistik qolishi mumkinmi?

[Manba: 2-hafta amaliy mashg'uloti, 14-mashq]

Yechish. (1) $P(0) = 800$. (2) $P(5) = 800e^{1.75} \approx 800 \cdot 5.755 \approx 4603$. (3) $T_d = \frac{\ln 2}{0.35} \approx 1.98$ soat. (4) Yo'q: eksponensial o'sish ozuqa tugashi va makonni hisobga olmaydi; real sharoitda model faqat madaniyat boshida yaqin amal qiladi.

3.3 Math I: 2-hafta mustaqil o'quv topshiriqlari (ISA) — savollar va to'liq yechimlar

1. Har bir bog'lanish y ni x ning funksiyasi sifatida aniqlaydimi: a. $y = 3x - 2$; b. $x^2 + y^2 = 9$; c. $y^2 = x + 1$; d. $x = y^3 - 2$.

[Manba: 2-hafta ISA, 2-topshiriq]

Yechish. a. Funksiya. b. Funksiya emas (aylana: har x ga ikkita y). c. $y = \pm\sqrt{x+1}$: $x > -1$ da ikkita qiymat, funksiya emas. d. y ga nisbatan yechamiz: $y = \sqrt[3]{x+2}$; kub ildiz har bir haqiqiy x uchun yagona, demak bu funksiya (gorizontal testdan o'tmasa ham, vertikal testdan o'tadi).

2. $f(x) = 2x^2 - 3x + 1$ uchun $f(0), f(-2), f(a), f(a+h), \frac{f(a+h)-f(a)}{h}$ ni hisoblang.

[Manba: 2-hafta ISA, 3-topshiriq]

Yechish. Amaliy mashg'ulotning 2-mashqi bilan bir xil: 1; 15; $2a^2 - 3a + 1$; $2a^2 + 4ah + 2h^2 - 3a - 3h + 1$; ayirma nisbati $= 4a + 2h - 3$.

3. Tabiiy sohani toping: a. $f(x) = \frac{x-1}{x^2+9}$; b. $g(x) = \sqrt{5-2x}$; c. $h(x) = \frac{\sqrt{x-1}}{x-4}$; d. $p(x) = \sqrt{\frac{x+2}{x-1}}$.

[Manba: 2-hafta ISA, 4-topshiriq]

Yechish. a. $(-\infty, \infty)$. b. $(-\infty, \frac{5}{2}]$. c. $x-1 \geq 0$ va $x \neq 4$ kerak: $[1, 4) \cup (4, \infty)$. d. $\frac{x+2}{x-1} \geq 0$ va $x \neq 1$ kerak: kritik nuqtalar -2 va 1 ; nisbat $(-\infty, -2] \cup (1, \infty)$ da nomanfiy.

4. $f(x) = x^2 + 2$, $g(x) = -|x-1| + 4$, $h(x) = \sqrt{x+3} - 2$ ning qiymatlar to'plamini (range) toping.

[Manba: 2-hafta ISA, 5-topshiriq]

Yechish. $x^2 \geq 0 \Rightarrow f \geq 2$: range $[2, \infty)$. $|x-1| \geq 0 \Rightarrow -|x-1| \leq 0 \Rightarrow g \leq 4$: range $(-\infty, 4]$. $\sqrt{x+3} \geq 0 \Rightarrow h \geq -2$: range $[-2, \infty)$. Har bir javob ichki funksiyaning ekstremal qiymatidan keladi.

5. $f(x) = x^2 - 1$, $g(x) = \sqrt{x+2}$ berilgan. $f+g$, $f-g$, fg , f/g va har birining yangi sohasini aniqlang.

[Manba: 2-hafta ISA, 6-topshiriq]

Yechish. $(f \pm g)(x) = x^2 - 1 \pm \sqrt{x+2}$; $(fg)(x) = (x^2 - 1)\sqrt{x+2}$; $(f/g)(x) = \frac{x^2-1}{\sqrt{x+2}}$. Sohalar: $f+g$, $f-g$, fg uchun $x \geq -2$ ($\mathbb{R}$ va $[-2, \infty)$ kesishmasi), ya'ni $[-2, \infty)$. f/g uchun qo'shimcha $\sqrt{x+2} \neq 0$, ya'ni $x \neq -2$: $(-2, \infty)$.

6. $f(x) = 2x - 3$, $g(x) = x^2 + 1$ berilgan. $(f \circ g)(x)$, $(g \circ f)(x)$, $(f \circ f)(x)$, $(g \circ g)(x)$ ni toping.

[Manba: 2-hafta ISA, 7-topshiriq]

Yechish. $(f \circ g)(x) = 2x^2 - 1$; $(g \circ f)(x) = 4x^2 - 12x + 10$; $(f \circ f)(x) = 2(2x - 3) - 3 = 4x - 9$; $(g \circ g)(x) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$. Birinchi ikkisi farq qiladi, demak kompozitsiya almashinuvchi emas; oxirgi ikkisi o'z-o'ziga kompozitsiya to'la qonuniy ekanini ko'rsatadi.

7. Har bir funksiya juftmi, toqmi yoki na juft na toq: $x^4 - 2x^2$; $x^3 - 4x$; $x^3 + x^2$; $|x| + 1$; $\frac{x^2}{x+1}$.

[Manba: 2-hafta ISA, 8-topshiriq]

Yechish. $x^4 - 2x^2$: faqat juft darajalar — juft. $x^3 - 4x$: faqat toq darajalar — toq. $x^3 + x^2$: aralash — na juft na toq. $|x| + 1$: juft (Amaliy 5-mashq). $\frac{x^2}{x+1}$: surat juft, mahraj ishara almashtiradi: $\frac{x^2}{-x+1} \neq \pm \frac{x^2}{x+1}$ — na juft na toq.

8. Tasniflang: $3x - 7$; $x^4 - 2x + 1$; $\frac{x+1}{x^2+4}$; $\sqrt{x^2+1}$; 2^x ; $\sin x$.

[Manba: 2-hafta ISA, 9-topshiriq]

Yechish. $3x - 7$: chiziqli (ko'phad). $x^4 - 2x + 1$: ko'phad. $\frac{x+1}{x^2+4}$: ratsional. $\sqrt{x^2+1}$: algebraik (ildiz). 2^x : eksponensial — transsendent. $\sin x$: trigonometrik — transsendent.

9. Amaliy mashg'ulotning 10-mashqidagi chiziqli harorat modeli — barcha besh qism.

[Manba: 2-hafta ISA, 10-topshiriq]

Yechish. Amaliy mashg'ulotning 7-mashqi bilan bir xil model va javoblar: $T(t) = 20 + 4t$; qiyalik = isitish tezligi $4^\circ\text{C}/\text{min}$, kesishma = boshlang'ich 20°C ; $T(8) = 52^\circ\text{C}$; 60°C ga $t = 10$ min da; soha $[0, 15]$.

10. $p(x) = x^3 - 4x$ va $r(x) = \frac{x+1}{x-1}$ uchun soha, kesishmalar, simmetriya, chekka xatti-harakat, aniqlanmagan nuqtalarni aniqlang.

[Manba: 2-hafta ISA, 12-topshiriq]

Yechish. p : Amaliy 8-mashqdagi kabi (soha $\mathbb{R}$; kesishmalar $-2, 0, 2$; toq; $\pm\infty$ chekkalar; yo'q). r : soha $x \neq 1$, ya'ni $(-\infty, 1) \cup (1, \infty)$. Kesishmalar: $r(0) = -1$; surat $x = -1$ da nol. Simmetriya: $r(-x) = \frac{-x+1}{-x-1} = \frac{x-1}{x+1} \neq \pm r(x)$ — na juft na toq. Chekka xatti-harakat: $x \rightarrow \pm\infty$ da $r(x) \rightarrow 1$ (gorizontal asimptota $y = 1$); $x \rightarrow 1$ da $r(x) \rightarrow \pm\infty$ (vertikal asimptota $x = 1$). Faqat $x = 1$ da aniqlanmagan.

11. Amaliy mashg'ulotning 9-mashqidagi bo'lakli elektr tarifi — barcha besh qism.

[Manba: 2-hafta ISA, 13-topshiriq]

Yechish. Amaliy 9-mashq bilan bir xil javoblar: $C(150) = 15$, $C(200) = 20$, $C(350) = 42.5$; uzluksiz burilgan grafik; qiyaliklar chegaraviy narxlar; cheklov — 200 kWh chegarada rag'bat uzluksizligi.

12. $30^\circ, 135^\circ, 300^\circ$ ni radianga va $\frac{\pi}{6}, \frac{5\pi}{4}, \frac{7\pi}{3}$ ni darajaga o'tkazing.

[Manba: 2-hafta ISA, 14-topshiriq]

Yechish. $\pi/180$ ga ko'paytiramiz: $30^\circ = \frac{\pi}{6}$; $135^\circ = \frac{3\pi}{4}$; $300^\circ = \frac{5\pi}{3}$. $180/\pi$ ga ko'paytiramiz: $\frac{\pi}{6} = 30^\circ$; $\frac{5\pi}{4} = 225^\circ$; $\frac{7\pi}{3} = 420^\circ$.

13. O'ziga bog'lanmaganmi (one-to-one)? a. $f(x) = 3x - 5$, $x \in \mathbb{R}$; b. $g(x) = x^2$, $x \in \mathbb{R}$; c. $h(x) = x^2$, $x \geq 0$; d. $p(x) = \frac{1}{x}$, $x \neq 0$.

[Manba: 2-hafta ISA, 15-topshiriq]

Yechish. a. Ha: nolga teng bo'lmagan qiyalikli to'g'ri gorizontal testdan o'tadi. b. Yo'q: $g(-2) = g(2) = 4$. c. Ha: $[0, \infty)$ da kvadratga ko'tarirish inyektiv (cheqlash gorizontal testni tiklaydi). d. Ha: $\frac{1}{x_1} = \frac{1}{x_2} \Rightarrow x_1 = x_2$.

14. Teskari funksiyalar va ularning sohalari: a. $f(x) = 4x - 7$; b. $g(x) = \frac{x+3}{2}$; c. $h(x) = x^2 + 1$, $x \geq 0$; d. $p(x) = \frac{2x+3}{x-1}$, $x \neq 1$; kompozitsiya bilan tekshiring.

[Manba: 2-hafta ISA, 16-topshiriq]

Yechish. a. $f^{-1}(x) = \frac{x+7}{4}$, soha $\mathbb{R}$ (Amaliy 10-mashq). b. $y = \frac{x+3}{2} \Rightarrow x = 2y - 3$, demak $g^{-1}(x) = 2x - 3$, soha $\mathbb{R}$. c. $y = x^2 + 1 \Rightarrow x = \sqrt{y-1}$ (musbat ildiz, chunki $x \geq 0$): $h^{-1}(x) = \sqrt{x-1}$, soha $[1, \infty)$. d. $y = \frac{2x+3}{x-1} \Rightarrow y(x-1) = 2x+3 \Rightarrow x(y-2) = y+3 \Rightarrow x = \frac{y+3}{y-2}$: $p^{-1}(x) = \frac{x+3}{x-2}$, soha $x \neq 2$. Har biri $f(f^{-1}(x)) = x$ ni bajaradi.

15. A, B, C, D ning o'sish/pasayish tasnifi (Amaliy 11-mashq).

[Manba: 2-hafta ISA, 17-topshiriq]

Yechish. Bir xil: A o'sish ($a = 1.08$), B pasayish ($a = 0.85$), C o'sish ($k = 0.03$), D pasayish ($k = -0.4$).

16. Yeching, avval aniq javoblarni berib: a. $2^{x+1} = 16$; b. $5^{2x} = 125$; c. $e^{3x} = 7$; d. $100e^{-0.2t} = 20$.

[Manba: 2-hafta ISA, 18-topshiriq]

Yechish. a. $x + 1 = 4 \Rightarrow x = 3$. b. $2x = 3 \Rightarrow x = \frac{3}{2}$. c. $3x = \ln 7 \Rightarrow x = \frac{\ln 7}{3} \approx 0.6486$. d. $e^{-0.2t} = 0.2 \Rightarrow t = -\frac{\ln 0.2}{0.2} = 5 \ln 5 \approx 8.047$.

17. Akkumulyator bo'shashi (Amaliy 13-mashq) — barcha to'rt qism.

[Manba: 2-hafta ISA, 19-topshiriq]

Yechish. Bir xil: 10 soatdan keyin $e^{-0.25} \approx 77.9\%$; 50% $\frac{\ln 2}{0.025} \approx 27.7$ soatda; 10% $\frac{\ln 10}{0.025} \approx 92.1$ soatda; eksponensial chekli vaqtda nolgacha yetmaydi.

18. Bakteriya aholisi (Amaliy 14-mashq) — barcha to'rt qism.

[Manba: 2-hafta ISA, 20-topshiriq]

Yechish. Bir xil: $P_0 = 800$; $P(5) \approx 4603$; ikkilanish vaqti $\frac{\ln 2}{0.35} \approx 1.98$ soat; faqat eksponensial modellar uzoq muddatda ishlamaydi (resurslar, makon, chiqindilar).

3.4 Chiziqli algebra I, 2-hafta: formulalar katalogi

2-hafta vektor geometriyasining ikki ko'paytmasi va birinchi matritsa tushunchalarini qo'shadi.

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + \dots + u_n v_n = \mathbf{u}^T \mathbf{v}, \quad \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad (45)$$

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0, \quad \text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}, \quad (46)$$

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}, \quad \|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| |\sin \theta| = \text{yuza}, \quad (47)$$

$$\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u}), \quad (\mathbf{u} \times \mathbf{v}) \perp \mathbf{u}, \mathbf{v}, \quad (48)$$

$$A = [a_{ij}] \in \mathbb{R}^{m \times n}, \quad (A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c a_{ij}, \quad (49)$$

$$A\mathbf{x} = x_1 \mathbf{a}_1 + \dots + x_n \mathbf{a}_n \quad (\text{ustunlar rasmi}), \quad (50)$$

bunda $\mathbf{a}_j$ — A ning j -ustuni. O'lchamlar mos bo'lishi shart: $A\mathbf{x}$ faqat A $m \times n$ va $\mathbf{x} \in \mathbb{R}^n$ bo'lganda mavjud, natija $\mathbb{R}^m$ da yotadi. Diagonal matritsa $D = \text{diag}(d_1, \dots, d_n)$ har bir koordinatani masshtablaydi: $D\mathbf{x} = (d_1 x_1, \dots, d_n x_n)^T$.

3.5 Chiziqli algebra I, 2-hafta: ma'ruza darsidagi muammolar — savollar va to'liq yechimlar

1. $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ berilgan. $\mathbf{u} \cdot \mathbf{v}$ ni hisoblang; burchakni tasniflang.

[Manba: LA 2-hafta ma'ruza, 1-muammo]

Yechish. $\mathbf{u} \cdot \mathbf{v} = 3 \cdot 4 + (-2) \cdot 5 = 12 - 10 = 2 > 0$. Musbat skalyar ko'paytma $\cos \theta > 0$ anglatadi, demak burchak o'tkir.

2. $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}$ bilan: $\|\mathbf{u}\|$ va $\text{dist}(\mathbf{u}, \mathbf{v})$ ni toping.

[Manba: LA 2-hafta ma'ruza, 2-muammo]

Yechish. $\|\mathbf{u}\| = \sqrt{4+1+4} = 3$. $\mathbf{u} - \mathbf{v} = \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix}$, demak $\text{dist}(\mathbf{u}, \mathbf{v}) = \sqrt{9+16+4} = \sqrt{29}$.

3. $\mathbf{u} = \begin{bmatrix} 2 \\ k \\ -1 \end{bmatrix}$ vektor $\mathbf{v} = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$ ga orthogonal bo'ladigan k ni toping.

[Manba: LA 2-hafta ma'ruza, 3-muammo]

Yechish. Ortogonallik $\mathbf{u} \cdot \mathbf{v} = 0$: $6 + k - 4 = 0$, demak $k = -2$.

4. $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ va $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ orasidagi burchakni toping.

[Manba: LA 2-hafta ma'ruza, 4-muammo]

Yechish. $\mathbf{u} \cdot \mathbf{v} = 1 - 1 = 0$: vektorlar perpendikulyar, $\theta = 90^\circ$ — skalyar ko'paytma nol bo'lgach formulalar kerak emas.

5. $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ vektorni $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ga proyeksiya qiling.

[Manba: LA 2-hafta ma'ruza, 5-muammo]

Yechish. $\mathbf{u} \cdot \mathbf{v} = 6$, $\mathbf{v} \cdot \mathbf{v} = 2$, demak $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{6}{2} \mathbf{v} = 3\mathbf{v} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$. Qoldiq $\mathbf{u} - 3\mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ shartni bajaradi: $(\mathbf{u} - 3\mathbf{v}) \cdot \mathbf{v} = 0$: yoyilma parallel + perpendikulyar qismlarga bo'linadi.

6. $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$ uchun $\mathbf{u} \times \mathbf{v}$ ni hisoblang va ortogonallikni tekshiring.

[Manba: LA 2-hafta ma'ruza, 6-muammo]

Yechish. (47) bo'yicha: $\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 2 \cdot 2 - 0 \cdot (-1) \\ 0 \cdot 3 - 1 \cdot 2 \\ 1 \cdot (-1) - 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ -7 \end{bmatrix}$. Tekshirish: $1 \cdot 4 + 2 \cdot (-2) + 0 \cdot (-7) = 0$ va $3 \cdot 4 + (-1)(-2) + 2 \cdot (-7) = 0$ — ikkala kirishga ham ortogonal.

7. Yon tomonlari $\mathbf{a} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$ bo'lgan parallelogram yuzasini toping.

[Manba: LA 2-hafta ma'ruza, 7-muammo]

Yechish. $\mathbf{a} \times \mathbf{b} = \begin{bmatrix} 0 \cdot (-1) - 1 \cdot 3 \\ 1 \cdot 1 - 2 \cdot (-1) \\ 2 \cdot 3 - 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 3 \\ 6 \end{bmatrix}$, demak yuza $\|\mathbf{a} \times \mathbf{b}\| = \sqrt{9+9+36} = \sqrt{54} = 3\sqrt{6}$.

8. $A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}$ uchun: o'lchamini ko'rsating, a_{13} , a_{22} ni toping va $\mathbf{a}_1$, $\mathbf{a}_3$ ustunlarni yozing.

[Manba: LA 2-hafta ma'ruza, 8-muammo]

Yechish. A 2×3 (avval qatorlar). $a_{13} = 4$ (1-qator, 3-ustun), $a_{22} = 3$. $\mathbf{a}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $\mathbf{a}_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$.

9. $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 \\ -1 & 5 \end{bmatrix}$ berilgan. $2A - B$ ni toping.

[Manba: LA 2-hafta ma'ruza, 9-muammo]

Yechish. $2A = \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$; elementma-element ayiramiz: $2A - B = \begin{bmatrix} -2 & -5 \\ 7 & -5 \end{bmatrix}$.

10. A 3×2 , B 2×3 , $\mathbf{x} \in \mathbb{R}^2$, $\mathbf{y} \in \mathbb{R}^3$. $A\mathbf{x}$, $A\mathbf{y}$, $B\mathbf{x}$, $B\mathbf{y}$ dan qaysilari aniqlangan?

[Manba: LA 2-hafta ma'ruza, 10-muammo]

Yechish. $A\mathbf{x}$: $(3 \times 2)(2 \times 1)$ OK, natija $\mathbb{R}^3$ da. $B\mathbf{y}$: $(2 \times 3)(3 \times 1)$ OK, natija $\mathbb{R}^2$ da. $A\mathbf{y}$ mos kelmaydi ($2 \neq 3$) va $B\mathbf{x}$ mos kelmaydi ($3 \neq 2$): ikkalasi ham aniqlanmagan. Faqat ichki o'lchamlar mos bo'lishi shart.

11. $D = \text{diag}(2, -1, 3)$, $\mathbf{x} = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$ berilgan. $D\mathbf{x}$ va $I_3\mathbf{x}$ ni toping.

[Manba: LA 2-hafta ma'ruza, 11-muammo]

Yechish. $D\mathbf{x} = \begin{bmatrix} 2 \cdot 4 \\ -1 \cdot 5 \\ 3 \cdot (-2) \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ -6 \end{bmatrix}$: har bir koordinata mos diagonal element bilan masshtablanadi. $I_3\mathbf{x} = \mathbf{x} = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$: birlik matritsa hech narsa qilmaydi.

12. $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \\ 4 & 0 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ berilgan. $A\mathbf{x}$ ni ustunlar rasmi bilan va qatorlar skalyar ko'paytmasi orqali hisoblang.

[Manba: LA 2-hafta ma'ruza, 12-muammo]

Yechish. Ustunlar: $2 \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}$. Qatorlar: $[1 \ 2] \cdot \mathbf{x} = 0$, $[-1 \ 3] \cdot \mathbf{x} = -5$, $[4 \ 0] \cdot \mathbf{x} = 8$, ustiga yozsak bir xil $\begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}$. Ikkala rasim doim mos keladi; ustun ko'rinishi ko'paytma nima anglatishini tushuntiradi.

13. $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ va $A\mathbf{x} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$ berilgan. $\mathbf{x}$ ni toping.

[Manba: LA 2-hafta ma'ruza, 13-muammo]

Yechish. Komponentalar: $x_1 + 2x_2 = 7$, $3x_1 - x_2 = 5$. Ikkinchisidan $x_2 = 3x_1 - 5$; qo'ysak: $x_1 + 6x_1 - 10 = 7$, demak $7x_1 = 17$, $x_1 = \frac{17}{7}$, $x_2 = \frac{16}{7}$. Shunday qilib $\mathbf{x} = \begin{bmatrix} 17/7 \\ 16/7 \end{bmatrix}$.

14. Nolga teng bo'lmagan $\mathbf{u} + \mathbf{v}$ va $\mathbf{u} - \mathbf{v}$ vektorlari ortogonal. $\|\mathbf{u}\|$ va $\|\mathbf{v}\|$ haqida nima kelib chiqadi?

[Manba: LA 2-hafta ma'ruza, 14-muammo]

Yechish. Ortogonallik $0 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2$ beradi (o'rta hadlar yo'qoladi), demak $\|\mathbf{u}\| = \|\mathbf{v}\|$: uzunliklar teng, $\mathbf{u}$ va $\mathbf{v}$ — to'rtburchaksimon rasmdagi diagonal yarim-vektorlari.

3.6 Chiziqli algebra I, 2-hafta: ISA mashqlari — savollar va to'liq yechimlar

1. $\mathbf{u} = (-3, 2)^T$, $\mathbf{v} = (4, 5)^T$ berilgan. $\mathbf{u} \cdot \mathbf{v}$ ni hisoblang; burchak o'tkirmi, to'g'rimi yoki kengmi?

[Manba: LA ISA 2-hafta, 1-mashq]

Yechish. $\mathbf{u} \cdot \mathbf{v} = -12 + 10 = -2 < 0$, demak $\cos \theta < 0$: burchak keng.

2. $\mathbf{a} = (1, -2, 4)^T$, $\mathbf{b} = (3, 0, -1)^T$ berilgan. $\mathbf{a} \cdot \mathbf{b}$, $\|\mathbf{a}\|$, $\|\mathbf{b}\|$, $\|\mathbf{a} - \mathbf{b}\|$ ni toping.

[Manba: LA ISA 2-hafta, 2-mashq]

Yechish. $\mathbf{a} \cdot \mathbf{b} = 3 + 0 - 4 = -1$; $\|\mathbf{a}\| = \sqrt{1 + 4 + 16} = \sqrt{21}$; $\|\mathbf{b}\| = \sqrt{9 + 0 + 1} = \sqrt{10}$; $\mathbf{a} - \mathbf{b} = (-2, -2, 5)^T$, $\|\mathbf{a} - \mathbf{b}\| = \sqrt{4 + 4 + 25} = \sqrt{33}$.

3. $\mathbf{u} = (2, k, 3)^T$ vektor $\mathbf{v} = (-1, 4, 2)^T$ ga orthogonal bo'ladigan k ni toping.

[Manba: LA ISA 2-hafta, 3-mashq]

Yechish. $\mathbf{u} \cdot \mathbf{v} = -2 + 4k + 6 = 0 \Rightarrow 4k = -4 \Rightarrow k = -1$.

4. $\mathbf{u} = (3, 1)^T$, $\mathbf{v} = (-2, 4)^T$ uchun aniq $\cos \theta$ va keyin eng yaqin darajagacha θ ni toping.

[Manba: LA ISA 2-hafta, 4-mashq]

Yechish. $\mathbf{u} \cdot \mathbf{v} = -6 + 4 = -2$; $\|\mathbf{u}\| = \sqrt{10}$, $\|\mathbf{v}\| = \sqrt{20} = 2\sqrt{5}$; $\cos \theta = \frac{-2}{\sqrt{10} \cdot 2\sqrt{5}} = \frac{-2}{10\sqrt{2}} = -\frac{1}{5\sqrt{2}} \approx -0.1414$, demak $\theta \approx 98^\circ$.

5. $\mathbf{u} = (6, 1)^T$ vektorni $\mathbf{v} = (2, -1)^T$ ga proyeksiya qiling; $\mathbf{u}$ – proj ning $\mathbf{v}$ ga ortogonal ekanini tekshiring.

[Manba: LA ISA 2-hafta, 5-mashq]

Yechish. $\mathbf{u} \cdot \mathbf{v} = 12 - 1 = 11$, $\mathbf{v} \cdot \mathbf{v} = 5$: $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{11}{5}(2, -1)^T = (\frac{22}{5}, -\frac{11}{5})^T$. Qoldiq: $\mathbf{u} - \text{proj} = (\frac{8}{5}, \frac{16}{5})^T$; $\mathbf{v}$ bilan skalyar ko'paytma: $\frac{16}{5} - \frac{16}{5} = 0$. Tasdiqlandi.

6. Nolga teng bo'lmagan $\mathbf{p}, \mathbf{q}$ shartni qondiradi: $\mathbf{p} \cdot \mathbf{q} = \|\mathbf{p}\| \|\mathbf{q}\|$. Yo'nalishlari haqida nima xulosa qilish mumkin?

[Manba: LA ISA 2-hafta, 6-mashq]

Yechish. (45) dan $\cos \theta = 1$, demak $\theta = 0$: vektorlar aynan bir yo'nalishda (biri ikkinchisining musbat ko'paytmasi). Burchak formulasidagi tenglik Cauchy-Schwarz g'oyasining chegara holi.

7. $\mathbf{a} = (2, 1, -1)^T$, $\mathbf{b} = (0, 3, 2)^T$ uchun $\mathbf{a} \times \mathbf{b}$ va $\mathbf{b} \times \mathbf{a}$ ni hisoblang; ishora bog'lanishini tekshiring.

[Manba: LA ISA 2-hafta, 7-mashq]

Yechish. $\mathbf{a} \times \mathbf{b} = \begin{bmatrix} 1 \cdot 2 - (-1) \cdot 3 \\ (-1) \cdot 0 - 2 \cdot 2 \\ 2 \cdot 3 - 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ 6 \end{bmatrix}$, va $\mathbf{b} \times \mathbf{a} = \begin{bmatrix} -5 \\ 4 \\ -6 \end{bmatrix} = -(\mathbf{a} \times \mathbf{b})$, (48) ga mos.

8. 7-mashqdagi $\mathbf{a} \times \mathbf{b}$ vektor $\mathbf{a}$ va $\mathbf{b}$ ga ortogonaligini tekshiring.

[Manba: LA ISA 2-hafta, 8-mashq]

Yechish. $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 2 \cdot 5 + 1 \cdot (-4) + (-1) \cdot 6 = 10 - 4 - 6 = 0$; $\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = 0 \cdot 5 + 3 \cdot (-4) + 2 \cdot 6 = -12 + 12 = 0$. Ikkalasi ham nol, (47) dan keyin kafolatlanganidek.

9. Yon tomonlari $\mathbf{u} = (1, 2, 2)^T$, $\mathbf{v} = (3, -1, 0)^T$ bo'lgan parallelogram yuzasini toping.

[Manba: LA ISA 2-hafta, 9-mashq]

Yechish. $\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 2 \cdot 0 - 2 \cdot (-1) \\ 2 \cdot 3 - 1 \cdot 0 \\ 1 \cdot (-1) - 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ -7 \end{bmatrix}$; yuza = $\sqrt{4 + 36 + 49} = \sqrt{89}$.

10. $P = (1, 0, 1)$, $Q = (3, 1, 0)$, $R = (0, 2, 4)$: PQR uchburchak yuzasini toping.

[Manba: LA ISA 2-hafta, 10-mashq]

Yechish. P dan qirralar: $\mathbf{PQ} = (2, 1, -1)^T$, $\mathbf{PR} = (-1, 2, 3)^T$. Ko'paytma:

$$\begin{bmatrix} 1 \cdot 3 - (-1) \cdot 2 \\ (-1)(-1) - 2 \cdot 3 \\ 2 \cdot 2 - 1 \cdot (-1) \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 5 \end{bmatrix}, \text{ norma } \sqrt{75} = 5\sqrt{3}; \text{ uchburchak yuzasi } = \frac{1}{2} \cdot 5\sqrt{3} = \frac{5\sqrt{3}}{2}$$

 (parallelogramning yarmi).

11. $\mathbf{a} = (1, t, 2)^T$ va $\mathbf{b} = (3, 6, 6)^T$ shartni qondiradigan barcha haqiqiy t ni toping: $\mathbf{a} \times \mathbf{b} = \mathbf{0}$; bo'lmasa, tushuntiring.

[Manba: LA ISA 2-hafta, 11-mashq]

Yechish. Parallel vektorlar nol ko'paytma beradi, demak $\mathbf{a} = c\mathbf{b}$ kerak: $1 = 3c \Rightarrow c = \frac{1}{3}$; u holda $t = 6c = 2$; uchinchisini tekshiramiz: $2 = 6c = 2$ — mos. Demak $t = 2$ yagona qiymat (u holda $\mathbf{a} = \frac{1}{3}\mathbf{b}$).

12. $\mathbf{a} = (1, 0, 2)^T$, $\mathbf{b} = (2, 1, -1)^T$, $\mathbf{c} = (0, 3, 1)^T$ berilgan. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ ni hisoblang; nol qiymatni izohlang.

[Manba: LA ISA 2-hafta, 12-mashq]

Yechish. $\mathbf{b} \times \mathbf{c} = \begin{bmatrix} 1 \cdot 1 - (-1) \cdot 3 \\ (-1) \cdot 0 - 2 \cdot 1 \\ 2 \cdot 3 - 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}$; $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 4 + 0 + 12 = 16$. Bu uch vektor span qilgan parallelepipedning ishorali hajmi; nol qiymat vektorlar bir tekislikda (3D hajm yo'q) anglatardi. Bu yerda $|16| = 16$ kub birlik.

13. $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \end{bmatrix}$ berilgan. O'lchamini ko'rsating; a_{12} , a_{23} ni toping; har bir qator va ustunni vektor sifatida yozing.

[Manba: LA ISA 2-hafta, 13-mashq]

Yechish. O'lcham 2×3 ; $a_{12} = -1$, $a_{23} = 5$. Qatorlar: $[2 \ -1 \ 0]$, $[4 \ 3 \ 5]$; ustunlar: $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$, $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 5 \end{bmatrix}$.

14. $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$ berilgan. $A + B$, $A - B$, $-2A$, $3A - 2B$ ni hisoblang.

[Manba: LA ISA 2-hafta, 14-mashq]

Yechish. $A + B = \begin{bmatrix} 6 & 2 \\ -2 & 6 \end{bmatrix}$; $A - B = \begin{bmatrix} -4 & 4 \\ -2 & 2 \end{bmatrix}$; $-2A = \begin{bmatrix} -2 & -6 \\ 4 & -8 \end{bmatrix}$;

$$3A - 2B = \begin{bmatrix} 3 - 10 & 9 + 2 \\ -6 - 0 & 12 - 4 \end{bmatrix} = \begin{bmatrix} -7 & 11 \\ -6 & 8 \end{bmatrix}.$$

15. $A \ 4 \times 3$, $B \ 3 \times 2$, $\mathbf{x} \in \mathbb{R}^2$, $\mathbf{y} \in \mathbb{R}^3$, $\mathbf{z} \in \mathbb{R}^4$. $A\mathbf{x}$, $A\mathbf{y}$, $A\mathbf{z}$, $B\mathbf{x}$, $B\mathbf{y}$, $B\mathbf{z}$ dan qaysilari aniqlangan va qanday chiqish o'lchamlari bilan?

[Manba: LA ISA 2-hafta, 15-mashq]

Yechish. $A\mathbf{y}$: $(4 \times 3)(3 \times 1)$ — aniqlangan, natija $\mathbb{R}^4$. $B\mathbf{x}$: $(3 \times 2)(2 \times 1)$ — aniqlangan, natija $\mathbb{R}^3$. Qolganlari ($A\mathbf{x}$, $A\mathbf{z}$, $B\mathbf{y}$, $B\mathbf{z}$) ichki o'lchamlari mos kelmasligi tufayli aniqlanmagan.

16. $D = \text{diag}(-2, 3, 1)$, $\mathbf{x} = (4, -1, 5)^T$ berilgan. $D\mathbf{x}$ ni hisoblang; D har bir koordinataga nima qilishini tavsiflang.

[Manba: LA ISA 2-hafta, 16-mashq]

Yechish. $D\mathbf{x} = (-8, 3, 5)^T$: birinchi koordinata -2 ga masshtablanadi (cho'zish va aks ettirish), ikkinchisi 3 ga, uchinchi 1 ga (o'zgarmagan). Diagonal matritsa — koordinata o'qlarining uchta mustaqil masshtablashi.

17. $A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & -1 \end{bmatrix}$, $\mathbf{x} = (2, -1, 3)^T$ berilgan. $A\mathbf{x}$ ni (a) ustunlar birikmasi sifatida, (b) qatorlar skalyar ko'paytmasi orqali hisoblang.

[Manba: LA ISA 2-hafta, 17-mashq]

Yechish. (a) $2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} - 1 \begin{bmatrix} -2 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 + 2 + 9 \\ 8 + 0 - 3 \end{bmatrix} = \begin{bmatrix} 13 \\ 5 \end{bmatrix}$. (b) 1-qator: $1 \cdot 2 + (-2)(-1) + 3 \cdot 3 = 13$; 2-qator: $4 \cdot 2 + 0 + (-1) \cdot 3 = 5$; ustiga yozsak: $\begin{bmatrix} 13 \\ 5 \end{bmatrix}$. Bir xil natija, bo'lishi kerakganidek.

18. $\mathbf{a}_1 = (1, 2)^T$, $\mathbf{a}_2 = (3, -1)^T$ berilgan. $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = (11, 1)^T$ bo'ladigan $\mathbf{x} = (x_1, x_2)^T$ ni toping.

[Manba: LA ISA 2-hafta, 18-mashq]

Yechish. Komponentalar: $x_1 + 3x_2 = 11$, $2x_1 - x_2 = 1$. Ikkinchisidan $x_2 = 2x_1 - 1$; u holda $x_1 + 6x_1 - 3 = 11 \Rightarrow 7x_1 = 14 \Rightarrow x_1 = 2$, $x_2 = 3$.

19. $A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}$ berilgan. $A\mathbf{x} = \mathbf{0}$ bo'ladigan barcha $\mathbf{x}$ ni tavsiflang; ustunlar nima uchun qaram?

[Manba: LA ISA 2-hafta, 19-mashq]

Yechish. $2x_1 + 4x_2 = 0$ va $-x_1 - 2x_2 = 0$ bir xil tenglama: $x_1 = -2x_2$. Yechimlar: $\mathbf{x} = t \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $t \in \mathbb{R}$ — butun to'g'ri yechimlar. Notrivial birikma $-2\mathbf{a}_1 + 1 \cdot \mathbf{a}_2 = \mathbf{0}$ mavjud, bu aynan ustunlarning chiziqli qaramligi (2-ustun 1-ustunning -2 barobar).

20. A 3×2 , bunda $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (2, -1, 4)^T$ va $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (3, 2, 0)^T$. A ni yozing va $A \begin{bmatrix} -2 \\ 5 \end{bmatrix}$ ni qator skalyar ko'paytmalarisiz hisoblang.

[Manba: LA ISA 2-hafta, 20-mashq]

Yechish. Ikkala tasvir A ning ustunlari: $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \\ 4 & 0 \end{bmatrix}$. Chiziqlik: $A \begin{bmatrix} -2 \\ 5 \end{bmatrix} = -2A\mathbf{e}_1 + 5A\mathbf{e}_2 = -2 \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + 5 \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 11 \\ 12 \\ -8 \end{bmatrix}$.

21. Bir xil o'lchamli vektorlar uchun $(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v})$ ni yoying; $\mathbf{u} + \mathbf{v} \perp \mathbf{u} - \mathbf{v}$ aynan $\|\mathbf{u}\| = \|\mathbf{v}\|$ bo'lganda o'rinli ekanini isbotlang.

[Manba: LA ISA 2-hafta, 21-mashq]

Yechish. Taqsimlab: $(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \mathbf{u} \cdot \mathbf{u} - \mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{u} - \mathbf{v} \cdot \mathbf{v} = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2$ (o'rta hadlar yo'qoladi). Ortogonallik buning nolga tengligini anglatadi, ya'ni $\|\mathbf{u}\|^2 = \|\mathbf{v}\|^2$, va normalar nomanfiy bo'lgani uchun $\|\mathbf{u}\| = \|\mathbf{v}\|$.

22. O'z-o'zini tekshirish: (1) skalyar ko'paytma qanday obyekt beradi, vektor ko'paytma-chi? (2) $\mathbf{u} \cdot \mathbf{v}$ ishorasi burchakni qanday tasniflaydi? (3) Nega 2×3 matritsa $\mathbb{R}^3$ dagi vektorga ko'payadi, lekin $\mathbb{R}^2$ dagiga yo'q? (4) $A\mathbf{x}$ da $\mathbf{x}$ elementlari qanday ro'l o'ynaydi? (5) Qaysi 2-hafta amallari teng o'lchamlar talab qiladi va qaysilari aynan $\mathbb{R}^3$ dagi vektorlarni?

[Manba: LA ISA 2-hafta, o'z-o'zini tekshirish]

Yechish. (1) Skalyar; vektor ($\mathbb{R}^3$ da). (2) Musbat $\Rightarrow$ o'tkir, nol $\Rightarrow$ to'g'ri, manfiy $\Rightarrow$ keng. (3) $A\mathbf{x}$ ko'paytmasi ichki o'lchamlar mosligini talab qiladi: 2×3 ga 3×1 mos keladi; 2×1 ga yo'q. (4) Ustunlarning $x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$ chiziqli birikmasida vazn sifatida. (5) Skalyar ko'paytma, qo'shish va ayirish teng o'lchamlar talab qiladi; vektor ko'paytma va uning geometriyasi (yuzalar, normal) faqat $\mathbb{R}^3$ da mavjud.

3.7 Chiziqli algebra I, 2-hafta: TA sessiyasi (15 muammo) — savollar va to'liq yechimlar

1. $\mathbf{a} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$: $\mathbf{a} \cdot \mathbf{b}$ ni hisoblang; burchakni tasniflang.

[Manba: LA 2-hafta TA, 1-muammo]

Yechish. $\mathbf{a} \cdot \mathbf{b} = -10 + 4 = -6 < 0$: keng.

2. $\mathbf{v} = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$ vektorning normasini va uning yo'nalishidagi birlik vektorni toping.

[Manba: LA 2-hafta TA, 2-muammo]

Yechish. $\|\mathbf{v}\| = \sqrt{4 + 1 + 4} = 3$; birlik vektor $\hat{\mathbf{v}} = \begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$.

3. $\mathbf{p} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$ va $\mathbf{q} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$ orasidagi masofani toping.

[Manba: LA 2-hafta TA, 3-muammo]

Yechish. $\mathbf{p} - \mathbf{q} = \begin{bmatrix} 3 \\ -4 \\ 3 \end{bmatrix}$; masofa $= \sqrt{9 + 16 + 9} = \sqrt{34}$.

4. $\begin{bmatrix} 1 \\ t \\ 2 \end{bmatrix}$ vektor $\begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}$ ga orthogonal bo'lsa, t ni toping.

[Manba: LA 2-hafta TA, 4-muammo]

Yechish. $2 - 3t + 8 = 0 \Rightarrow t = \frac{10}{3}$.

5. $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ uchun $\cos \theta$ ni toping.

[Manba: LA 2-hafta TA, 5-muammo]

Yechish. $\mathbf{u} \cdot \mathbf{v} = 1$, $\|\mathbf{u}\| = \sqrt{5}$, $\|\mathbf{v}\| = \sqrt{10}$; $\cos \theta = \frac{1}{\sqrt{50}} = \frac{\sqrt{2}}{10}$.

6. $\mathbf{u} = (5, -1)^T$ vektorni $\mathbf{v} = (2, 1)^T$ ga proyeksiya qiling; $\mathbf{u}$ ni parallel + ortogonal qismlar sifatida yozing.

[Manba: LA 2-hafta TA, 7-muammo]

Yechish. $\mathbf{u} \cdot \mathbf{v} = 9$, $\mathbf{v} \cdot \mathbf{v} = 5$: $\text{proj} = \frac{9}{5} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 18/5 \\ 9/5 \end{bmatrix}$. Ortogonal qism: $\mathbf{u} - \text{proj} = \begin{bmatrix} 7/5 \\ -14/5 \end{bmatrix}$. Yoyilma: $\begin{bmatrix} 5 \\ -1 \end{bmatrix} = \begin{bmatrix} 18/5 \\ 9/5 \end{bmatrix} + \begin{bmatrix} 7/5 \\ -14/5 \end{bmatrix}$.

7. $\mathbf{a} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$ uchun $\mathbf{a} \times \mathbf{b}$ ni hisoblang.

[Manba: LA 2-hafta TA, 8-muammo]

Yechish. $\mathbf{a} \times \mathbf{b} = \begin{bmatrix} (-1) \cdot 1 - 3 \cdot 4 \\ 3 \cdot 0 - 2 \cdot 1 \\ 2 \cdot 4 - (-1) \cdot 0 \end{bmatrix} = \begin{bmatrix} -13 \\ -2 \\ 8 \end{bmatrix}$.

8. $A = (0, 0, 0)$, $B = (2, 1, 0)$, $C = (1, 0, 3)$: uchburchak yuzasini toping.

[Manba: LA 2-hafta TA, 9-muammo]

Yechish. $\mathbf{AB} = (2, 1, 0)^T$, $\mathbf{AC} = (1, 0, 3)^T$; ko'paytma $= \begin{bmatrix} 1 \cdot 3 - 0 \cdot 0 \\ 0 \cdot 1 - 2 \cdot 3 \\ 2 \cdot 0 - 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \\ -1 \end{bmatrix}$, norma $= \sqrt{46}$; yuza $= \frac{\sqrt{46}}{2}$.

9. $M = \begin{bmatrix} 1 & 0 & -2 \\ 4 & 3 & 5 \end{bmatrix}$ uchun: o'lcham, m_{21} , barcha ustunlar.

[Manba: LA 2-hafta TA, 10-muammo]

Yechish. 2×3 ; $m_{21} = 4$; ustunlar $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$.

10. $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 2 & -2 \end{bmatrix}$ berilgan. $3A + 2B$ ni toping.

[Manba: LA 2-hafta TA, 11-muammo]

Yechish. $3A = \begin{bmatrix} 6 & 3 \\ -3 & 9 \end{bmatrix}$, $2B = \begin{bmatrix} 0 & 8 \\ 4 & -4 \end{bmatrix}$; yig'indi $= \begin{bmatrix} 6 & 11 \\ 1 & 5 \end{bmatrix}$.

11. $A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ berilgan. $A\mathbf{x}$ ni ustunlar birikmasi sifatida hisoblang.

[Manba: LA 2-hafta TA, 12-muammo]

Yechish. $1 \begin{bmatrix} 2 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 4 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 + 2 + 9 \\ 0 - 8 + 3 \end{bmatrix} = \begin{bmatrix} 13 \\ -5 \end{bmatrix}$.

12. $\mathbf{a}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ berilgan. $[\mathbf{a}_1 \ \mathbf{a}_2]\mathbf{x} = \mathbf{b}$ bo'ladigan barcha $\mathbf{x}$ ni toping.

[Manba: LA 2-hafta TA, 13-muammo]

Yechish. Ikkala tenglama $x_1 + 2x_2 = 3$ ga qisqaradi (sistema mos, lekin ortiqcha, chunki $\mathbf{a}_2 = 2\mathbf{a}_1$ va $\mathbf{b} = 3\mathbf{a}_1$). Umumiy yechim: $\mathbf{x} = \begin{bmatrix} 3 - 2t \\ t \end{bmatrix}$, $t \in \mathbb{R}$ — bir xil birikma uchun cheksiz ko'p vaznlar.

13. Koordinatalarni yoymasdan, $(\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = 0$ ekanini tushuntiring.

[Manba: LA 2-hafta TA, 14-muammo]

Yechish. Vektor ko'paytma ikkala kirishiga ortogonal, demak ularning har bir birikmasiga ham ortogonal: $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$ va $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0$; qo'shsak, $\mathbf{u} + \mathbf{v}$ bilan skalyar ko'paytma $0 + 0 = 0$.

14. $A 2 \times 2$, bunda $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. A va $A \begin{bmatrix} 5 \\ -2 \end{bmatrix}$ ni toping.

[Manba: LA 2-hafta TA, 15-muammo]

Yechish. Standart baza vektorlarining tasvirlari ustunlardir: $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$. U holda $A \begin{bmatrix} 5 \\ -2 \end{bmatrix} = 5 \begin{bmatrix} 2 \\ -1 \end{bmatrix} - 2 \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -13 \end{bmatrix}$; chiziqlik hisobni ikkita masshtablashga aylantiradi va skalyar ko'paytmalar kerak bo'lmaydi.

4 3-hafta — Chegaralar (Math I); Matritsalar algebrasi, determinantlar, aralash ko'paytma (Chiziqli algebra I)

4.1 Math I: 3-hafta formulalar katalogi

4.1.1 O'zgarish tezliklari va ikki asoschi muammo

Chegaralar kursga ikki savol orqali kirib keladi: kattalik qanchalik tez o'zgarayotgani (urinma/tezzlik) va funksiya qiymatlari nuqta yaqinida qanday xulq atayotgani. Qisqarayotgan oraldagi o'rtacha tezlik oniy tezlikni aniqlaydi.

$$m_{\text{sec}} = \frac{f(x) - f(a)}{x - a} \quad ((a, f(a)) \text{ dan o'tuvchi sekantning qiyaligi}), \quad (51)$$

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a} \quad (a, t] \text{ oraldagi o'rtacha tezzlik}, \quad (52)$$

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = f'(a) \quad (\text{oniy tezlik; limit sifatida aniqlanadi}) \quad (53)$$

4.1.2 Limit tushunchasi, bir tomonlama va cheksiz chegaralar

Intuativ ta'rif (OpenStax §2.2) va uning bir tomonlama variantlari haftaning amaliy yuragi; ikki tomonlama limit aynan ikkala bir tomonlama limit mos kelganda mavjud.

$$\lim_{x \rightarrow a} f(x) = L \iff \text{barcha } f(x) \text{ qiymatlar } (x \neq a) \text{ } x \rightarrow a \text{ da } L \text{ ga intiladi}, \quad (54)$$

$$\lim_{x \rightarrow a^-} f(x) = L, \quad \lim_{x \rightarrow a^+} f(x) = L \quad (\text{chap/o'ng chegaralar}), \quad (55)$$

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L \quad (\text{mavjudlik mezon}), \quad (56)$$

$$\lim_{x \rightarrow a^-} f(x) = \pm\infty, \quad \lim_{x \rightarrow a^+} f(x) = \pm\infty, \quad \lim_{x \rightarrow a} f(x) = \pm\infty \quad (\text{cheksiz chegaralar}), \quad (57)$$

$$\lim_{x \rightarrow a^-} \frac{1}{(x-a)^n} = -\infty, \quad \lim_{x \rightarrow a^+} \frac{1}{(x-a)^n} = +\infty \quad (n \text{ juft}); \quad (58)$$

$$\lim_{x \rightarrow a^-} \frac{1}{(x-a)^n} = -\infty, \quad \lim_{x \rightarrow a^+} \frac{1}{(x-a)^n} = +\infty \quad (n \text{ toq}) : x = a \text{ vertikal asimptota}. \quad (59)$$

$x = a$ to'g'ri f ning vertikal asimptotasi, agar (57) dagi bir tomonlama cheksiz chegiralarning kamida bittasi o'sha yerda bajarilsa.

4.1.3 Aniq ta'rif (§2.5, ma'ruzada ta'kidlangan)

ε - δ ta'rifi "ixtiyoriy darajada yaqin" ni miqdorlashtirilgan masofalarga aylantiradi; ekvivalent modul tengsizliklar — ish vositasi.

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \varepsilon > 0 \exists \delta > 0 : 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon, \quad (60)$$

$$|f(x) - L| < \varepsilon \iff L - \varepsilon < f(x) < L + \varepsilon, \quad (61)$$

$$0 < |x - a| < \delta \iff a - \delta < x < a + \delta \text{ va } x \neq a. \quad (62)$$

4.1.4 Bu haftada ko'rilgan ikki standart chegara

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (\text{ma'ruza 2.2-jadvali: qiymatlar } 0.9983, 0.99998, 0.9999998 \dots), \quad (63)$$

$$\lim_{x \rightarrow 0} \sin \frac{1}{x} \text{ mavjud emas (tebranish),} \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty, \quad \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty. \quad (64)$$

4.2 Math I: 3-hafta ISA — savollar va to'liq yechimlar

1. Ma'ruzada va Amaliy mashg'ulotda keltirilgan barcha muammolar, mashqlar va misollarni o'rganing; tez va xatosiz javob berishga tayyor bo'ling.

[Manba: 3-hafta ISA, 2-topshiriq]

Yechish. Bu hisoblash emas, o'rganish ko'rsatmasi; tegishli material ma'ruza misollarida, quyidagi Amaliy mashg'ulot blokida (2-8 mashqlar) va tayinlangan darslik masalalarida (keyingi bandlar) to'liq yechilgan.

2. $f(x) = x^3$ uchun $x = 1$ dagi urinma qiyaligini $m_{\text{sec}} = \frac{f(1+h)-f(1)}{h}$ orqali $h = -0.1, -0.01, -0.001, 0.001, 0.01, 0.1$ qiymatlarida baholang. Chapdan va o'ngdan olingan baholar bir xil songa intiladimi?

[Manba: 3-hafta ISA, 3-topshiriq]

Yechish. Bu yerda $f(1) = 1$ va $f(1+h) = (1+h)^3 = 1 + 3h + 3h^2 + h^3$, demak $m_{\text{sec}} = 3 + 3h + h^2$. Qiymatlar: $h = -0.1 \Rightarrow 2.71$; $-0.01 \Rightarrow 2.9701$; $-0.001 \Rightarrow 2.997$; $0.001 \Rightarrow 3.003$; $0.01 \Rightarrow 3.0301$; $0.1 \Rightarrow 3.31$. Ikkala tomon ham 3 ga intiladi; haqiqatdan $\lim_{h \rightarrow 0} (3 + 3h + h^2) = 3$ — aniq urinma qiyaligi ($x = 1$ da $3x^2$ hosila).

3. OpenStax Calculus 1 ning "1-bob" (2.2-bo'lim mashqlar ro'yxati) dagi 46-49 va 60-64 muammolarini yeching.

[Manba: 3-hafta ISA, 4-topshiriq]

Yechish. Bu grafik asosidagi mashqlar quyida §4.4 da to'liq yozilgan va yechilgan; kurs varaqasidagi raqamlash o'sha mashqlar ro'yxatiga tegishli.

4. Harorat regulyatori $T < 50$ uchun $C(T) = 0$ va $T \geq 50$ uchun $C(T) = 10$ (volt) beradi. $\lim_{T \rightarrow 50^-} C(T)$, $\lim_{T \rightarrow 50^+} C(T)$, $\lim_{T \rightarrow 50} C(T)$ ni aniqlang; kalitlash amali sifatida izohlang.

[Manba: 3-hafta ISA, 5-topshiriq]

Yechish. Chapdan regulyator "o'chirilgan" shoxni o'qiydi: $\lim_{T \rightarrow 50^-} C(T) = 0$. O'ngdan "yoqilgan" shoxni o'qiydi: $\lim_{T \rightarrow 50^+} C(T) = 10$. Ikkala bir tomonlama limit farq qilgani uchun ikki tomonlama limit mavjud emas (56). Izoh: $T = 50$ da keskin kalitlanish — chiqish 0 V dan 10 V ga sakraydi, termostat isitgichni aynan shunday boshqaradi.

5. Robot aravachasining pozitsiyasi: $s(t) = 2t^2 + 3t$ (m, s). (1) $h = 1, 0.1, 0.01, -0.01$ uchun o'rtacha tezlik $\frac{s(2+h)-s(2)}{h}$; (2) ifodani algebraik soddalashtiring; (3) $\lim_{h \rightarrow 0} \frac{s(2+h)-s(2)}{h}$ ni hisoblang; (4) izohlang.

[Manba: 3-hafta ISA, 6-topshiriq]

Yechish. (1) $s(2) = 14$. Qiymatlar: $h = 1$: $\frac{s(3)-14}{1} = \frac{27-14}{1} = 13$; $h = 0.1$: $\frac{14.72-14}{0.1} = 7.2$; $h = 0.01$: $\frac{14.0702-14}{0.01} = 7.02$; $h = -0.01$: $\frac{13.9302-14}{-0.01} = 6.98$. (2) $s(2+h) = 2(2+h)^2 + 3(2+h) = 8 + 8h + 2h^2 + 6 + 3h = 14 + 11h + 2h^2$, demak $\frac{s(2+h)-s(2)}{h} = \frac{11h+2h^2}{h} = 11 + 2h$. (3) $\lim_{h \rightarrow 0} (11 + 2h) = 11$ m/s. (4) $t = 2$ sdagi oniy tezlik 11 m/s; o'rtachalar ikkala tomondan unga intiladi.

6. Ikkala sistema chiqishlari $A(t) = (1.08)^t$ va $B(t) = e^{0.03t}$. (1) $\lim_{t \rightarrow \infty} A(t)$ va $\lim_{t \rightarrow \infty} B(t)$ ni hisoblang; (2) $A(t)$ ni e^{kt} ko'rinishiga keltiring; (3) qaysi biri tezroq o'sadi? (4) $\lim_{t \rightarrow \infty} \frac{B(t)}{A(t)}$ ni hisoblang; (5) izohlang.

[Manba: 3-hafta ISA, 7-topshiriq]

Yechish. (1) Ikkala asos ham 1 dan katta, demak ikkala limit ham $+\infty$. (2) $A(t) = e^{t \ln 1.08}$, $\ln 1.08 \approx 0.0770$, demak $A(t) = e^{0.0770t}$. (3) A tezroq o'sadi, chunki uning tezlik konstantasi $0.0770 > 0.03$. (4) $\frac{B(t)}{A(t)} = e^{(0.03-0.0770)t} = e^{-0.047t} \rightarrow 0$, $t \rightarrow \infty$ da. (5) Ikkalasi ham cheksizlikka intilsa-da, nisbat nolga intiladi: B A yonida ahamiyatsiz bo'lib qoladi — 4-haftaning “kichik-o” taqqoslash urug'i.

4.3 Math I: 3-hafta amaliy mashg'uloti (muhandislik chegaralari) — savollar va to'liq yechimlar

Bu sessiya uchun kurs kelishuvlari: matematik protsedurani ko'rsatish; birliklarni kiritish; muhandislik atamalarida izohlash; limit mavjud bo'lmagan holatni ko'rsatish.

1. Darslikning 2.2-bo'lim mashqlar ro'yxatidagi 50-54 muammolarni yeching.

[Manba: 3-hafta amaliy mashg'uloti, 1-mashq]

Yechish. Quyida §4.4 da yozilgan va to'liq yechilgan (50-54 bandlar).

2. Bir tomonlama chegaralar: $x < 2$ uchun $f(x) = 2x + 1$, $x \geq 2$ uchun $f(x) = 7 - x$. $\lim_{x \rightarrow 2^-} f$, $\lim_{x \rightarrow 2^+} f$, $\lim_{x \rightarrow 2} f$, $f(2)$ ni toping; ikki tomonlama limit mavjudmi, tushuntiring.

[Manba: 3-hafta amaliy mashg'uloti, 2-mashq]

Yechish. Chap shox: $\lim_{x \rightarrow 2^-} (2x + 1) = 5$. O'ng shox: $\lim_{x \rightarrow 2^+} (7 - x) = 5$. Teng, demak (56) bo'yicha $\lim_{x \rightarrow 2} f(x) = 5$. $f(2) = 7 - 2 = 5$ ham — funksiya qiymati limit bilan mos, demak grafikda $x = 2$ da uzilish yo'q.

3. O'lchov yo'qligi yaqinida kalibrlash: $V(T) = \frac{T^2 - 2500}{T - 50}$ mV, $T \neq 50$ (T °C da). (1) $V(50)$ nega aniqlanmagan? (2) $\lim_{T \rightarrow 50} V(T)$ ni baholang; (3) limit yetishmayotgan kalibrlash qiymatini qanday almashtiradi?

[Manba: 3-hafta amaliy mashg'uloti, 3-mashq]

Yechish. (1) Formula $T - 50$ ga bo'ladi, u esa $T = 50$ da 0; sensor modeli bu yerda shunchaki qiymatga ega emas. (2) Kvadratlar ayirmasini ko'paytuvchilarga ajratamiz: $T^2 - 2500 = (T - 50)(T + 50)$, demak $T \neq 50$ uchun $V(T) = T + 50$ va $\lim_{T \rightarrow 50} V(T) = 100$ mV. (3) Limit sensor 50°C da qanday o'qishini tasvirlaydi; kalibrlash jadvalini o'sha nuqtada 100 mV bilan to'ldirish asoslanadi, chunki chiqish ikkala tomondan shu qiymatga intiladi — bu “olib tashlanadigan” xulq, sakrash emas.

4. Kalitlangan boshqaruv sistemasi: $t < 4$ uchun $V(t) = 2t + 1$ va $t \geq 4$ uchun $V(t) = 12 - t$ (V). Bir tomonlama chegaralar, ikki tomonlama limit va $V(4)$ ni aniqlang; $t = 4$ da keskin kuchlanish o'zgarishi bormi?

[Manba: 3-hafta amaliy mashg'uloti, 4-mashq]

Yechish. Chap: $\lim_{t \rightarrow 4^-} (2t + 1) = 9$ V. O'ng: $\lim_{t \rightarrow 4^+} (12 - t) = 8$ V. $V(4) = 12 - 4 = 8$ V. Bir tomonlama chegaralar farq qiladi, demak $\lim_{t \rightarrow 4} V(t)$ mavjud emas: $t = 4$ da quvvat 9 V dan 8 V ga keskin tushadi. Muhandislik ma'nosi: 1 V bosqich boshqariladigan qurilmada o'tkazinchali jarayonlarni qo'zg'atishi mumkin; bosqich zararli bo'lsa, kirish filtrlari yoki yumshoq ishga tushirish mantiqi ishlatiladi.

5. Kuchlanish-deformatsiya: $\varepsilon < 0.002$ uchun $\sigma(\varepsilon) = 200000\varepsilon$ va $\varepsilon \geq 0.002$ uchun $\sigma(\varepsilon) = 400 + 10000(\varepsilon - 0.002)$ (σ MPa da; Hook sohasi, so'ng mustahkamlanish). $\varepsilon = 0.002$ dagi bir tomonlama chegaralarni toping; ikki tomonlama limit mavjudmi? Model sakrashni bashorat qiladimi? O'tish yaqinida grafikni eskizing.

[Manba: 3-hafta amaliy mashg'uloti, 5-mashq]

Yechish. Elastik tomon: $\lim_{\varepsilon \rightarrow 0.002^-} 200000\varepsilon = 200000(0.002) = 400$ MPa. Plastik tomon: $\lim_{\varepsilon \rightarrow 0.002^+} (400 + 10000(\varepsilon - 0.002)) = 400$ MPa. Ikkalasi 400 MPa ga teng, demak ikki tomonlama limit mavjud va 400 MPa ga teng; sakrash yo'q — bo'lakli model hosil chegarasida uzluksiz, faqat qiyaligini o'zgartiradi (200 000 dan 10 000 MPa gacha birlik deformatsiyaga). Grafik (0.002, 400) da uchrashgan ikki to'g'ri bo'lak, aniq burilish bilan: avval elastik yuklash, keyin aniq tekisroq plastik shox.

6. Uzoq muddatli RC javob: $V_s = 12$ V, $R = 10000 \Omega$, $C = 100 \mu\text{F}$ bilan $V(t) = V_s(1 - e^{-t/(RC)})$. (1) $\lim_{t \rightarrow 0^+} V(t)$ ni hisoblang; (2) $\lim_{t \rightarrow \infty} V(t)$; (3) ikkalasini fizik izohlang.

[Manba: 3-hafta amaliy mashg'uloti, 6-mashq]

Yechish. (1) $t = 0^+$ da eksponensial $e^0 = 1$, demak $V(0^+) = 12(1 - 1) = 0$ V — boshlang'ich zaryadsiz kondensator. (2) $t \rightarrow \infty$ da $e^{-t/(RC)} \rightarrow 0$, demak $V \rightarrow V_s = 12$ V. (3) Kondensator 0 V dan quvvat manbai 12 V tomon zaryadlanadi va unga asimptotik yaqinlashadi; vaqt konstantasi $\tau = RC = 10^4 \cdot 10^{-4} = 1$ s tempni belgilaydi (1 s dan keyin taxminan 63%, 5τ dan keyin 99% zaryad). " $t \rightarrow \infty$ da $V \rightarrow V_s$ " limit tili asimptotik zaryadlanishni aynan shunday bayon qiladi.

7. Terminal tezzlik: $v_T > 0$, $k > 0$ bilan $v(t) = v_T(1 - e^{-kt})$. (1) $v(0)$? (2) $\lim_{t \rightarrow \infty} v(t)$? (3) Bu model cheksiz tezlanishdan nega realistikroq?

[Manba: 3-hafta amaliy mashg'uloti, 7-mashq]

Yechish. (1) $v(0) = v_T(1 - 1) = 0$: jism tinch holatdan boshlaydi. (2) $e^{-kt} \rightarrow 0$, demak $v \rightarrow v_T$: tezzlik terminal tezzlikka yaqinlashadi. (3) Havo qarshiligi tezzlik bilan o'sib, gravitatsiyani muvozanatlaydi; shundan keyin net kuch nol va tezzlik endi o'smaydi — tushayotgan jism cheksiz tezlanish o'rniga chekli limitga haqiqatdan ham yaqinlashadi.

8. Ma'lumot uzatish tezligi: $R(n) = \frac{100n}{n+20}$ Mbit/s. (1) $R(10)$, $R(100)$, $R(1000)$ ni hisoblang; (2) $\lim_{n \rightarrow \infty} R(n)$; (3) n o'sishi nega oxir-oqibat kamayuvchi yaxshilanishlar beradi? (4) maksimal nazariy tezlik?

[Manba: 3-hafta amaliy mashg'uloti, 8-mashq]

Yechish. (1) $R(10) \approx 33.3$, $R(100) \approx 83.3$, $R(1000) \approx 98.0$ Mbit/s. (2) n ga bo'lib: $R(n) = \frac{100}{1+20/n} \rightarrow 100$ Mbit/s. (3) Resursning har bir qo'shimcha birligi tobora kattaroq bazaga bo'linadi, demak chegaraviy foyda kamayadi — kamayuvchi rentabellik. (4) Gorizontaal asimptota 100 Mbit/s — modelning shipi, unga yaqinlashiladi, lekin hech qachon oshib ketilmaydi.

4.4 Math I: tayinlangan darslik mashqlari (OpenStax §2.2, 46-54 va 60-64 bandlar)

Bu grafik o'qish mashqlari mashqlar to'plami bilan chop etilgan to'rtta rasmga tayanadi; kerakli qiymatlar grafiklardan o'qiladi, darsda qilinganidek. Avval mashq bayonlari, so'ng yechimlar.

1. $y = f(x)$ ning birinchi grafigi berilgan, hal qiling: (46) $\lim_{x \rightarrow 10^-} f(x) = 0$; (47) $\lim_{x \rightarrow -2^+} f(x) = 3$; (48) $\lim_{x \rightarrow -8} f(x) = f(-8)$; (49) $\lim_{x \rightarrow 6} f(x) = 5$ — to'g'rimi yoki noto'g'rimi?

[Manba: OpenStax §2.2, 46-49 mashqlar]

Yechish. (46) To'g'ri: x chapdan 10 ga yaqinlashganda egri chiziqli to'ldirilgan $(10, 0)$ nuqtasiga tushadi, demak chap limit 0. (47) To'g'ri: -2 dan ozgina o'ngda grafik 3 balandlikda turadi $(-2, 3)$ dagi to'ldirilgan nuqta. (48) Noto'g'ri: $x = -8$ yaqinida egri chiziqli -6 balandlikdagi ochiq doiraga intiladi, demak $\lim_{x \rightarrow -8} f(x) = -6$, holbuki $f(-8) = -3$ (to'ldirilgan nuqta); limit va qiymat mos kelmasligi mumkin. (49) Noto'g'ri: chapdan egri chiziqli $(6, 2)$ dagi ochiq doiraga tushadi, demak $\lim_{x \rightarrow 6^-} = 2$, holbuki o'ng shox to'ldirilgan $(6, 5)$ nuqtasidan chiqadi, demak $\lim_{x \rightarrow 6^+} = 5$; bir tomonlama chegaralar farq qiladi $\Rightarrow$ ikki tomonlama limit mavjud emas, demak u 5 ga teng bo'lolmaydi.

2. Bo'lakli-chiziqli grafik uchun ($y = x + 1$ to'g'ri ochiq $(1, 2)$ nuqttagacha; to'ldirilgan $(1, 1)$ nuqta; keyin $y = 3 - x$ to'g'ri), toping: (50) $\lim_{x \rightarrow 1^-} f$; (51) $\lim_{x \rightarrow 1^+} f$; (52) $\lim_{x \rightarrow 1} f$; (53) $\lim_{x \rightarrow 2} f$; (54) $f(1)$.

[Manba: OpenStax §2.2, 50-54 mashqlar]

Yechish. (50) Chap shox $y = x + 1 \rightarrow 2$. (51) O'ng shox $y = 3 - x \rightarrow 2$. (52) Ikkala tomon ham 2 ga teng, demak $\lim_{x \rightarrow 1} f(x) = 2 - f(1) = 1$ bo'lsa ham; limit nuqtaning o'zini hisobga olmaydi. (53) $x = 2$ da faqat o'ng shox yashaydi: $3 - 2 = 1$. (54) $f(1) = 1$ (to'ldirilgan nuqta).

3. Uchinchi grafik uchun (chap: ochiq $(0, 1)$ nuqtaga tushayotgan to'g'ri; o'ng: to'ldirilgan $(0, -4)$ nuqtadan ko'tarilib, $x = 2$ yaqinida noldan o'tuvchi egri chiziqli), toping: (55) $\lim_{x \rightarrow 0^-} f$; (56) $\lim_{x \rightarrow 0^+} f$; (57) $\lim_{x \rightarrow 0} f$; (58) $\lim_{x \rightarrow 2} f$.

[Manba: OpenStax §2.2, 55-58 mashqlar]

Yechish. (55) 1. (56) -4 . (57) $1 \neq -4$, demak ikki tomonlama limit mavjud emas (56). (58) O'ng shox $x = 2$ da o'qdan o'tadi, demak limit 0.

4. To'rtinchi grafik uchun (uch bo'lak: ochiq $(-2, 0)$ nuqtagacha ko'tarilayotgan to'g'ri; to'ldirilgan $(-2, 2)$ nuqtadan boshlanib origin orqali $(2, -2)$ ga tushayotgan kesma; ochiq $(2, 2)$ nuqtadan ko'tarilayotgan to'g'ri), toping: (59) $\lim_{x \rightarrow -2^-} f$; (60) $\lim_{x \rightarrow -2^+} f$; (61) $\lim_{x \rightarrow -2} f$; (62) $\lim_{x \rightarrow 2^-} f$; (63) $\lim_{x \rightarrow 2^+} f$; (64) $\lim_{x \rightarrow 2} f$.

[Manba: OpenStax §2.2, 59-64 mashqlar]

Yechish. (59) 0. (60) 2. (61) $0 \neq 2$: mavjud emas. (62) -2 . (63) 2. (64) $-2 \neq 2$: mavjud emas. Ikkala tugun ham haqiqiy sakrash: teng bo'lmagan bir tomonlama chegaralarni o'rtacha hech narsa tuzatmaydi.

4.5 Chiziqli algebra I: 3-hafta formulalar katalogi

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c a_{ij}, \quad (A^T)_{ij} = a_{ji}, \quad (A^T)^T = A, \quad (65)$$

$$(A + B)^T = A^T + B^T, \quad (cA)^T = cA^T, \quad A^T = A \text{ (simmetrik)}, \quad (66)$$

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{n \times p} \Rightarrow AB \in \mathbb{R}^{m \times p}, \quad (67)$$

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \quad (68)$$

$$A(BC) = (AB)C, \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC, \quad (69)$$

$$(AB)^T = B^T A^T, \quad I_m A = A = A I_n, \quad (70)$$

$$AB = [A\mathbf{b}_1 \cdots A\mathbf{b}_p] \text{ (ustunlar qoidasi)}, \quad (71)$$

$$\mathbf{u}^T \mathbf{v} = \text{skalyar}, \quad \mathbf{u} \mathbf{v}^T = \text{matritsa (tashqi ko'paytma)}. \quad (72)$$

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc \quad (\text{ishorali yuza}), \quad (73)$$

$$C_{ij} = (-1)^{i+j} \det M_{ij} \text{ (kofaktor)}, \quad \det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}, \quad (74)$$

bunda M_{ij} — minori (i -qator va j -ustunni o'chirilgan). Har qanday qator yoki ustun bo'ylab yoyish mumkin; nollari bor qator yoki ustunlar hisobni kamaytiradi. Uch-burchak matritsalarida determinant diagonal elementlar ko'paytmasi. Boshlang'ich qator amallari qoidalari 1-jadvalda jamlangan.

Table 1: Kvadrat matritsalar uchun determinant qoidalari (qator versiyalari; ustun versiyalari ham o'rinli).

A ustidagi amal	$\det A$ ga ta'siri
Ikkala qatorni almashtirish	ishorasini almashtiradi
Bitta qatorni c ga ko'paytirish	determinantni c ga ko'paytiradi
Bitta qatorning ko'paytmasini boshqasiga qo'shish	o'zgarish yo'q
Nol qator yoki ikki proporsional qator	determinant 0
Transponirlash	$\det(A^T) = \det A$
Bir xil o'lchamli kvadrat matritsalar ko'paytmasi	$\det(AB) = \det A \det B$

Butun $n \times n$ matritsani masshtablash n ta qatorning hammasini masshtablaydi: $\det(cA) = c^n \det A$. Determinantlar qo'shishga taqsimlanmaydi: $A = B = I_2$ uchun $\det(A + B) = 4$, holbuki $\det A + \det B = 2$.

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}] \quad (\text{A.K}), \quad (75)$$

$$V_{\text{parallelepiped}} = |(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|, \quad h = \frac{|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|}{\|\mathbf{u} \times \mathbf{v}\|}, \quad (76)$$

$$V_{\text{tetrahedron}} = \frac{1}{6} |\det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]|, \quad T = 0 \iff \mathbf{u}, \mathbf{v}, \mathbf{w} \text{ komplanar}. \quad (77)$$

4.6 Chiziqli algebra I, 3-hafta ma'ruza muammolari (15 ta) — savollar va to'liq yechimlar

1. $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$ berilgan. $2A - B$ ni hisoblang va $(2, 1)$ elementini aniqlang; A ni 2×3 matritsaga qo'shish mumkinmi?

[Manba: LA 3-hafta ma'ruza, 1-muammo]

Yechish. $2A = \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$; $2A - B = \begin{bmatrix} 0 & -5 \\ 7 & -4 \end{bmatrix}$; $(2, 1)$ element $2(3) - (-1) = 7$. 2×3 matritsaga qo'shish aniqlanmagan — ikkala o'lcham ham mos bo'lishi shart, faqat elementlar soni emas.

2. $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$ berilgan. C^T ni o'lchami bilan yozing; CC^T ni hisoblang; u simmetrikmi?

[Manba: LA 3-hafta ma'ruza, 2-muammo]

Yechish. $C^T = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ -1 & 2 \end{bmatrix}$ (3×2). $CC^T = \begin{bmatrix} 1+4+1 & 0+6-2 \\ 0+6-2 & 0+9+4 \end{bmatrix} = \begin{bmatrix} 6 & 4 \\ 4 & 13 \end{bmatrix}$. Diagonal elementlar qatorlarning o'zlari bilan skalyar ko'paytmasi; diagonaldan tashqarisi — qatorlar skalyar ko'paytmalari. Har doim simmetrik, chunki $(CC^T)^T = (C^T)^T C^T = CC^T$.

3. C va $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ uchun $C\mathbf{x}$ ni ikkala usul bilan hisoblang.

[Manba: LA 3-hafta ma'ruza, 3-muammo]

Yechish. Qatorlar: $1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 1 = -1$; $0 + 3(-1) + 2 = -1$; demak $C\mathbf{x} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$.

Ustunlar: $2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. $(2 \times 3)(3 \times 1)$ o'lchamlar 2×1 chiqish beradi.

4. 1-muammodagi A, B bilan AB va BA ni hisoblang. Tengmi?

[Manba: LA 3-hafta ma'ruza, 4-muammo]

Yechish. $AB = \begin{bmatrix} 1 \cdot 2 + (-2)(-1) & 1 \cdot 1 + (-2) \cdot 4 \\ 3 \cdot 2 + 0 & 3 \cdot 1 + 0 \cdot 4 \end{bmatrix} = \begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}$; $BA = \begin{bmatrix} 5 & -4 \\ 11 & 2 \end{bmatrix}$. Har xil; ikkala tartib mavjud bo'lsa ham, tenglik kafolatlanmaydi.

5. $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 1 & -1 \\ 3 & 2 \end{bmatrix}$ berilgan. CD va DC o'lchamlarini bashoratlang, so'ng ikkalasini hisoblang.

[Manba: LA 3-hafta ma'ruza, 5-muammo]

Yechish. CD : $(2 \times 3)(3 \times 2) = 2 \times 2$; DC : $(3 \times 2)(2 \times 3) = 3 \times 3$.

$$CD = \begin{bmatrix} 2+2-3 & 0-2-2 \\ 0+3+6 & 0-3+4 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 9 & 1 \end{bmatrix}, \quad DC = \begin{bmatrix} 2 & 4 & 0 \\ 1 & -1 & -3 \\ 3 & 12 & 1 \end{bmatrix}.$$

Ko'paytmalar o'lchamlari har xil va hech qachon taqqoslanmaydi.

6. $S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ (silliyish), $H = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ (cho'zish), $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ berilgan. $HS\mathbf{x}$ va $SH\mathbf{x}$ ni hisoblang; har bir transformatsiyani tavsiflang.

[Manba: LA 3-hafta ma'ruza, 6-muammo]

Yechish. $S\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, u holda $HS\mathbf{x} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$; shuningdek $H\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, u holda $SH\mathbf{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. HS da silliyish hissasi keyin cho'ziladi; SH da cho'zish avval sodir bo'ladi va silliyish ustiga qo'shiladi. Bir xil ikkita amal, boshqacha tartib, boshqacha natija: matritsalar almashinuvchi emas.

7. $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$ uchun: $\mathbf{u}^T\mathbf{v}$ va $\mathbf{u}\mathbf{v}^T$ ni hisoblang; $(\mathbf{u}\mathbf{v}^T)^T = \mathbf{v}\mathbf{u}^T$ ni tekshiring.

[Manba: LA 3-hafta ma'ruza, 7-muammo]

Yechish. Ichki: $\mathbf{u}^T\mathbf{v} = 2 + 0 - 3 = -1$ (skalyar). Tashqi: $\mathbf{u}\mathbf{v}^T = \begin{bmatrix} 2 & 0 & 3 \\ 4 & 0 & 6 \\ -2 & 0 & -3 \end{bmatrix}$ (3×3). Elementma-element transponirlab, $(\mathbf{u}\mathbf{v}^T)_{ij}^T = u_j v_i = v_i u_j = (\mathbf{v}\mathbf{u}^T)_{ij}$, demak $(\mathbf{u}\mathbf{v}^T)^T = \mathbf{v}\mathbf{u}^T = \begin{bmatrix} 2 & 4 & -2 \\ 0 & 0 & 0 \\ 3 & 6 & -3 \end{bmatrix}$: yozilgan matritsalar bir-birining transponiridir.

8. 1-muammodagi A, B dan foydalanib, $(AB)^T$ va $B^T A^T$ ni alohida hisoblang; $A^T B^T$ mos keladimi?

[Manba: LA 3-hafta ma'ruza, 8-muammo]

Yechish. $AB = \begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}$, demak $(AB)^T = \begin{bmatrix} 4 & 6 \\ -7 & 3 \end{bmatrix}$. Avval transponirlar: $A^T = \begin{bmatrix} 1 & 3 \\ -2 & 0 \end{bmatrix}$, $B^T = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}$; u holda $B^T A^T = \begin{bmatrix} 2 \cdot 1 + (-1)(-2) & 2 \cdot 3 + 0 \\ 1 \cdot 1 + 4(-2) & 1 \cdot 3 + 0 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ -7 & 3 \end{bmatrix} = (AB)^T$, (70) tasdiqlanadi. Lekin $A^T B^T = \begin{bmatrix} -1 & 3 \\ -2 & -6 \end{bmatrix}$ farq qiladi: to'g'ri ayniyat koeffitsientlar tartibini teskari aylantiradi.

9. $E = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ berilgan. $\det E$ ni hisoblang; ustunlarning parallelogram va uchburchak yuzalarini toping; yo'nalish saqlanadimi?

[Manba: LA 3-hafta ma'ruza, 9-muammo]

Yechish. $\det E = 3 \cdot 2 - 1 \cdot 1 = 5$. Parallelogram yuzasi $= |5| = 5$; uchburchak $= \frac{5}{2}$. Determinant musbat: yo'nalish saqlanadi (yuza o'zi har doim nomanfiy).

10. $F = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ uchun $\det F$ ni birinchi qator kofaktorlari orqali hisoblang; birinchi ustun bo'ylab tekshiring.

[Manba: LA 3-hafta ma'ruza, 10-muammo]

Yechish. Birinchi qator: $\det F = 2 \det \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} - 1 \det \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} + 0 = 2(6) - 1(-1) = 13$.

Birinchi ustun: $2 \det \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} + 1 \det \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = 12 + 1 = 13$ ((3,1) kofaktor ishorasi $(-1)^{3+1} = +$). Har qanday qator yoki ustun bir xil qiymat beradi.

11. 3×3 matritsa M uchun $\det M = 5$. N ni shakllantiring: 1,2-qatorlarni almashtirib; 3-qatorni 2 ga ko'paytiring; 2-qatorga $3 \times$ (yangi 1-qator) qo'shing. $\det N$, $\det(2M)$, $\det(M^T M)$ ni toping.

[Manba: LA 3-hafta ma'ruza, 11-muammo]

Yechish. Almashtirish: ishora aylanadi (-5) ; 3-qator masshtabi: yana 2 ko'effitsient; qator qo'shish: o'zgarish yo'q. $\det N = -2 \cdot 5 = -10$. Barcha qatorlarni masshtablash: $\det(2M) = 2^3 \cdot 5 = 40$. Ko'paytma qoidasi: $\det(M^T M) = \det M^T \det M = 25$.

12. $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ berilgan. $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ ni, parallelepiped va tetraedr hajmlarini hisoblang; $\mathbf{u}, \mathbf{v}$ almashtirilsa nima o'zgaradi?

[Manba: LA 3-hafta ma'ruza, 12-muammo]

Yechish. $\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 0 \cdot 0 - 1 \cdot 2 \\ 1 \cdot 1 - 2 \cdot 0 \\ 2 \cdot 2 - 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}$; $\mathbf{w}$ bilan skalyar ko'paytma: $-2 + 1 + 12 =$

13. Parallelepiped hajmi 13; tetraedr $\frac{13}{6}$. $\mathbf{u}$ va $\mathbf{v}$ ni almashtirish ishorani -13 ga aylantiradi, lekin ikkala hajm ham 13 va $\frac{13}{6}$ qoladi (modul qiymatlar).

13. $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ k \end{bmatrix}$ uchun: vektorlarni bir tekislikda joylashtiradigan

barcha k ni toping; o'sha k da $\mathbf{w}$ ni $\mathbf{u}, \mathbf{v}$ orqali ifodalang.

[Manba: LA 3-hafta ma'ruza, 13-muammo]

Yechish. Bir tekislikda bo'lish $\iff T = 0$. Bu yerda $\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 0 \cdot 1 - 1 \cdot 2 \\ 1 \cdot 0 - 1 \cdot 1 \\ 1 \cdot 2 - 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$,

demak $T = -2 - 1 + 2k = 2k - 3 = 0 \Rightarrow k = \frac{3}{2}$. U holda $\mathbf{w} = \alpha \mathbf{u} + \beta \mathbf{v}$: birinchi ikki koordinata $\alpha = 1$, $2\beta = 1 \Rightarrow \beta = \frac{1}{2}$ beradi; uchinchisini tekshiramiz: $1 + \frac{1}{2} = \frac{3}{2}$ — mos, $\mathbf{w} = \mathbf{u} + \frac{1}{2}\mathbf{v}$.

14. $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ berilgan. $DX = XD$ bo'ladigan barcha X ni toping; D ning ikkala diagonal elementi teng bo'lsa-chi?

[Manba: LA 3-hafta ma'ruza, 14-muammo]

Yechish. $DX = \begin{bmatrix} 2a & 2b \\ 3c & 3d \end{bmatrix}$, $XD = \begin{bmatrix} 2a & 3b \\ 2c & 3d \end{bmatrix}$. Tenglik $b = c = 0$ ni, a, d erkin bo'lishini majbur qiladi: aynan diagonal matritsalar bu D bilan almashinadi. Agar $D = \lambda I$ bo'lsa, $DX = \lambda X = XD$ har bir X uchun bajariladi.

15. $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ va $L = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ berilgan. Qirralarni qayta hisoblamasdan, har bir qirraga L qo'llangandan keyingi ishorali aralash ko'paytma va hajmni toping; hajm koeffitsienti nega universal?

[Manba: LA 3-hafta ma'ruza, 15-muammo]

Yechish. $U = [\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$ deb olamiz; 12-muammodan $\det U = 13$. Transformatsiya qilingan qirralar LU hosil qiladi, demak $\det(LU) = \det L \det U = (-2)(13) = -26$; yangi hajm 26. Koeffitsient $|\det L| = 2$ qanday bo'lishidan qat'i nazar U , chunki $\det(LU) = \det L \det U$ har bir U uchun — determinantlar ko'paytma qoidasining geometrik ma'nosi bor: $|\det L|$ transformatsiyaning hajm masshtablash koeffitsienti.

5 4-hafta — Chegaralar II: Chegara qonunlari, standart chegaralar, aniqlanmagan holdlar, asimptotika, Landau belgilari (Math I)

4-hafta Test 1 takroridan oldingi tayanch hafta: 3-hafta limit mexanikasini ishlaydigan asboblarni yig'indisiga aylantiradi. Majburiy o'qish — OpenStax §2.3 (Chegara qonunlari), Prof. Neofotistosning 4-hafta Konspekti esa umumiy limit xossalari, standart chegaralar, aniqlanmagan holdlar, asimptotik ekvivalentlik va O va o Landau belgilarini umumlashtiradi. Bu qism ikkala manbadagi har bir tenglamani ro'yxatlaydi, so'ng §2.3 to'plamining (83-130 bandlar) har bir mashqini va Konspektning har bir yechilgan savolini hal qiladi.

5.1 4-hafta formulalar katalogi

5.1.1 Chegaralarning umumiy xossalari (4-hafta konspekti)

$$\lim_{x \rightarrow a} f(x) = L : f(x) \text{ ni } L \text{ ga ixtiyoriy darajada yaqinlashtirish mumkin } (x \rightarrow a, x \neq a) \quad (78)$$

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L \quad (\text{ikki tomonlama mezon}), \quad (79)$$

limit, mavjud bo'lganda, yagona.

5.1.2 Chegara qonunlari (OpenStax Teorema 2.5)

Agar $\lim_{x \rightarrow a} f(x) = L$ va $\lim_{x \rightarrow a} g(x) = M$ (ikkalasi ham mavjud) va c konstanta bo'lsa:

$$\text{Yig'indi: } \lim_{x \rightarrow a} [f(x) + g(x)] = L + M; \quad \text{Ayirma: } \lim_{x \rightarrow a} [f(x) - g(x)] = L - M; \quad (80)$$

$$\text{Konstanta ko'paytma: } \lim_{x \rightarrow a} c f(x) = cL; \quad \text{Ko'paytma: } \lim_{x \rightarrow a} [f(x)g(x)] = LM; \quad (81)$$

$$\text{Bo'linma: } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M} \quad (M \neq 0); \quad \text{Daraja: } \lim_{x \rightarrow a} [f(x)]^n = L^n \quad (n \in \mathbb{N}); \quad (82)$$

$$\text{Ildiz: } \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{L} \quad (n \text{ toq: istalgan } L; n \text{ juft: } L \geq 0); \quad (83)$$

$$\text{Asosiy: } \lim_{x \rightarrow a} x = a, \quad \lim_{x \rightarrow a} c = c. \quad (84)$$

Ko'phadlar/ratsional funksiyalar: $\lim_{x \rightarrow a} p(x) = p(a), \quad \lim_{x \rightarrow a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)} \quad (q(a) \neq 0).$ (85)

Agar a yaqinida $x \neq a$ bo'lgan barcha joyda $f(x) = g(x)$ bo'lsa, u holda $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$ — qo'yishdan oldin ko'paytuvchilarni qisqartirish litsenziyasi.

5.1.3 Siqish (Pinchlovchi) teoremasi (OpenStax Teorema 2.7)

$$f(x) \leq g(x) \leq h(x) \quad (a \text{ yaqinida, } x \neq a), \quad \lim_{x \rightarrow a} f(x) = L = \lim_{x \rightarrow a} h(x) \implies \lim_{x \rightarrow a} g(x) = L. \quad (86)$$

5.1.4 Standart chegaralar va ulardan olinadigan trigonometrik chegaralar (4-hafta konspekti)

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}, \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \quad (87)$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1, \quad \lim_{x \rightarrow 0} (1+x)^{1/x} = e, \quad \lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0 \quad (n \in \mathbb{N}). \quad (88)$$

Trigonometrik chegiralarda burchaklar radian bo'lishi shart. (87) ni almashtirish orqali moslashtiramiz, masalan

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{x} = 5 \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} = 5, \quad \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \implies \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0 \quad (\text{qo'shma son}). \quad (89)$$

5.1.5 Aniqlanmagan holdlar (4-hafta konspekti)

$$\frac{0}{0}, \frac{\infty}{\infty}, 0 \cdot \infty, \infty - \infty, 0^0, 1^\infty, \infty^0 \quad (90)$$

“Aniqlanmagan” so'zi limit mavjud emas degani emas; bu qo'shimcha tahlil kerak degani. Bir xil boshlang'ich hold har xil natijaga ega bo'lishi mumkin:

$$\lim_{x \rightarrow 0} \frac{x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{x^2}{x} = 0, \quad \lim_{x \rightarrow 0} \frac{x}{x^2} \text{ mavjud emas (chekli)}, \quad (91)$$

holbuki ko'paytuvchilarga ajratgandan keyin $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2$. Hal qilish vositalari: ko'paytuvchilarga ajratish, ratsionallashuv, standart chegaralar, asimptotik ekvivalentlik, o'zgaruvchini almashtirish, algebraik qayta tuzish, Siqish teoremasi, Taylor yoyilmalari va (keyinroq) L'Hôpital qoidasi.

5.1.6 Asimptotik ekvivalentlik (4-hafta konspekti)

$$f(x) \sim g(x) \quad (x \rightarrow a) \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 1, \quad (92)$$

$$f(x) \sim g(x) \iff f(x) = g(x)(1 + \varepsilon(x)), \quad \varepsilon(x) \rightarrow 0, \quad (93)$$

$$f \sim g \iff f = g + o(g). \quad (94)$$

$$x \rightarrow 0 \text{ da: } \sin x \sim x, \quad \tan x \sim x, \quad 1 - \cos x \sim \frac{x^2}{2}, \quad e^x - 1 \sim x, \quad (95)$$

$$\ln(1+x) \sim x, \quad (1+x)^\alpha - 1 \sim \alpha x \quad (\alpha \text{ konstanta}). \quad (96)$$

Raqamli jihatdan $\frac{\sin(0.01)}{0.01} \approx 0.999983$ “yaqin, lekin teng emas”ni illyustratsiya qiladi. Cheksizlikda eng yuqori darajali had hukmron:

$$3x^2 + 5x - 1 \sim 3x^2 \quad (x \rightarrow \infty), \quad \frac{2x^3 + x}{5x^2 - 1} \sim \frac{2x^3}{5x^2} = \frac{2}{5}x \quad (x \rightarrow \infty). \quad (97)$$

Ekvivalentliklar ko'paytirilishi va bo'linishi mumkin, umuman qo'shilmasligi mumkin.

5.1.7 Landau belgilari: katta- O va kichik- o (4-hafta konspekti)

$$f(x) = O(g(x)) \quad (x \rightarrow a) \iff \frac{|f(x)|}{|g(x)|} \text{ } a \text{ yaqinida chegaralangan qoladi,} \quad (98)$$

$$f(x) = o(g(x)) \quad (x \rightarrow a) \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0, \quad (99)$$

$$f = o(g) \implies f = O(g); \quad f = O(g) \not\Rightarrow f = o(g). \quad (100)$$

$$x^2 = o(x), \quad x^3 = o(x^2) \quad (x \rightarrow 0); \quad (101)$$

$$x = o(x^2), \quad x^{10} = o(e^x) \quad (x \rightarrow \infty). \quad (102)$$

$$3x = O(x) \text{ lekin } 3x \neq o(x) \quad (x \rightarrow 0) \text{ chunki } \frac{3x}{x} = 3. \quad (103)$$

$$\sin x = x + o(x) \quad (x \rightarrow 0); \quad e^x = 1 + x + o(x); \quad e^x = 1 + x + \frac{x^2}{2} + o(x^2). \quad (104)$$

$$2x^2 + 3x + 1 = 2x^2 + O(x) \quad (x \rightarrow \infty). \quad (105)$$

So'zlarda: katta- O — f, g ning tartibiga ko'pi bilan teng kattalik tartibiga ega; kichik- o — f, g bilan solishtirganda ahamiyatsiz bo'lib qoladi. Ehtiyot bo'ling: $f = O(g)$ oddiy ma'noda tenglama emas — $O(g)$ butun funksiyalar sinfini bildiradi.

5.2 OpenStax §2.3, Checkpoint 2.11-2.20 muammolari — savollar va to'liq yechimlar

1. Chegara qonunlaridan foydalanib $\lim_{x \rightarrow -3} (2x - 1)\sqrt{x + 4}$ ni hisoblang, har bir qadamda ishlatilgan qonunni ko'rsatib.

[Manba: OpenStax §2.3, Checkpoint 2.11]

Yechish. Ikkala koeffitsient limiti ham mavjud, demak ko'paytma qoidasi qo'llaniladi: $\lim(2x - 1) = 2(-3) - 1 = -7$ (yig'indi, ayirma, asosiy chegalar); $\lim \sqrt{x + 4} = \sqrt{-3 + 4} = 1$ (ildiz qoidasi). Ko'paytma: $-7 \cdot 1 = -7$.

2. $\lim_{x \rightarrow -2} (3x^3 - 2x + 7)$ ni hisoblang.

[Manba: Checkpoint 2.12]

Yechish. To'g'ridan-to'g'ri qo'yish (85) bilan oqlanadi: $3(-8) - 2(-2) + 7 = -24 + 4 + 7 = -13$.

3. $\lim_{x \rightarrow -3} \frac{x^2 + 4x + 3}{x^2 - 9}$ ni hisoblang.

[Manba: Checkpoint 2.13]

Yechish. 0/0 hold. Ikkalasini ajratamiz: $x \neq -3$ uchun $\frac{(x+1)(x+3)}{(x+3)(x-3)} = \frac{x+1}{x-3}$. Limit: $\frac{-1-1}{-3-3} = \frac{-2}{-6} = \frac{1}{3}$.

4. $\lim_{x \rightarrow 5} \frac{\sqrt{x-1}-2}{x-5}$ ni hisoblang.

[Manba: Checkpoint 2.14]

Yechish. $0/0$ hold. Qo'shma son $\sqrt{x-1}+2$ ga ko'paytiramiz:

$$\frac{(x-1)-4}{(x-5)(\sqrt{x-1}+2)} = \frac{x-5}{(x-5)(\sqrt{x-1}+2)} = \frac{1}{\sqrt{x-1}+2} \quad (x \neq 5),$$

demak limit $\frac{1}{\sqrt{4+2}} = \frac{1}{4}$.

5. $\lim_{x \rightarrow -3} \frac{\frac{1}{x+2}+1}{x+3}$ ni hisoblang.

[Manba: Checkpoint 2.15]

Yechish. Murakkab kasrni soddalashtiramiz: $\frac{1}{x+2}+1 = \frac{x+3}{x+2}$, demak ifoda $x \neq -3$ uchun $\frac{x+3}{(x+2)(x+3)} = \frac{1}{x+2}$ ga teng. Limit: $\frac{1}{-3+2} = -1$.

6. $\lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{4}{x^2-2x-3} \right)$ ni hisoblang.

[Manba: Checkpoint 2.16]

Yechish. $x^2-2x-3 = (x-3)(x+1)$ ko'paytuvchilarga ajratamiz va birlashtiramiz:

$$\frac{1}{x-3} - \frac{4}{(x-3)(x+1)} = \frac{(x+1)-4}{(x-3)(x+1)} = \frac{x-3}{(x-3)(x+1)} = \frac{1}{x+1} \quad (x \neq 3),$$

demak limit $\frac{1}{4}$.

7. Bo'lakli checkpoint: $x \leq 3$ uchun $f(x) = \begin{cases} x^2 \\ x+4 \end{cases}$, $x > 3$ uchun; tugun nuqtasidagi bir tomonlama chegaralarni hisoblang.

[Manba: Checkpoint 2.17]

Yechish. Chapdan parabola shoxi $\lim_{x \rightarrow 3^-} f(x) = 9$ beradi; o'ngdan to'g'ri chiziq $\lim_{x \rightarrow 3^+} (x+4) = 7$ beradi. Teng emas, demak $\lim_{x \rightarrow 3} f(x)$ mavjud emas.

8. $\lim_{x \rightarrow 1} \frac{x+2}{(x-1)^2}$ ni hisoblang (2.23 misol usuli).

[Manba: Checkpoint 2.18]

Yechish. Surat $\rightarrow 3$ (noldan farqli), mahraj ikkala tomondan $\rightarrow 0^+$ (kvadrat), demak nisbat $\rightarrow +\infty$: cheksiz limit, va $x = 1$ vertikal asimptota.

9. Siqish teoremasidan foydalanib $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$ ni hisoblang.

[Manba: Checkpoint 2.19]

Yechish. $-1 \leq \sin \frac{1}{x} \leq 1$ bo'lgani uchun $x^2 \geq 0$ ga ko'paytirib $-x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$ olamiz. Ikkala chegara ham $\rightarrow 0$; (86) bo'yicha limit 0.

10. $\lim_{\theta \rightarrow 0} \frac{1-\cos \theta}{\sin \theta}$ ni hisoblang.

[Manba: Checkpoint 2.20]

Yechish. Surat va mahrajni θ ga bo'lamiz (limit ichida $\theta \neq 0$ bo'lgani uchun qonuniy):

$$\frac{1-\cos \theta}{\sin \theta} = \frac{\frac{1-\cos \theta}{\theta}}{\frac{\sin \theta}{\theta}} \rightarrow \frac{0}{1} = 0,$$

$\frac{1-\cos \theta}{\theta} \rightarrow 0$ (2.25 misol) va $\frac{\sin \theta}{\theta} \rightarrow 1$ (standart chegara) dan foydalanib.

5.3 OpenStax §2.3 mashqlari 83-130 — savollar va to'liq yechimlar

5.3.1 Chegara qonunlaridan foydalanish (83-86)

1. $\lim_{x \rightarrow 0} (4x^2 - 2x + 3).$

[Manba: §2.3, 83-mashq]

Yechish. (85) bo'yicha qo'yish: $4(0)^2 - 2(0) + 3 = 3.$

2. $\lim_{x \rightarrow 1} \frac{x^3 + 3x^2 + 5}{4 - 7x}.$

[Manba: §2.3, 84-mashq]

Yechish. 1 dagi mahraj $-3 \neq 0$, demak bo'linma qoidasi va qo'yish $\frac{1+3+5}{-3} = \frac{9}{-3} = -3$ beradi.

3. $\lim_{x \rightarrow -2} \sqrt{x^2 - 6x + 3}.$

[Manba: §2.3, 85-mashq]

Yechish. Ildiz ostidagi ifoda -2 da uzluksiz va musbat; ildiz qoidasi: $\sqrt{4 + 12 + 3} = \sqrt{19}.$

4. $\lim_{x \rightarrow -1} (9x + 1)^2.$

[Manba: §2.3, 86-mashq]

Yechish. Daraja qoidasi, so'ng qo'yish: $(9(-1) + 1)^2 = (-8)^2 = 64.$

5.3.2 To'g'ridan-to'g'ri qo'yish (87-92)

1. $\lim_{x \rightarrow 7} x^2.$

[Manba: §2.3, 87-mashq]

Yechish. 49.

2. $\lim_{x \rightarrow -2} (4x^2 - 1).$

[Manba: §2.3, 88-mashq]

Yechish. $4(4) - 1 = 15.$

3. $\lim_{x \rightarrow 0} \frac{1}{1 + \sin x}.$

[Manba: §2.3, 89-mashq]

Yechish. Mahraj $1 + \sin 0 = 1 \neq 0$: bo'linma qoidasi $\frac{1}{1} = 1$ beradi.

4. $\lim_{x \rightarrow 2} e^{2x - x^2}.$

[Manba: §2.3, 90-mashq]

Yechish. Daraja: $2(2) - 4 = 0$; $e^0 = 1.$

5. $\lim_{x \rightarrow 1} \frac{2+7x}{x+6}.$

[Manba: §2.3, 91-mashq]

Yechish. $\frac{2+7}{1+6} = \frac{9}{7}.$

6. $\lim_{x \rightarrow 3} \ln e^{3x}.$

[Manba: §2.3, 92-mashq]

Yechish. $\ln e^{3x} = 3x$ ayniyat (logarifm va eksponensial teskari), demak limit $3 \cdot 3 = 9.$

5.3.3 Aniqlanmagan 0/0 holdi (93-102)

1. $\lim_{x \rightarrow 4} \frac{x^2-16}{x-4}.$

[Manba: §2.3, 93-mashq]

Yechish. 0/0 hold. Ajratamiz: $x \neq 4$ uchun $\frac{(x-4)(x+4)}{x-4} = x+4$; limit = 8.

2. $\lim_{x \rightarrow 2} \frac{x-2}{x^2-2x}.$

[Manba: §2.3, 94-mashq]

Yechish. Mahrajni ajratamiz: $x \neq 2$ uchun $\frac{x-2}{x(x-2)} = \frac{1}{x}$; limit = $\frac{1}{2}$.

3. $\lim_{x \rightarrow 6} \frac{3x-18}{2x-12}.$

[Manba: §2.3, 95-mashq]

Yechish. $x \neq 6$ uchun $\frac{3(x-6)}{2(x-6)} = \frac{3}{2}$; limit = $\frac{3}{2}$.

4. $\lim_{h \rightarrow 0} \frac{(1+h)^2-1}{h}.$

[Manba: §2.3, 96-mashq]

Yechish. $\frac{1+2h+h^2-1}{h} = \frac{h(2+h)}{h} = 2+h$; limit = 2 (bu x^2 ning 1 dagi hosilasi).

5. $\lim_{t \rightarrow 9} \frac{t-9}{\sqrt{t}-3}.$

[Manba: §2.3, 97-mashq]

Yechish. 0/0 hold. Mahrajning qo'shma soniga ko'paytiramiz:

$$\frac{t-9}{\sqrt{t}-3} \cdot \frac{\sqrt{t}+3}{\sqrt{t}+3} = \frac{(t-9)(\sqrt{t}+3)}{t-9} = \sqrt{t}+3 \quad (t \neq 9),$$

demak limit $\sqrt{9}+3=6$.

6. $\lim_{h \rightarrow 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h}$, bunda a — noldan farqli haqiqiy konstanta.

[Manba: §2.3, 98-mashq]

Yechish. Suratni birlashtiramiz: $\frac{a-(a+h)}{a(a+h)} = \frac{-h}{a(a+h)}$. h ga bo'lamiz: $h \neq 0$ uchun $\frac{-1}{a(a+h)}$. Limit: $-\frac{1}{a^2}$ ($1/x$ ning a dagi hosilasi).

7. $\lim_{\theta \rightarrow \pi} \frac{\sin \theta}{\tan \theta}.$

[Manba: §2.3, 99-mashq]

Yechish. π yaqinida $\theta \neq \pi$ uchun: $\frac{\sin \theta}{\tan \theta} = \sin \theta \cdot \frac{\cos \theta}{\sin \theta} = \cos \theta$ (qisqartirish qonuniy, chunki u yerda $\sin \theta \neq 0$). Limit: $\cos \pi = -1$.

8. $\lim_{x \rightarrow 1} \frac{x^3-1}{x^2-1}.$

[Manba: §2.3, 100-mashq]

Yechish. Ikkalasini ajratamiz: $\frac{(x-1)(x^2+x+1)}{(x-1)(x+1)} = \frac{x^2+x+1}{x+1}$; limit = $\frac{3}{2}$.

9. $\lim_{x \rightarrow 1/2} \frac{2x^2+3x-2}{2x-1}.$

[Manba: §2.3, 101-mashq]

Yechish. $2x^2+3x-2 = (2x-1)(x+2)$; $x+2$ ga qisqartiramiz; limit = $\frac{1}{2} + 2 = \frac{5}{2}$.

10. $\lim_{x \rightarrow -3} \frac{\sqrt{x+4}-1}{x+3}.$

[Manba: §2.3, 102-mashq]

Yechish. Qo'shma son: $\frac{\sqrt{x+4}+1}{\sqrt{x+4}+1}$ ga ko'paytiramiz:

$$\frac{(x+4)-1}{(x+3)(\sqrt{x+4}+1)} = \frac{1}{\sqrt{x+4}+1} \quad (x \neq -3);$$

limit = $\frac{1}{1+1} = \frac{1}{2}$.

5.3.4 $K/0$ holdlar: bir tomonlama cheksiz chegaralar (103-106)

1. $\lim_{x \rightarrow -2^-} \frac{2x^2+7x-4}{x^2+x-2}.$

[Manba: §2.3, 103-mashq]

Yechish. Ajratamiz: surat = $(2x-1)(x+4)$, mahraj = $(x+2)(x-1)$. $x = -2$ da surat $(-5)(2) = -10 \neq 0$, holbuki mahraj $\rightarrow 0$: $K/0$ hold. -2 yaqinidagi ishora: $x = -2.1$ olamiz: $(x+2) < 0$, $(x-1) < 0$, demak mahraj musbat; surat $-10 < 0$ ga yaqin. Nisbat $\rightarrow -\infty$.

2. $\lim_{x \rightarrow -2^+} \frac{2x^2+7x-4}{x^2+x-2}.$

[Manba: §2.3, 104-mashq]

Yechish. Xuddi shu surat; endi $(x+2) > 0$, $(x-1) < 0$: mahraj manfiy. Nisbat $\rightarrow +\infty$.

3. $\lim_{x \rightarrow 1^-} \frac{2x^2+7x-4}{x^2+x-2}.$

[Manba: §2.3, 105-mashq]

Yechish. 1 dagi surat: $(1)(5) = 5 > 0$. $x \rightarrow 1^-$ uchun: $(x+2) > 0$, $(x-1) < 0$, mahraj manfiy: nisbat $\rightarrow -\infty$.

4. $\lim_{x \rightarrow 1^+} \frac{2x^2+7x-4}{x^2+x-2}.$

[Manba: §2.3, 106-mashq]

Yechish. Mahraj 1 ning o'ng tomonida musbat: nisbat $\rightarrow +\infty$. (-2 va 1 dagi ikki tomonlama limitlar ham mavjud emas; $x = -2$ va $x = 1$ vertikal asimptotalar.)

5.3.5 Berilgan chegaralar bilan limit qonunlari (107-114)

Butun bo'limda: $\lim_{x \rightarrow 6} f(x) = 4$, $\lim_{x \rightarrow 6} g(x) = 9$, $\lim_{x \rightarrow 6} h(x) = 6$.

1. $\lim_{x \rightarrow 6} 2f(x)g(x).$

[Manba: §2.3, 107-mashq]

Yechish. Ko'paytma, so'ng konstanta ko'paytma: $2 \cdot 4 \cdot 9 = 72$.

2. $\lim_{x \rightarrow 6} \frac{g(x)-1}{f(x)}.$

[Manba: §2.3, 108-mashq]

Yechish. Bo'linma qoidasi: $\frac{9-1}{4} = 2$.

3. $\lim_{x \rightarrow 6} (f(x) + \frac{1}{3}g(x)).$

[Manba: §2.3, 109-mashq]

Yechish. Yig'indi va konstanta ko'paytma: $4 + \frac{9}{3} = 7$.

4. $\lim_{x \rightarrow 6} (h(x))^3.$

[Manba: §2.3, 110-mashq]

Yechish. Daraja qoidasi: $6^3 = 216$.

5. $\lim_{x \rightarrow 6} \sqrt{g(x) - f(x)}.$

[Manba: §2.3, 111-mashq]

Yechish. Ayirma, so'ng ildiz: $\sqrt{9-4} = \sqrt{5}$.

6. $\lim_{x \rightarrow 6} x h(x).$

[Manba: §2.3, 112-mashq]

Yechish. $\lim x = 6$ bilan ko'paytma qoidasi: $6 \cdot 6 = 36$.

7. $\lim_{x \rightarrow 6} [(x+1)f(x)]$.
[Manba: §2.3, 113-mashq]

Yechish. $(6+1) \cdot 4 = 28$.

8. $\lim_{x \rightarrow 6} (f(x)g(x) - h(x))$.
[Manba: §2.3, 114-mashq]

Yechish. $4 \cdot 9 - 6 = 30$.

5.3.6 Bo'lakli funksiyalar va grafik asosidagi chegaralar (115-125)

1. $x \leq 3$ uchun $f(x) = x^2$, $x > 3$ uchun $f(x) = x + 4$: toping a. $\lim_{x \rightarrow 3^-} f$, b. $\lim_{x \rightarrow 3^+} f$.
[Manba: §2.3, 115-mashq]

Yechish. a. Parabola shoxi: 9. b. To'g'ri shoxi: 7. (Ikki tomonlama limit mavjud emas.)

2. $x \leq 0$ uchun $g(x) = x^3 - 1$, $x > 0$ uchun $g(x) = 1$: toping a. $\lim_{x \rightarrow 0^-} g$, b. $\lim_{x \rightarrow 0^+} g$.
[Manba: §2.3, 116-mashq]

Yechish. a. Kub shoxi: $0^3 - 1 = -1$. b. Konstanta shoxi: 1. (Ikki tomonlama mavjud emas.)

3. $x < 2$ uchun $h(x) = x^2 - 2x + 1$, $x \geq 2$ uchun $h(x) = 3 - x$: toping a. $\lim_{x \rightarrow 2^-} h$, b. $\lim_{x \rightarrow 2^+} h$.
[Manba: §2.3, 117-mashq]

Yechish. a. $4 - 4 + 1 = 1$. b. $3 - 2 = 1$. Ikkala tomon mos: $\lim_{x \rightarrow 2} h(x) = 1 = h(2)$ — bo'lakli ta'rifga qaramay, o'sha yerda uzluksiz.

4. f va g — grafik berilgan ikki funksiya (V shaklidagi $f(x) = |x + 2|$; chiziqli chap shox $g(x) = -x - 6$ va egri o'ng shoxli bo'lakli g), hisoblang $\lim_{x \rightarrow -3^+} (f(x) + g(x))$.
[Manba: §2.3, 118-mashq]

Yechish. Grafikdan: $f(-3) = |-3 + 2| = 1$ va egri shoxda $g(-3^+) = 1$. Yig'indi qoidasi: $1 + 1 = 2$.

5. $\lim_{x \rightarrow -3^-} (f(x) - 3g(x))$.
[Manba: §2.3, 119-mashq]

Yechish. Chapdan chiziqli shox $g(-3^-) = -(-3) - 6 = -3$ beradi... grafikdan tugunni o'qib, -3 dagi chap shox qiymati -1 ; grafik qiymat $g(-3^-) = -1$ ni olamiz: $f(-3) - 3g(-3^-) = 1 - 3(-1) = 4$.

6. $\lim_{x \rightarrow 0} \frac{f(x)g(x)}{3}$.
[Manba: §2.3, 120-mashq]

Yechish. $f(0) = |2| = 2$; $g(0) = -2$ (egri chiziq $(0, -2)$ dan o'tadi). Ko'paytma va bo'linma: $\frac{2 \cdot (-2)}{3} = -\frac{4}{3}$.

7. $\lim_{x \rightarrow -3} \frac{2+g(x)}{f(x)}$.
[Manba: §2.3, 121-mashq]

Yechish. Chap: $\frac{2+g(-3^-)}{f(-3)} = \frac{2+(-1)}{1} = 1$; o'ng: $\frac{2+1}{1} = 3$. Bir tomonlama chegaralar farq qiladi, demak ikki tomonlama limit mavjud emas.

8. $\lim_{x \rightarrow 1} (f(x))^2$.
[Manba: §2.3, 122-mashq]

Yechish. $f(1) = |3| = 3$; daraja qoidasi: 9.

9. $\lim_{x \rightarrow 1} \sqrt[3]{f(x) - g(x)}.$

[Manba: §2.3, 123-mashq]

Yechish. Grafikdan $g(1) = -5$ va $f(1) = 3$: ayirma 8; kub ildiz: $\sqrt[3]{8} = 2$.

10. $\lim_{x \rightarrow -7} x g(x).$

[Manba: §2.3, 124-mashq]

Yechish. $x = -7$ da chiziqli shox $g(-7) = -(-7) - 6 = 1$ beradi; ko'paytma: $(-7)(1) = -7$.

11. $\lim_{x \rightarrow -9} [x f(x) + 2 g(x)].$

[Manba: §2.3, 125-mashq]

Yechish. $f(-9) = |-7| = 7$; $g(-9) = 9 - 6 = 3$; $(-9)(7) + 2(3) = -63 + 6 = -57$.

5.3.7 Siqish teoremasi va qo'llanmalar (126-130)

T To'g'rimi yoki noto'g'rimi: agar $x = 2$ yaqinida $2x - 1 \leq g(x) \leq x^2 - 2x + 3$ bo'lsa, u holda $\lim_{x \rightarrow 2} g(x) = 0$.

[Manba: §2.3, 126-mashq]

Yechish. Noto'g'ri. $x = 2$ da ikkala chegara ham $2(2) - 1 = 3 = 4 - 4 + 3$ ga teng, demak siqish teoremasi bo'yicha $\lim_{x \rightarrow 2} g(x) = 3$, nol emas. (Siqish teoremasi chegaralarning umumiy limitini aynan beradi.)

T $\lim_{\theta \rightarrow 0} \theta^2 \cos \frac{1}{\theta}.$

[Manba: §2.3, 127-mashq]

Yechish. $-1 \leq \cos \frac{1}{\theta} \leq 1$ dan $-\theta^2 \leq \theta^2 \cos \frac{1}{\theta} \leq \theta^2$; ikkala chegara ham $\rightarrow 0$; siqish: limit = 0.

1. $\lim_{x \rightarrow 0} f(x)$, bunda x ratsional bo'lsa $f(x) = 0$, x irratsional bo'lsa $f(x) = x^2$.

[Manba: §2.3, 128-mashq]

Yechish. Barcha x uchun: $0 \leq f(x) \leq x^2$ (ratsional x beradi 0; irratsional x beradi $x^2 \geq 0$). $0 \rightarrow 0$ va $x^2 \rightarrow 0$ bo'lgani uchun siqish teoremasi $\lim_{x \rightarrow 0} f(x) = 0$ ni majbur qiladi — f har bir $x \neq 0$ da qattiq uzluksiz bo'lsa ham.

T Kulon qonuni: $q = 10^{-19}$ bilan $E(r) = \frac{q}{4\pi\epsilon_0 r^2}$; $\lim_{r \rightarrow 0^+} E(r)$ ni hisoblang va izohlang; nega o'ngdan?

[Manba: §2.3, 129-mashq]

Yechish. Surat musbat konstanta; $r \rightarrow 0^+$ da mahraj $\rightarrow 0^+$, demak $E(r) \rightarrow +\infty$: sinov nuqtasi zaryadga yaqinlashganda maydon kattaligi chegarasiz o'sadi. Faqat $r \rightarrow 0^+$ ma'noga ega, chunki r — masofa va manfiy bo'la olmaydi (va $r = 0$ — zaryadning o'zi).

T Zichlik $\rho = m/V$ dan $V(\rho) = \frac{m}{\rho}$; $m = 8$ kg bilan $\lim_{\rho \rightarrow 0^+} V(\rho)$ ni hisoblang va fizik ma'nosini tushuntiring.

[Manba: §2.3, 130-mashq]

Yechish. $V(\rho) = \frac{8}{\rho}$; $\rho \rightarrow 0^+$ da hajm $\rightarrow +\infty$: massasi qat'iy, zichligi nolga intilayotgan jism chegarasiz hajm egallashi kerak ("suyulib borayotgan ideal gaz" ning formal bayoni). O'ngdan yaqinlashish yagona fizik variant, chunki zichlik musbat.

5.4 4-hafta konspektining yechilgan misollari — savollar va to'liq yechimlar

1. $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$ ning $0/0$ holdiga ega ekanini ko'rsating va uni hisoblang; $\lim_{x \rightarrow 0} \frac{x}{x}$, $\lim_{x \rightarrow 0} \frac{x^2}{x}$, $\lim_{x \rightarrow 0} \frac{x}{x^2}$ bilan taqqoslang.

[Manba: 4-hafta konspekti, aniqlanmagan holdlar]

Yechish. $x = 1$ qo'ysak $\frac{0}{0}$ — aniqlanmagan. Ko'paytuvchilarga ajratamiz: $x \neq 1$ uchun $\frac{(x-1)(x+1)}{x-1} = x+1$, demak limit 2. Uchta taqqoslash bir xil $0/0$ holdini uch xil natijaga olib kelishini ko'rsatadi: $\frac{x}{x} = 1 \rightarrow 1$; $\frac{x^2}{x} = x \rightarrow 0$; $\frac{x}{x^2} = \frac{1}{x} \rightarrow \pm\infty$ (chekli ikki tomonlama limit yo'q). Shuning uchun $0/0$ avtomatik qiymat bermaydi: natija surat va mahrajning nolga qanday yaqinlashishiga bog'liq.

2. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ standart chegarasi yordamida $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$ ni hisoblang.

[Manba: 4-hafta konspekti, standart chegaralar]

Yechish. $u = 5x$ almashtiramiz ($x \rightarrow 0$ da $u \rightarrow 0$): $\frac{\sin 5x}{x} = 5 \cdot \frac{\sin 5x}{5x} = 5 \cdot \frac{\sin u}{u} \rightarrow 5 \cdot 1 = 5$.

3. Asimptotik ekvivalentlik yordamida $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ va $\lim_{x \rightarrow 0} \frac{1-\cos x}{x^2}$ ni hisoblang.

[Manba: 4-hafta konspekti, asimptotika]

Yechish. $x \rightarrow 0$ da $\sin 3x \sim 3x$ bo'lgani uchun $\frac{\sin 3x}{5x} \sim \frac{3x}{5x} = \frac{3}{5}$, demak limit $\frac{3}{5}$. $1 - \cos x \sim \frac{x^2}{2}$ bo'lgani uchun $\frac{1-\cos x}{x^2} \sim \frac{x^2/2}{x^2} = \frac{1}{2}$, demak limit $\frac{1}{2}$ — ekvivalentlik orqali (87) standart chegarasini qayta chiqaradi.

4. $x \rightarrow \infty$ da $3x^2 + 5x - 1 \sim 3x^2$ ekanini ko'rsating va $\lim_{x \rightarrow \infty} \frac{2x^3+x}{5x^2-1}$ ni ekvivalentliklar orqali hisoblang.

[Manba: 4-hafta konspekti, cheksizlikdagi ekvivalentlik]

Yechish. $\frac{3x^2+5x-1}{3x^2} = 1 + \frac{5}{3x} - \frac{1}{3x^2} \rightarrow 1 + 0 - 0 = 1$, ekvivalentlik o'rnatildi: eng yuqori darajali had uzoq muddatli xatti-harakatni boshqaradi. U holda $\frac{2x^3+x}{5x^2-1} \sim \frac{2x^3}{5x^2} = \frac{2}{5}x$, bu esa $\rightarrow +\infty$; aniqrog'i, asl kasrning $\frac{2}{5}x$ ga nisbati 1 ga intiladi.

5. $x \rightarrow 0$ da $\sin x = O(x)$ ekanini va $3x = O(x)$ lekin $3x \neq o(x)$ ekanini ko'rsating.

[Manba: 4-hafta konspekti, Landau belgilari]

Yechish. $|\sin x| \leq |x|$ tengsizligi barcha haqiqiy x uchun o'rinli (standart geometrik fakt), bu aynan $\frac{|\sin x|}{|x|}$ ning chegaralanganligi: $\sin x = O(x)$. $3x$ uchun: $\frac{|3x|}{|x|} = 3$ chegaralangan, demak $3x = O(x)$; lekin $\frac{3x}{x} = 3 \not\rightarrow 0$, demak $3x \neq o(x)$. Bu O va o farqining eng sodda illyustratsiyasi: katta- O har qanday chegaralangan nisbatga ruxsat beradi, kichik- o nisbat nolga intilishini majbur qiladi.

6. $x \rightarrow 0$ da $x^2 = o(x)$ ni va $x \rightarrow \infty$ da $x = o(x^2)$, $x^{10} = o(e^x)$ ni tekshiring.

[Manba: 4-hafta konspekti, Landau belgilari]

Yechish. $\frac{x^2}{x} = x \rightarrow 0$: nol yaqinida $x^2 = o(x)$. Cheksizlikda $\frac{x}{x^2} = \frac{1}{x} \rightarrow 0$: $x = o(x^2)$; va $\lim_{x \rightarrow \infty} \frac{x^{10}}{e^x} = 0$ (88) standart chegara bo'yicha $\frac{x^{10}}{e^x} \rightarrow 0$: eksponensial o'sish oxir-oqibat har bir qat'iy darajadan ustun keladi.

7. $\sin x \sim x$ va $e^x = 1 + x + \frac{x^2}{2} + o(x^2)$ ni kichik- o tilida yozing. Nega $f(x) = O(g(x))$ oddiy tenglama emas?

[Manba: 4-hafta konspekti, bog'lanishlar]

Yechish. (94) bo'yicha $f \sim g$ agar faqat agar $f = g + o(g)$: demak $\sin x \sim x$ ni $\sin x = x + o(x)$ deb yozish mumkin — xato x bilan solishtirganda ahamiyatsiz. Eksponensial uchun $e^x = 1 + x + o(x)$ bajariladi, chunki $\frac{e^x-1-x}{x} \rightarrow 0$, va yana bir hadni saqlash $e^x = 1 + x + \frac{x^2}{2} + o(x^2)$ beradi. Nihoyat, $f = O(g)$ tenglama emas, chunki $O(g)$ g ning konstanta ko'paytmalari bilan chegaralangan funksiyalarning butun sinfini bildiradi; $2x^2 + 3x + 1 = 2x^2 + O(x)$ yozuvi aynan qoldiq $3x + 1$ ning x ning konstanta ko'paytmasidan tezroq o'smasligini anglatadi.

6 4-hafta — Matritsa rangi va teskari matritsa (Chiziqli algebra I)

Chiziqli algebra I ning 4-haftasi matritsa rangi va Gauss-Jordan yo'q qilish orqali teskari matritsa mavzularini o'z ichiga oladi va 1-3 haftalar material bo'yicha Test 1 bilan yakunlanadi. Repozitoriyda alohida 4-hafta mashqlar varaqasi yo'q, shuning uchun bu qism (i) ikki mavzuning to'liq formulalar katalogini — tayinlangan Strang materialidan tuzilgan va (ii) kurs TA varaqalari uslubidagi, to'liq yechimli darajalashtirilgan mashqlar to'plamini beradi. Ular kurs boshqa joylarda ishlatadigan bir xil matematik tuzilmalardan foydalanadi, demak ular matritsa algebra qismlari bo'yicha Test-1 tayyorgarligi vazifasini ham bajaradi.

6.1 4-hafta formulalar katalogi

6.1.1 Rang va yo'q qilish

$$\text{rank } A \leq \min(m, n), \quad (106)$$

$$A \in \mathbb{R}^{n \times n} : \text{rank } A = n \iff A\mathbf{x} = \mathbf{0} \text{ faqat } \mathbf{x} = \mathbf{0} \text{ yechimga ega,} \quad (107)$$

$$A \text{ teskarililigi} \iff \text{rank } A = n \iff n \text{ ta pivot} \iff \det A \neq 0, \quad (108)$$

$$A \text{ teskarililigi} \iff A\mathbf{x} = \mathbf{b} \text{ har bir } \mathbf{b} \text{ uchun aynan bitta yechimga ega.} \quad (109)$$

6.1.2 Teskari matritsa (Strang, Introduction to Linear Algebra 4e, §2.5)

$$AA^{-1} = I = A^{-1}A; \quad \mathbf{x} = A^{-1}\mathbf{b} \text{ bo'lsa, } A\mathbf{x} = \mathbf{b} \text{ yechim,} \quad (110)$$

$$A^{-1} \text{ mavjud} \iff \text{yo'q qilish } n \text{ ta pivot beradi (qator almashtirishlar mumkin),} \quad (111)$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (ad-bc \neq 0), \quad (112)$$

$$(AB)^{-1} = B^{-1}A^{-1}; \quad (ABC)^{-1} = C^{-1}B^{-1}A^{-1} \quad (\text{teskari tartib}), \quad (113)$$

$$\text{diag}(d_1, \dots, d_n)^{-1} = \text{diag}\left(\frac{1}{d_1}, \dots, \frac{1}{d_n}\right) \quad (d_i \neq 0), \quad (114)$$

$$\mathbf{x} \neq \mathbf{0}, A\mathbf{x} = \mathbf{0} \implies A \text{ teskarili emas,} \quad (115)$$

$$BA = I, AC = I \implies B = B(AC) = (BA)C = C \quad (\text{yagonalik}). \quad (116)$$

(115) ning izohi: hech bir matritsa $\mathbf{0}$ ni orqaga $\mathbf{x}$ ga qaytara olmaydi.

6.1.3 Gauss-Jordan yo'q qilish

$$[A \ I] \xrightarrow{\text{qator amallari}} [I \ A^{-1}] \quad (117)$$

Bir xil qator amallari A ni I ga va I ni A^{-1} ga aylantiradi. Qator amallari: ikkala qatorni almashtirish; qatorni noldan farqli konstantaga ko'paytirish; bitta qatorning ko'paytmasini boshqasiga qo'shish. Har biri teskari oson (amalni bekor qilish) bo'lgan boshlang'ich E matritsaga mos keladi, shuning uchun $(E_2E_1A)^{-1} = A^{-1}E_1^{-1}E_2^{-1}$ teskari tartibda chiqadi.

6.2 4-hafta mashq muammolari (rang, teskari, Gauss-Jordan) — savollar va to'liq yechimlar

1. $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 5 \\ 3 & 6 & 4 \end{bmatrix}$ uchun $\text{rank } A$ ni yo'q qilish orqali hisoblang.

[Manba: 4-hafta LA mashqi, 1-muammo]

Yechish. Birinchi pivot (1) ostini yo'q qilamiz: $R_2 - 2R_1 \rightarrow (0, 0, 3)$; $R_3 - 3R_1 \rightarrow (0, 0, 1)$. Ikkinchi ustun elementlari yo'qolgan, demak ikkinchi pivot 3-ustundan kelishi kerak: pivot = 3. Uni ostidan yo'q qilamiz: $R_3 - \frac{1}{3}R_2 \rightarrow (0, 0, 0)$. Ikki noldan farqli pivot: $\text{rank } A = 2$. Boshidan ko'ringan $\text{col}_2 = 2\text{col}_1$ bog'lanishiga e'tibor bering — rang yetishmovchiligi belgisi.

2. $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ teskarilimi, uch usulda aniqlang: pivotlar, determinant va noldan farqli nol-vektor.

[Manba: 4-hafta LA mashqi, 2-muammo]

Yechish. (Pivotlar) $R_2 - 2R_1 \rightarrow (0, 0)$: faqat bitta pivot — singulyar. (Determinant) $ad - bc = 2 \cdot 2 - 1 \cdot 4 = 0$ — (112) bo'yicha singulyar. (Nol-vektor) $B(1, -2)^T = (2 - 2, 4 - 4)^T = \mathbf{0}$ — $B\mathbf{x} = \mathbf{0}$ ning noldan farqli yechimi, demak (115) bo'yicha B teskarililmaydi. Uchala test ham mos keladi.

3. $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ ning teskarisini 2×2 formula bilan toping va $AA^{-1} = I$ ni tekshiring.

[Manba: 4-hafta LA mashqi, 3-muammo]

Yechish. $\det A = 6 - 5 = 1$, demak (112) bo'yicha $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$. Tekshirish:

$$AA^{-1} = \begin{bmatrix} 2 \cdot 3 + 1 \cdot (-5) & 2(-1) + 1 \cdot 2 \\ 5 \cdot 3 + 3(-5) & 5(-1) + 3 \cdot 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \checkmark$$

4. $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ ni teskarilash uchun $[A \ I]$ ustida Gauss-Jordan qo'llang.

[Manba: 4-hafta LA mashqi, 4-muammo]

Yechish.

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array} \right] \xrightarrow{R_2 - 3R_1} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array} \right] \xrightarrow{R_1 - 2R_2} \left[\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array} \right].$$

Demak $A^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$; tekshirish: $AA^{-1} = \begin{bmatrix} 1 \cdot 7 + 2(-3) & 1(-2) + 2(1) \\ 3 \cdot 7 + 7(-3) & 3(-2) + 7(1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$
✓

5. $K = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ ni Gauss-Jordan bilan teskarilang va pivotlar ko'paytmasi

$\det K$ ga tengligini tekshiring.

[Manba: 4-hafta LA mashqi, 5-muammo (Strang'ning sevimli misoli)]

Yechish. $[K \ I]$ ustida ishlaymiz: birinchi pivot 2 bilan yo'q qilamiz ($R_2 + \frac{1}{2}R_1$), so'ng ikkinchi pivot $\frac{3}{2}$ bilan ($R_3 + \frac{2}{3}R_2$), pivotlari 2, $\frac{3}{2}$, $\frac{4}{3}$ bo'lgan yuqori uchburchak shaklga yetamiz. Keyin yuqoriga qarab tozalaymiz va har bir qatorni o'z pivotiga bo'lamiz; o'ng blok

$$K^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

bo'ladi. Pivotlar ko'paytmasi $2 \cdot \frac{3}{2} \cdot \frac{4}{3} = 4 = \det K$, va haqiqatdan K^{-1} butun matritsani $\det K$ ga bo'ladi — (112) dan keyin qayd etilgan qolip. (K simmetrik va tridiagonal, lekin K^{-1} zich.)

6. $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ uchun $A\mathbf{x} = \mathbf{b}$ ni 4-muammodagi A^{-1} yordamida yeching.

[Manba: 4-hafta LA mashqi, 6-muammo]

Yechish. $\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$. Tekshirish: $1 \cdot 5 + 2(-2) = 1$ va $3 \cdot 5 + 7(-2) = 1$. ✓

7. $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ va $B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$ berilgan. $(AB)^{-1}$ ni oldin ko'paytirishsiz hisoblang, so'ng tekshiring.

[Manba: 4-hafta LA mashqi, 7-muammo]

Yechish. (113) bo'yicha $(AB)^{-1} = B^{-1}A^{-1}$. Bu yerda $\det A = 2$ va $\det B = 2$, demak

$$A^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ -\frac{1}{2} & 1 \end{bmatrix}, \quad B^{-1}A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{2} & \frac{3}{4} \end{bmatrix}.$$

Tekshirish: $AB = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$, $\det(AB) = 4$, demak to'g'ridan-to'g'ri $(AB)^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{2} & \frac{3}{4} \end{bmatrix}$ — bir xil matritsa. Va $(AB)^{-1}(AB) = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{2} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. ✓ Teskari tartib qonuni tasdiqlandi.

8. (Konseptual, Test-1 uslubida.) To'g'rimi yoki noto'g'rimi, asoslar bilan: (a) A teskarili bo'lsa, A^T ham teskarili. (b) $\mathbf{x} \neq \mathbf{0}$ va $\lambda = 0$ bilan $A\mathbf{x} = \lambda\mathbf{x}$ bo'lsa, A teskarili emas. (c) Teskarili 3×3 matritsalar ko'paytmasi singulyar bo'lishi mumkin. (d) Har bir 2×3 matritsa uchun $\text{rank}(A) \leq 2$.

[Manba: 4-hafta LA mashqi, 8-muammo]

Yechish. (a) To'g'ri: A da n ta pivot $\iff A^T$ da n ta pivot (qator rangi = ustun rangi); formal ravishda $(A^T)^{-1} = (A^{-1})^T$. (b) To'g'ri: $A\mathbf{x} = \mathbf{0} \cdot \mathbf{x} = \mathbf{0}$ bo'lgan noldan farqli $\mathbf{x}$ (115) ni buzadi. (c) Noto'g'ri: (113) bo'yicha teskarili matritsalar ko'paytmasi teskarili. (d) To'g'ri: rang qatorlar sonidan oshmaydi (har bir pivot bitta qator ishlatadi), demak $\text{rank} \leq 2 < 3$.

9. (Amaliy, Test-1 uslubida.) Qanday c qiymat(lar)ida $A = \begin{bmatrix} 1 & 2 \\ 3 & c \end{bmatrix}$ singulyar? Qolgan

qiymatlar uchun $A\mathbf{x} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$ ni teskari formula bilan yeching.

[Manba: 4-hafta LA mashqi, 9-muammo]

Yechish. $\det A = c - 6$; singulyar aynan $c = 6$ da. Aks holda $A^{-1} = \frac{1}{c-6} \begin{bmatrix} c & -2 \\ -3 & 1 \end{bmatrix}$ va $\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{c-6} \begin{bmatrix} 4c-20 \\ -2 \end{bmatrix}$, masalan $c = 7$ uchun: $\mathbf{x} = (8, -2)^T$.

10. (Murakkabroq.) A teskarili bo'lsa, mos keladigan har bir B uchun $\text{rank}(AB) = \text{rank}(B)$ ekanini ko'rsating.

[Manba: 4-hafta LA mashqi, 10-muammo]

Yechish. Teskarili matritsaga ko'paytirish pivotlar sonini o'zgartirmaydi: standart dalil: $\text{rank}(AB) \leq \text{rank } B$ (AB qatorlari B qatorlarining birikmalari), va xuddi shuni $B = A^{-1}(AB)$ ga qo'llash $\text{rank } B \leq \text{rank}(AB)$ beradi; ikkala tengsizlik tenglikni majbur qiladi. Intuitsiya: teskarili ko'paytuvchi fazolarni faqat cho'zadi/aylanadi/silliyadi, tekislamaydi, demak rang yo'qolmaydi va ortmaydi.

7 Darslik qo'shimchalari — majburiy o'qishning qo'shimcha mashqlari to'liq yechimlar bilan

Bu qism savollar bazasini kurs ko'rsatmalariga muvofiq darsliklarning o'ziga kengaytiradi. OpenStax Calculus 1 dan 1.2 va 1.5 bo'lim mashqlarining (GUIDANCE belgilagan 2-hafta bo'limlari) namunaviy ammo to'liq gradatsiyasini va jadval asosidagi 2.2 bo'lim mashqlarini (42-45) yechamiz. Strang'ning Introduction to Linear Algebra (4-nashr) kitobidan haftalik o'qishlar ko'rsatgan ikki manzilga — skalyar ko'paytma va proyeksiyalar (§1.2) hamda teskarilar (§2.5) bo'yicha muammolarni yechamiz.

7.1 OpenStax §1.2 (Funksiyalarning asosiy sinflari) — tanlangan mashqlar yechilgan

1. Har bir nuqtalar juftligi uchun ulardan o'tuvchi to'g'ri chiziqlarning qiyaligini toping va chiziq o'suvchi, kamayuvchi, gorizontali yoki vertikaligini ayting: (59) $(-2, 4)$ va $(1, 1)$; (61) $(3, 5)$ va $(-1, 2)$; (63) $(2, 3)$ va $(5, 7)$; (65) $(2, 4)$ va $(1, 4)$; (66) $(1, 4)$ va $(1, 0)$.

[Manba: OpenStax §1.2, 59-66 mashqlar]

Yechish. Qiyalik $m = \frac{y_2 - y_1}{x_2 - x_1}$. (59) $m = \frac{1-4}{1+2} = -1$: kamayuvchi. (61) $m = \frac{2-5}{-1-3} = \frac{3}{4}$: o'suvchi. (63) $m = \frac{7-3}{5-2} = \frac{4}{3}$: o'suvchi. (65) $m = \frac{4-4}{1-2} = 0$: gorizontali. (66) $\Delta x = 0$: vertikal, qiyalik aniqlanmagan.

2. Tenglamani qiyalik-kesishma shaklida yozing: (67) qiyalik -6 , $(1, 3)$ dan o'tadi; (70) qiyalik $\frac{2}{5}$, x -kesishma 8 ; (71) $(2, 1)$ va $(-2, -1)$ dan o'tadi; (73) x -kesishma 5 va y -kesishma -3 .

[Manba: OpenStax §1.2, 67-73 mashqlar]

Yechish. (67) $y - 3 = -6(x - 1) \Rightarrow y = -6x + 9$. (70) x -kesishma 8 degani $(8, 0)$ nuqta: $y - 0 = \frac{2}{5}(x - 8) \Rightarrow y = \frac{2}{5}x - \frac{16}{5}$. (71) $m = \frac{-1-1}{-2-2} = \frac{-2}{-4} = \frac{1}{2}$; $y - 1 = \frac{1}{2}(x - 2) \Rightarrow y = \frac{1}{2}x$. (73) $(5, 0)$ va $(0, -3)$ dan o'tadi: $m = \frac{-3-0}{0-5} = \frac{3}{5}$; $y = \frac{3}{5}x - 3$.

3. (75-76) Har bir chiziqli tenglama uchun qiyalik m va y -kesishma b ni bering: $y = 2x - 3$; $y = -\frac{1}{7}x + 1$.

[Manba: OpenStax §1.2, 75-76 mashqlar]

Yechish. (75) $m = 2$, $b = -3$. (76) $m = -\frac{1}{7}$, $b = 1$.

4. (81-82) Qiyalik va kesishma bilan qayta yozing: $2x + 3y = 6$; $6x - 5y + 15 = 0$.

[Manba: OpenStax §1.2, 81-82 mashqlar]

Yechish. (81) $y = 2 - \frac{2}{3}x$: $m = -\frac{2}{3}$, $b = 2$. (82) $y = \frac{6}{5}x + 3$: $m = \frac{6}{5}$, $b = 3$.

5. (83–86) Har bir ko'phad uchun: daraja; nollar; y -kesishma; chekka xatti-harakat; juftlik. a. $f(x) = 2x^2 - 3x - 5$; c. $f(x) = x^3 + 3x^2 - x - 3$.

[Manba: OpenStax §1.2, 83-86 mashqlar]

Yechish. a. Daraja 2; $2x^2 - 3x - 5 = (2x - 5)(x + 1)$: nollar $\frac{5}{2}$ va -1 ; y -kesishma -5 ; yuqoriga ochiladi (bosh koeffitsient $2 > 0$), demak ikkala chetda $f \rightarrow +\infty$; faqat juft darajalar yo'q — na juft na toq ($-3x$ toq darajali hadi va konstanta tufayli). c. Daraja 3; $x = -1$ sinovimiz: $-1 + 3 + 1 - 3 = 0$, demak $(x + 1)$ ko'paytuvchi: $x^3 + 3x^2 - x - 3 = (x + 1)(x^2 - 1) = (x + 1)(x - 1)(x + 1) = (x - 1)(x + 1)^2$: nollar 1 (oddiy) va -1 (ikki karra); y -kesishma -3 ; bosh koeffitsienti musbat kub: $x \rightarrow +\infty$ da $f \rightarrow +\infty$, $x \rightarrow -\infty$ da $f \rightarrow -\infty$; na juft na toq.

6. (88–91) g grafigini asosiy grafikdan boshlab tasvirlang: (88) $g(x) = x^2 - 1$; (89) $g(x) = (x + 3)^2 + 1$; (90) $g(x) = \sqrt{x + 2}$; (91) $g(x) = -\sqrt{x} - 1$.

[Manba: OpenStax §1.2, 88-91 mashqlar]

Yechish. (88) $y = x^2$ parabolasi 1 ga pastga surilgan. (89) Parabola 3 ga chapga va 1 ga yuqoriga surilgan (uch $(-3, 1)$). (90) Kvadrat ildiz grafigi 2 ga chapga surilgan (soha $[-2, \infty)$). (91) Kvadrat ildiz grafigi x -o'qqa aks ettirilgan va 1 ga pastga surilgan (soha $[0, \infty)$, qiymatlar ≤ -1).

7. (94) $x \leq 0$ uchun $f(x) = 4x + 3$, $x > 0$ uchun $f(x) = -x + 1$: $f(-3)$, $f(0)$, $f(2)$ ni hisoblang; eskizing.

[Manba: OpenStax §1.2, 94-mashq]

Yechish. $f(-3) = 4(-3) + 3 = -9$ (chap shox); $f(0) = 3$; $f(2) = -2 + 1 = -1$ (o'ng shox). Grafik: $(0, 3)$ gacha ko'tarilayotgan to'g'ri, so'ng $(0, 1)$ dagi ochiq nuqtadan tushayotgan to'g'ri — 0 da sakrash uzilishi.

8. (97) $x \neq 2$ uchun $g(x) = \frac{3}{x-2}$, $x = 2$ uchun $g(x) = 4$: $g(0)$, $g(-4)$, $g(2)$ ni hisoblang.

[Manba: OpenStax §1.2, 97-mashq]

Yechish. $g(0) = \frac{3}{-2} = -\frac{3}{2}$; $g(-4) = \frac{3}{-6} = -\frac{1}{2}$; $g(2) = 4$ maxsus ta'rif bo'yicha (formula bu yerda aniqlanmagan, aynan shuning uchun bo'lakli qiymat berilgan).

9. To'g'rimi yoki noto'g'rimi (98–103): (98) $f(x) = \frac{4x+1}{7x-2}$ transsendent; (99) $g(x) = \sqrt[3]{x}$ toq ildiz funksiyasi; (100) logarifmik funksiya algebraik; (101) haqiqiy konstanta b bilan $f(x) = x^b$ eksponensial funksiya; (102) juft ildiz funksiyaning sohasi barcha haqiqiy sonlar; (103) kompaniya uskunaga \$20,500 sarfladi, 3 yildan keyin qiymati \$12,300 — $V(t)$ ni toping va izohlang.

[Manba: OpenStax §1.2, 98-103 mashqlar]

Yechish. (98) Noto'g'ri: ko'phadlar nisbati ratsional, demak algebraik, transsendent emas. (99) To'g'ri: kub ildizlar barcha haqiqiy sonlar uchun aniqlangan va funksiya toq. (100) Noto'g'ri: logarifmlar transsendent. (101) Noto'g'ri: x^b — daraja funksiyasi; eksponensialda o'zgaruvchi darajada, b^x . (102) Noto'g'ri: juft ildizlar nomanfiy ildiz ostini talab qiladi, demak soha manfiylarni istisno qiladi (toq ildizlar barcha haqiqiy sonlarni qabul qiladi). (103) Chiziqli eskirish: $V(t) = 20500 + \frac{12300-20500}{3}t = 20500 - \frac{8200}{3}t \approx 20500 - 2733.33t$. $V(0) = 20500$ — xarid narxi; t -kesishma $t \approx 7.5$ yil — qiymat nolga yetgan vaqt; $V(5) = 20500 - \frac{41000}{3} \approx \6833 .

7.2 OpenStax §1.5 (Eksponensial va logarifmik funksiyalar) — tanlangan mashqlar yechilgan

1. (239-245) Eskizing; soha, qiymatlar to'plami, gorizontaal asimptotani bering: a. $f(x) = e^x + 2$; b. $f(x) = -2^x$; c. $f(x) = 1 - 2^{-x}$; d. $f(x) = e^{-x} - 1$.

[Manba: OpenStax §1.5, 239-245 mashqlar]

Yechish. a. e^x 2 ga yuqoriga surilgan: soha $\mathbb{R}$, range $(2, \infty)$, asimptota $y = 2$. b. x -o'qqa aks ettirilgan: soha $\mathbb{R}$, range $(-\infty, 0)$, asimptota $y = 0$. c. $x \rightarrow +\infty$ da $2^{-x} \rightarrow 0$, demak $f \rightarrow 1$ pastdan; $x \rightarrow -\infty$ da $2^{-x} \rightarrow \infty$, demak $f \rightarrow -\infty$: o'suvchi, range $(-\infty, 1)$, asimptota $y = 1$. d. e^x ning ko'zgusi (o'ngga pasayish): soha $\mathbb{R}$, range $(-1, \infty)$, asimptota $y = -1$.

2. (246-253) Ekvivalent eksponensial shaklga o'tkazing: a. $\log_3 81 = 4$; b. $\log_8 2 = \frac{1}{3}$; c. $\log_5 1 = 0$; d. $\ln \frac{1}{e^3} = -3$.

[Manba: OpenStax §1.5, 246-253 mashqlar]

Yechish. (39) bo'yicha: a. $3^4 = 81$. b. $8^{1/3} = 2$. c. $5^0 = 1$. d. $e^{-3} = \frac{1}{e^3}$.

3. (254-262) Ekvivalent logarifmik shaklga o'tkazing: a. $2^3 = 8$; b. $4^{-2} = \frac{1}{16}$; c. $10^2 = 100$; d. $(\frac{1}{3})^3 = \frac{1}{27}$; e. $\sqrt[3]{64} = 4$; f. $9^y = 150$.

[Manba: OpenStax §1.5, 254-262 mashqlar]

Yechish. a. $\log_2 8 = 3$. b. $\log_4 \frac{1}{16} = -2$. c. $\log 100 = 2$. d. $\log_{1/3} \frac{1}{27} = 3$. e. $64^{1/3} = 4$ demak $\log_{64} 4 = \frac{1}{3}$. f. $\log_9 150 = y$.

4. (264-266) Eskizing; soha, qiymatlar to'plami, vertikal asimptotani bering: a. $f(x) = 3 + \ln x$; b. $f(x) = \ln(x - 1)$; c. $f(x) = \ln(-x)$.

[Manba: OpenStax §1.5, 264-266 mashqlar]

Yechish. a. $\ln x$ 3 ga yuqoriga surilgan: soha $(0, \infty)$, range $\mathbb{R}$, vertikal asimptota $x = 0$. b. 1 ga o'ngga surilgan: soha $(1, \infty)$, range $\mathbb{R}$, asimptota $x = 1$. c. y -o'qqa aks ettirilgan: soha $(-\infty, 0)$, range $\mathbb{R}$, asimptota $x = 0$.

5. (270-275) Logarifm qonunlari bilan yoying: a. $\log x^4$; b. $\log \frac{9a^3}{b}$; c. $\ln a \sqrt[3]{b}$; d. $\log_5 \sqrt{125xy^3}$.

[Manba: OpenStax §1.5, 270-275 mashqlar]

Yechish. a. $4 \log x$. b. $\log 9 + 3 \log a - \log b = 2 \log 3 + 3 \log a - \log b$. c. $\ln a + \frac{1}{3} \ln b$. d. $\frac{1}{5} \log_5 (125xy^3) = \frac{1}{5} (3 + \log_5 x + 3 \log_5 y)$.

6. (276-283) Aniq yeching: a. $5^x = 125$; b. $e^{3x} - 15 = 0$; c. $8^x = 4$; d. $4^{x+1} - 32 = 0$; e. $3^{x/14} = \frac{1}{10}$; f. $10^x = 7.21$.

[Manba: OpenStax §1.5, 276-283 mashqlar]

Yechish. a. $x = 3$ ($125 = 5^3$). b. $3x = \ln 15$, $x = \frac{\ln 15}{3} \approx 0.9027$. c. $2^{3x} = 2^2$, $x = \frac{2}{3}$. d. $4^{x+1} = 2^5 \Rightarrow 2^{2x+2} = 2^5 \Rightarrow x = \frac{3}{2}$. e. $\frac{x}{14} \ln 3 = -\ln 10$, $x = -\frac{14 \ln 10}{\ln 3} \approx -29.32$. f. $x = \log 7.21 \approx 0.8580$.

7. (284-291) Mumkin bo'lsa aniq yeching: a. $\log_3 x = 0$; b. $\log_5 x = -2$; c. $\log_3(x + 5) = 0$; d. $\log_2(2x - 7) = 0$; e. $\ln \sqrt{x + 3} = 2$; f. $\log_6(x + 9) + \log_6 x = 2$; g. $\log_4(x + 2) - \log_4(x - 1) = 0$.

[Manba: OpenStax §1.5, 284-291 mashqlar]

Yechish. a. $x = 3^0 = 1$. b. $x = 5^{-2} = \frac{1}{25}$. c. $x + 5 = 3^0 = 1$, $x = -4$; sohani tekshiramiz $x + 5 > 0$: $1 > 0$ OK, demak $x = -4$ yaramadi deyilmaydi, ishlaydi ($\log_3 1 = 0$). d. $2x - 7 = 1$, $x = 4$. e. $\sqrt{x + 3} = e^2$, $x = e^4 - 3 \approx 51.60$. f. Birlashtiramiz: $\log_6(x(x + 9)) = 2 \Rightarrow x(x + 9) = 36 \Rightarrow x^2 + 9x - 36 = 0 \Rightarrow (x + 12)(x - 3) = 0$; soha $x > 0$ talab qiladi, demak $x = 3$. g. $\log_4 \frac{x+2}{x-1} = 0 \Rightarrow \frac{x+2}{x-1} = 1 \Rightarrow x + 2 = x - 1$: imkonsiz; yechim yo'q.

8. (292-297) Asosni almashtirib, to'rt xonagacha taxminlang: a. $\log_5 47$; b. $\log_2 82$; c. $\log_{0.5} 211$; d. $\log_{0.2} 0.452$.

[Manba: OpenStax §1.5, 292-297 mashqlar]

Yechish. (41) bo'yicha: a. $\frac{\ln 47}{\ln 5} \approx 2.3922$. b. $\frac{\ln 82}{\ln 2} \approx 6.3576$. c. $\frac{\ln 211}{\ln 0.5} \approx -7.7181$. d. $\frac{\ln 0.452}{\ln 0.2} \approx 0.3536$.

9. (307) Bakteriya koloniyasi 12 soatda ikkilanadi, boshlanishda 1000. a. $Q = Q_0 e^{kt}$ yordamida k ni toping; b. 200,000 ga yetishi qancha vaqt oladi?

[Manba: OpenStax §1.5, 307-mashq]

Yechish. a. Ikkilanish: $2 = e^{12k}$, demak $k = \frac{\ln 2}{12} \approx 0.0578$ soatiga. b. $200000 = 1000e^{kt}$: $e^{kt} = 200$, $t = \frac{\ln 200}{k} = \frac{12 \ln 200}{\ln 2} \approx 92.1$ soat.

10. (308) Quyvon populyatsiyasi har 6 oyda ikkilanadi, boshlanishda 120. a. $P = P_0 a^t$ yordamida a ni toping; b. qachon 3500 ga yetadi?

[Manba: OpenStax §1.5, 308-mashq]

Yechish. a. 6 oylik davr uchun $a = 2$ (yiliga o'tkazsa, yil birligida $a = 4$; 6 oyda $a = 2$). b. $3500 = 120 \cdot 2^t$ (t 6 oylik birlarda): $2^t = \frac{175}{6} \approx 29.17$, $t = \log_2 29.17 = \frac{\ln 29.17}{\ln 2} \approx 4.87$ davr, ya'ni taxminan 2.4 yil.

11. (309) 1906-yilgi San-Fransisko zilzilasi magnitudasi 8.3; o'sha yilgi yapon zilzilasi 4.9. San-Fransisko qanchalik kuchliroq bo'lgan (Rixter shkalasi)?

[Manba: OpenStax §1.5, 309-mashq]

Yechish. Rixter intensivlik nisbati: $10^{8.3-4.9} = 10^{3.4} \approx 2512$: San-Fransisko zilzilasi taxminan 2500 barobar kuchliroq bo'lgan — (40) logarifmik shkalasining jonli ko'rgazmasi.

7.3 OpenStax §2.2 — jadval asosidagi mashqlar (42-45) yechilgan

1. Jadvaldan ($t = 0.1, 0.01, 0.001, 0.0001$) $\lim_{t \rightarrow 0^+} \frac{\cos t}{t}$ ni baholang.

[Manba: OpenStax §2.2, 42-mashq]

Yechish. Qiymatlar: $t = 0.1$: $\frac{0.995004}{0.1} = 9.95$; $t = 0.01$: 99.95; $t = 0.001$: 999.95; $t = 0.0001$: 9999.95. Qiymatlar chegarasiz o'sadi: $\lim_{t \rightarrow 0^+} \frac{\cos t}{t} = +\infty$ (surat $\rightarrow 1$, holbuki mahraj $\rightarrow 0^+$ — (57) dagi kabi $K/0$ hold).

2. Jadvaldan $\lim_{x \rightarrow 2} \frac{1 - \frac{2}{x}}{x^2 - 4}$ ni baholang.

[Manba: OpenStax §2.2, 43-mashq]

Yechish. Avval soddalashtiramiz: $x \neq 2$ uchun $\frac{1 - \frac{2}{x}}{x^2 - 4} = \frac{\frac{x-2}{x}}{(x-2)(x+2)} = \frac{1}{x(x+2)}$. Jadval qiymatlari: $x = 1.99$: $1.99 \cdot 3.99 = 7.9401$, $\frac{1}{7.9401} \approx 0.1259$; $x = 2.01$: $2.01 \cdot 4.01 = 8.0601$, ≈ 0.1241 — $0.125 = \frac{1}{8}$ ga intiladi. Algebra tasdiqlaydi: limit $\frac{1}{2 \cdot 4} = \frac{1}{8}$.

- T Jadval taxmini, so'ng grafik tekshiruvi: $\lim_{\theta \rightarrow 0} \sin \frac{\pi}{\theta}$.

[Manba: OpenStax §2.2, 44-mashq]

Yechish. Qiymatlar: $\theta = 0.1$: $\sin(10\pi) = 0$; 0.01 : $\sin(100\pi) = 0$; 0.001 : $\sin(1000\pi) = 0$; 0.0001 : 0. Jadval 0 ni maslahat beradi — lekin grafik 0 yaqinida -1 va 1 orasida cheksiz tez tebranadi (masalan $\theta = \frac{2}{\pi}$ beradi $\sin \frac{\pi^2}{2} \approx 1$). Limit mavjud emas; jadval ishlamadi, chunki $\frac{1}{n}$ namunaviy nuqtalari aynan sinusning nollari. Bu jadval usuli haqidagi ogohlantiruvchi mashq.

T Jadval taxmini, so'ng grafik tekshiruvi: $\lim_{\alpha \rightarrow 0^+} \frac{1}{\alpha} \cos \frac{\pi}{\alpha}$.

[Manba: OpenStax §2.2, 45-mashq]

Yechish. $\alpha = \frac{1}{n}$ da: $\frac{1}{\alpha} \cos(n\pi) = n(-1)^n$ — qiymatlar $\pm n$ kattalikda o'sadi va ishora almashtiradi. Grafik tobora katta musbat va manfiy qiymatlar orasida tebranadi: limit mavjud emas ($+\infty$ yoki $-\infty$ ham emas — ishoralar almashadi). Omadli nuqtalardagi ($\alpha = \frac{1}{2n}$, beradi $+2n \rightarrow +\infty$) jadval teskari yo'nalishda chalg'itardi.

7.4 Strang, Introduction to Linear Algebra 4e — tanlangan muammolar yechilgan

1. (§1.1) $\mathbf{u} = (2, 1)$ uchun barcha $c\mathbf{u}$ ni va $\mathbf{u} = (2, 1)$, $\mathbf{v} = (1, 2)$ uchun barcha $c\mathbf{u} + d\mathbf{v}$ birikmalarni tavsiflang; $c\mathbf{u} + d\mathbf{v}$ birikmalar qachon tekislikni to'ldiradi?

[Manba: Strang Intro 4e, §1.1 g'oyalar to'plami]

Yechish. $c\mathbf{u}$ $(2, 1)$ dan o'tuvchi to'g'ri ni to'ldiradi: undagi har bir vektor ko'paytma. $c\mathbf{u} + d\mathbf{v}$ birikmalari $\mathbf{u}$ va $\mathbf{v}$ kollinear bo'lmasa bosh nuqtadan o'tuvchi tekislikni to'ldiradi; bu yerda $\det \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = 3 \neq 0$, demak ikki yo'nalish mustaqil va birikmalar butun $\mathbb{R}^2$ ni to'ldiradi. Aynan shu mezon — noldan farqli determinant — 4-haftada teskarilik sifatida qayt keladi (108).

2. (§1.2) Skalyar ko'paytmalarni hisoblang: $\mathbf{u} = (1, 2, 1)$, $\mathbf{v} = (3, 0, 2)$ uchun $\mathbf{u} \cdot \mathbf{v}$; ikkalasiga ham perpendikulyar $\mathbf{w}$ vektorni toping.

[Manba: Strang Intro 4e, §1.2 muammolar to'plami]

Yechish. $\mathbf{u} \cdot \mathbf{v} = 3 + 0 + 2 = 5$. Perpendikulyar vektor — vektor ko'paytma: $\mathbf{w} = \mathbf{u} \times \mathbf{v} = \begin{bmatrix} 2 \cdot 2 - 1 \cdot 0 \\ 1 \cdot 3 - 1 \cdot 2 \\ 1 \cdot 0 - 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ -6 \end{bmatrix}$; tekshirish: $\mathbf{w} \cdot \mathbf{u} = 4 + 2 - 6 = 0$ va $\mathbf{w} \cdot \mathbf{v} = 12 + 0 - 12 = 0$.
✓

3. (§1.2) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ qachon manfiy bo'ladi va bu burchak haqida nima deydi? Qachon nol?

[Manba: Strang Intro 4e, §1.2 tushuncha]

Yechish. Aynan $\cos \theta < 0$ bo'lganda manfiy, ya'ni burchak 90° dan oshadi (keng); $\theta = 90^\circ$ da nol — perpendikulyarlik. Bu 2-hafta TA muammolarida ishlatilgan tasnif.

4. (§2.5) $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $\mathbf{b} = (3, 4)^T$ bilan $A\mathbf{x} = \mathbf{b}$ ni A^{-1} orqali yeching.

[Manba: Strang Intro 4e, §2.5 muammolar to'plami]

Yechish. $\det A = 3$: $A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$; $\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{3} \begin{bmatrix} 2 \cdot 3 - 4 \\ -3 + 8 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 5/3 \end{bmatrix}$.
Tekshirish: $2 \cdot \frac{2}{3} + \frac{5}{3} = 3$, $\frac{2}{3} + 2 \cdot \frac{5}{3} = 4$. ✓

5. (§2.5) $A\mathbf{x} = \mathbf{0}$ ning noldan farqli yechimi teskari matritsani nega istisno qiladi? Konspekt “hech bir matritsa $\mathbf{0}$ ni orqaga $\mathbf{x}$ ga qaytara olmaydi” deb atagan 2 gaplik asosni bering.

[Manba: Strang Intro 4e, §2.5 4-eslatma]

Yechish. Agar $A\mathbf{x} = \mathbf{0}$ bo'lib $\mathbf{x} \neq \mathbf{0}$ va A ning teskarisi bo'lsa, u holda $\mathbf{x} = A^{-1}A\mathbf{x} = A^{-1}\mathbf{0} = \mathbf{0}$ — zidlik. Nol vektorning “oldindan tasviri muammosi” faqat yagona xaritalanishda bo'lmaydi; nol-vektor buni isbotlaydi.

6. (§2.5) A va C teskarili 3×3 matritsalar bo'lsa, $A + C$ teskarilimi? AC -chi? $A^{-1}C$ -chi?

[Manba: Strang Intro 4e, §2.5 muammolar to'plami]

Yechish. $A + C$: albatta emas (masalan $C = -A$ beradi 0); teskarililar yig'indisi singulyar bo'lishi mumkin. AC : ha, $(AC)^{-1} = C^{-1}A^{-1}$ bilan (113). $A^{-1}C$: ha, ikki teskarilining ko'paytmasi.

7.5 Strang, Linear Algebra and its Applications 4e, §1.6 (Teskari va transponirlar) — asosiy natijalar va bitta yechilgan misol

§1.6 dagi tayinlangan o'qish katalogga uch band qo'shadi: $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$ ning

Gauss-Jordan usuli bilan yechilgan teskarilishi, $\det A^{-1} = \frac{1}{\det A}$ ayniyati va transponir ayniyatlari.

1. $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$ uchun Gauss-Jordan bajaring va A^{-1} ni ko'rsating.

[Manba: Strang LAiA 4e, §1.6 1-misol]

Yechish. I bilan qo'shib yo'q qilamiz: pivotlar 2, -8 , 1 ($R_2 - 2R_1$, $R_3 + R_1$, so'ng qisqartirilgan qatorlarga $R_3 + \frac{3}{4}R_2$ dan keyin). Pivotlar ustidagilarni tozalashni davom ettirib, nihoyat qatorlarni 2, -8 , 1 ga bo'lsak

$$A^{-1} = \begin{bmatrix} 12/8 & -5/8 & -6/8 \\ 8/8 & -3/8 & -2/8 \\ -1/8 & 1/8 & 1/8 \end{bmatrix} = \frac{1}{8} \begin{bmatrix} 12 & -5 & -6 \\ 8 & -3 & -2 \\ -1 & 1 & 1 \end{bmatrix}$$

bo'ladi. Determinant pivotlar ko'paytmasiga teng $2(-8)(1) = -16$, va haqiqatdan $A^{-1} \det A$ ga bo'lishni ko'rsatadi (kofaktorlarning ishora joylashuvi bilan — ekvivalent ravishda $A^{-1} = \frac{\text{adj } A}{\det A}$).

2. §1.6 da ishlatilgan determinant-transponir ayniyatlarini ko'rsating: $\det A^T$, $\det(AB)$, $\det A^{-1}$.

[Manba: Strang LAiA 4e, §1.6]

Yechish. $\det A^T = \det A$; $\det(AB) = \det A \det B$ (1-jadval); va $B = A^{-1}$ qo'ysak, $\det(A) \det(A^{-1}) = \det I = 1$, demak

$$\det A^{-1} = \frac{1}{\det A},$$

bu aynan “teskarili $\iff \det A \neq 0$ ” bayonotini bildiradi va 3-hafta determinant katalogini 4-hafta teskari katalogi bilan bog'laydi.

3. $(A^{-1})^T = (A^T)^{-1}$ ni isbotlang va $A^T A$ nega manfiy bo'lmagan diagonal bilan simmetrik ekanini tushuntiring.

[Manba: Strang LAiA 4e, §1.6]

Yechish. $AA^{-1} = I$ ni transponirlash $(A^{-1})^T A^T = I$ beradi, bu aynan $(A^{-1})^T$ ning A^T teskarisi ekanini bildiradi. Simmetriya uchun: $(A^T A)^T = A^T (A^T)^T = A^T A$; diagonal elementlar $\sum_k a_{ki}^2 \geq 0$ — kvadratlar yig'indisi — 3-hafta ma'ruza muammolaridagi CC^T ortidagi fakt.